

● FCV-5-2VT

● Vertical Multi-Stage Pump

● Features :-

Stainless Steel 304

3 phase

220-380V

60HZ

IE3

2 pole

3500 rpm

Class 155(F)

Protection: IP55

● Product Photo :-



● Proformance Table :-

Model	Power		Max flow	Max head	Voltage	Inlet	Outlet
	KW	HP	m³/h	meters		mm	mm
FCV-5-2VT	0.55	0.75	10	19	220-380V	DN32	DN32

● Design

Vertical multistage in Line pump where Discharge and suction ports are located on one line to make it easier for connection access, it's either in Oval Victualic threaded to NPT cast iron discharge casing design or Slip on flanged and drilled to Universal standard which suite both DIN and ANSI standards.

● Applications: -

Water supply

Industry

Liquid transfer

Water treatment

Irrigation

● FCV Coding:

F: FUJIKA

C: Clean water

V: Vertical pump

● Classifications:

FCV: Standard SS304

FCV-N: SS316

V: Oval flange

R: Round flange

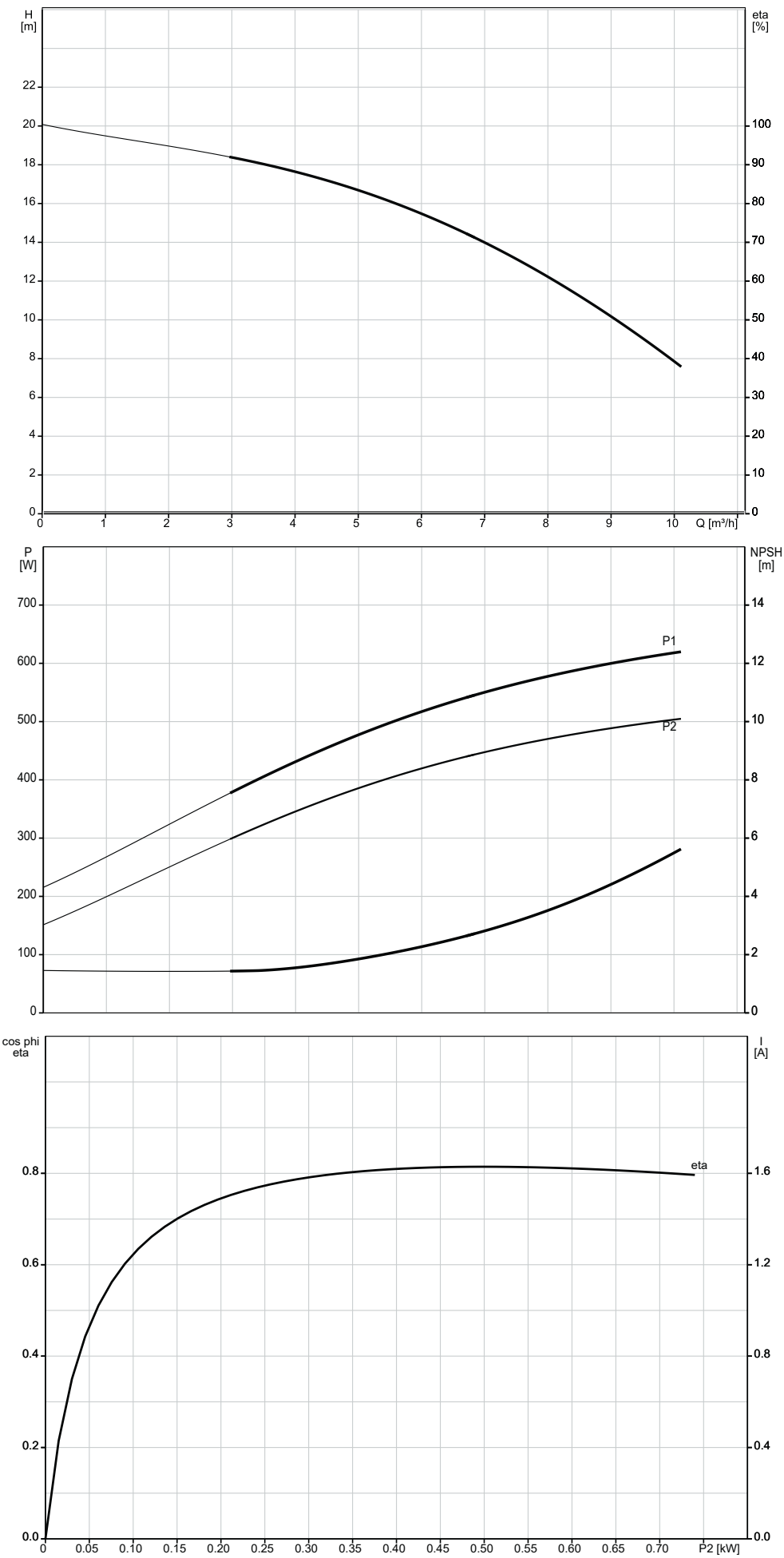
T: Three phase

● Working conditions: -

Temperature range: -20 Up to 104 °C

Ambient temperatue 50 °C

PERFORMANCE CURVE



Model: FCV-5-2VT



Design: Vertical multistage in Line pump where Discharge and suction ports are located on one line to make it easier for connection access, it's either in Oval Victualic threaded to NPT cast iron discharge casing design or Slip on flanged and drilled to Universal standard which suite both DIN and ANSI standards.

Construction of Material:

FCV basic design is made of :

Stainless steel imeller, sleeves and shaft , and cast iron discarge casing.

FCV-N

All wetted parts are made from SUS316 design.

The Shaft Mechanical seal is cartridge pull out type for easier replacement.

Mechanical Seal is EPDM /SUS304 for FCV and VITON / SUS316 for FCV-N Ensuring high reliability.

Coupling:

Shaft Coupling are rigid split type PJE Coupling.

Motor Type:

IE3 / IE2, 3-phase AC squirrel cage synchronous, high efficiency, TEFC fan cooled, IP55 Jetting protected.

General description :

The high efficiency ratings are in conformity to grade 3 efficiency value GB18613-2012 and IE of IEC60034-30 standard, but also the efficiency grade and installation dimensions are totally in conformity to IEC standard.

Motor shape is square motor as standard and round motor for larger sizes.

General specifications:

Operation factor: S1 (Continuous duty)

Insulation class: F

Protection class: IP55

Cooling ref.: IC411

Mount type: B35/B6

Ambient temperature: (-15 Up to 60 °C)

Altitude: ≤1000m

Motor paint finish is made with the most high technology with double painting and epoxy coating.

Motor housing construction are alaminum for motor rated up to 15HP and high grade cast iron for motors above rating.

The colour code: RAL 9005 deep jet black.

A standard split coupling connects the pump and motor shaft.

It's enclosed in the pump head/motor stool by means of two coupling guards.

The pump head and flange for motor mounting bracket is made in one piece (cast iron). the pump head cover is a seperate component (stainless steel).

Pump head cover stage has 1/2" priming plug and vent screw.

The pump is fitted with a balanced O-ring seal unit with a rigid torque-transmission system.

This seal type is assembled in a cartridge unit which makes replacement safe and easy.

In compliance with:

ENERGY EFFICIENCY ASIA JAPAN, IE3 IE2 IE1

MEPS (Minimum energy performance standard)

Current Regulation Energy saving act / top runner program standard

JIS C 4034-30

Power supply system

Three phase

Minimum energy performance IE3

Operated with inverter frequency: ok

Output (KW): 0.75 Up to 55KW

Number of poles: 2,4,6

Voltage (V): Up to 1000V

Frequency (HZ): 50HZ,60HZ,50/60HZ

Service duty: S1 ≥ 100%

Cooling method: tefc

Degree of protection: IP55

Area classification: safe area

Altitude: all

Ambient temperature: from (-20 °C and above)

Pump technical data

Rated flow: 5 m³/h
 Rated head: 15 meter
 Pump speed: 3450 rpm
 Nominal pressure PN: 25 bar
 Maximum pressure rating: 125% than shutoff pressure
 Pump mounting: vertical base mounting pump
 Shaft seal arrangement: single face cartridge type mechanical seal
 Seal code: Sic/Sic/Viton/SS316

Materials:

Wetted parts: Stainless steel EN 1.4408 SUS316
 Bearing: CU or SKF life sealed
 greased bearing
 Insulation: vertical
 Ambient temperature: 60 °C
 Maximum operating pressure: 25 bar
 Maximum pressure at stated temperature: 25 bar @ 0-90 °C

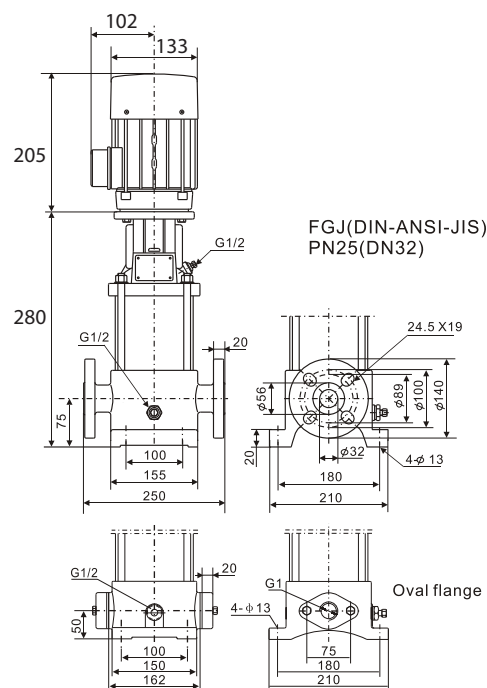
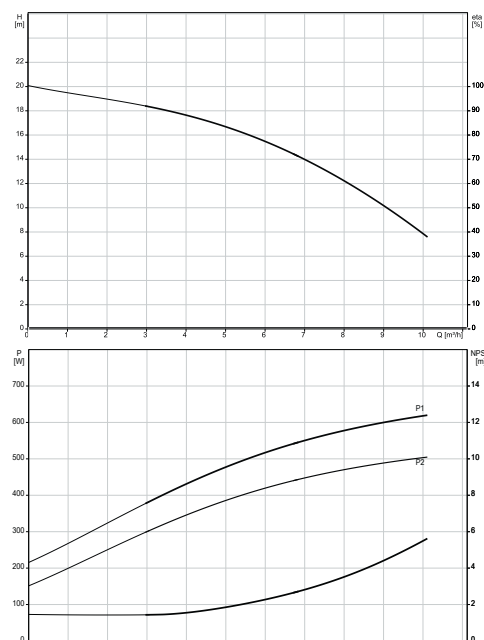
Motor type: by FUJIKA

IE efficiency class: IE3
 Rated power P2: 0.55 KW
 Power (P2) required by pump: 0.55 KW
 Frequency: 60HZ
 Rated voltage: 3 X 220-255V Δ / 380-440V Y
 Rated current: 2.4A / 1.4A
 Starting current: 500-600%
 Cos phi - power: 0.81
 Rotational speed: 3400-3450 rpm
 IE efficiency: IE3 76.8%
 Minimum efficiency: 71%
 Maximum efficiency: 83%
 Number of poles: 2
 Enclosure class: IP55
 Insulation class: F (IEC)

Dimensional:

Connection: UNIVERSAL flange drilled to both DIN & ANSI standards.
 Flange size for motor:
 Electrical data: IEC
 Motor standard: IEC DIN
 Approvals: CE
 Certified: IECEE, Coc
 Approvals for drinking water: WRAS
 Performance curve: ISO9906 2012

Gross weight: 25 KG
 Shipping volume: 0.054 m³
 Country of origin: PRC



Material resistivity to chemicals:

Stainless steel 316 material composition Up to :
18% chromium, 10% nickel, and 2% molybdenum.

Allows it to be used in variable applications and
that require superior corrosion resistance and
hygienic.

Applications such as:

Acidic water
Dissolved Chlorine
Sea water
Falls and Fountains
Water treatment
Diesel Fuel
Aggressive oil
Cooking oil
Drinking water
Juices
Water treatment
Food industry
Pharmaceutical Industry
Hygienic process
Fertilization
Fisheries
Irrigation

Characteristics:

The 316 stainless steel is used in salt and other
chlorides applications exposure and lasting
durability
steel over 304.

316 stainless steel's added molybdenum, makes it
much more corrosion resistant than 304 or other
stainless steel alloys. This allows 316 stainless steel
to be used in highly acidic environments that
other steels can't tolerate. 316's corrosion-resist-
ant properties also make it ideal in food process-
ing, pharmaceutical and medical applications due
to its ability to withstand the disinfectants and
high temperatures media.

Exposure to salt and chlorides can cause havoc on
304 stainless steel alloys. Salt can cause 304 stain-
less steel to rust by attacking the protective oxide
layer, and 304 steel is also more easily wear from
the effects of salt and chlorides. However, 316
stainless steel has all of the desirability properties of
304 stainless steel, but with the added benefit of
molybdenum, which allows it to withstand expo-
sure to salt and chlorides for more durability.

Austenitic stainless steels generally show high
corrosion rates when exposed to concentrated,
de-aerated alkaline aqueous environment at tem-
peratures above 60 °C.

Permissible corrosion rates of 0.05–2 mm/year.

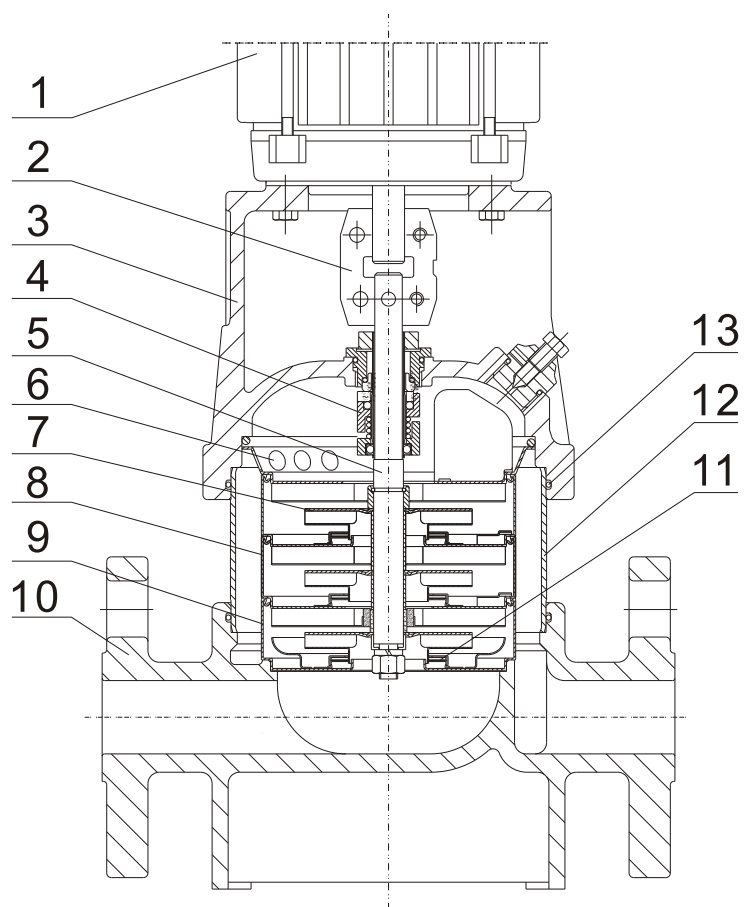
Caution:

The common stainless steel sheets types, 304 and
316 pumps should be considered non-resistant to
hydrochloric acid at any concentration and temper-
ature.

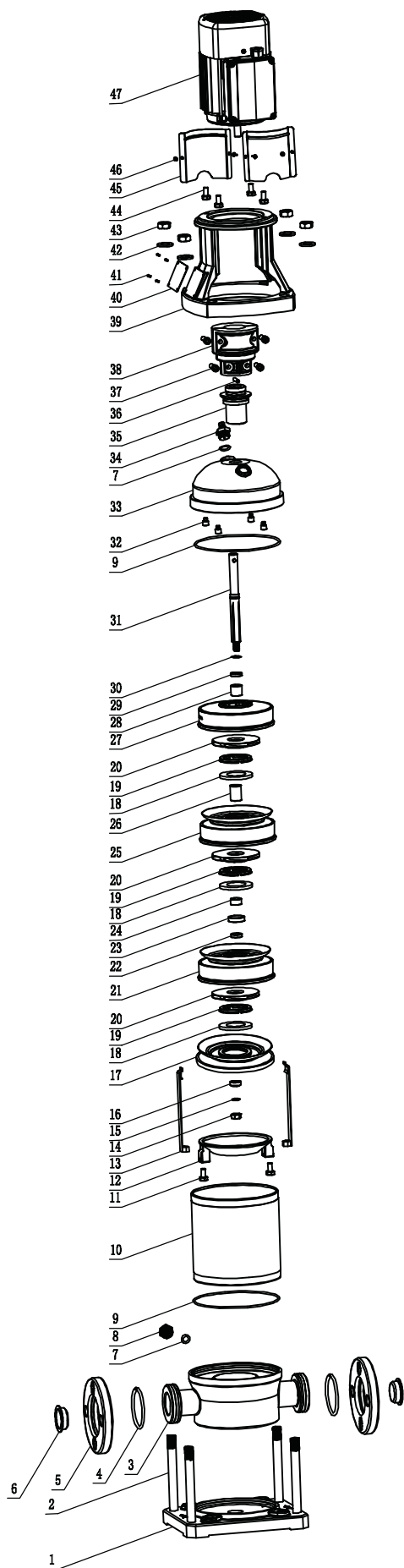
For Sulfuric acid:

SUS304 is only resistant up to 1% acid while type
316 resistant is up to 10% acid at 50° C temp.

SECTIONAL DRAWING



No	Description	Material	EN/DIN	AISI/ASTM
1	Motor	X	X	X
2	Shaft connector	X	X	X
3	Pump head	Cast iron	EN-JL 1030	ASTM25B
4	Mechanical seal	X	X	X
5	Shaft	S.S	X	AISI420
6	Outlet	S.S	1.4301	AISI304
7	Impeller	S.S	1.4301	AISI304
8	Hydraulic stack	S.S	1.4301	AISI304
9	Inlet	S.S	1.4301	AISI304
10	Pump body	Cast iron	EN.JL1030	ASTM25B
11	Neck ring	PTEF	X	X
12	Outer sleeve	S.S	1.4301	AISI316
13	O-ring	EDM/FKM	X	X



NO	Name	Spec.	Material	QTY
1	Base		HT200	1
2	Two-head bolt stud	M16XL	Chrome	4
3	Pump body		SUS304	1
4	Flange clasp		SUS304	2
5	Flange		SUS304	2
6	Dust cap	G1	PP	2
7	Ring	d18X2.4	NBR	2
8	Drain valve assembly		SUS304	1
9	Ring	d170X3.1	NBR	2
10	Housing		SUS316	1
11	Hexagon bolt	M8X20	SUS304	2
12	Fixed board		SUS316	1
13	Tension rod		SUS316	2
14	Type 1 non-metallic insert hexagon lock nut	M10	SUS316	1
15	Spring washer	ø10	SUS316	1
16	Impeller fix ring		SUS316	1
17	Inlet section		SUS316	1
18	Impeller fix ring		PTFE	N
19	Mouthpiece outer ring		SUS316	N
20	impeller		SUS316	N
21	Diffusor with bearing		SUS316	N
22	Short sleeve	L=4.5	SUS316	N
23	Bearing sleeve		Hard alloy	N
24	Short sleeve	L=10.5	SUS316	N
25	Diffusor		SUS316	N
26	Long sleeve	L=29	SUS316	N
27	Outlet section		SUS316	1
28	Short sleeve	L=19	SUS316	1
29	Snap ring seat		SUS316	1
30	Clasp		SUS316	1
31	Pump shaft		SUS316	1
32	Clasp seat shock colloidal		FI	4
33	Pump cover		SUS316	1
34	Discharge valve screw		SUS316	2
35	Mechanical seal	NJK-16-Q18FF	Carbon silicon carbide-FT	1
36	Cylindrical pin	ø6X25	SUS316	1
37	Hexagon socket head cap screw	M6X20	A3	4
38	Coupling			1
39	Motor base		HT200	1
40	Name plate		AL	1
41	Plate rivets	ø2.5	H62	4
42	Washer	ø16		4
43	Hexagonal nut	M16		4
44	Hexagonal bolt	M10X30	A3	4
45	Protective cover		SUS304	2
46	Cross recessed pan head screw and washer assemblies	M4X8	SUS304	4
47	Motor			1

● FCV-5-3VT

● Vertical Multi-Stage Pump

● Features :-

Stainless Steel 304

3 phase

220-380V

60HZ

IE3

2 pole

3500 rpm

Class 155(F)

Protection: IP55

● Product Photo :-



● Proformance Table :-

Model	Power		Max flow	Max head	Voltage	Inlet	Outlet
	KW	HP	m ³ /h	meters		mm	mm
FCV-5-3VT	1.1	1.5	10	28	220-380V	DN32	DN32

● Design

Vertical multistage in Line pump where Discharge and suction ports are located on one line to make it easier for connection access, it's either in Oval Victualic threaded to NPT cast iron discharge casing design or Slip on flanged and drilled to Universal standard which suite both DIN and ANSI standards.

● Applications: -

Water supply

Industry

Liquid transfer

Water treatment

Irrigation

● FCV Coding:

F: FUJIKA

C: Clean water

V: Vertical pump

● Classifications:

FCV: Standard SS304

FCV-N: SS316

V: Oval flange

R: Round flange

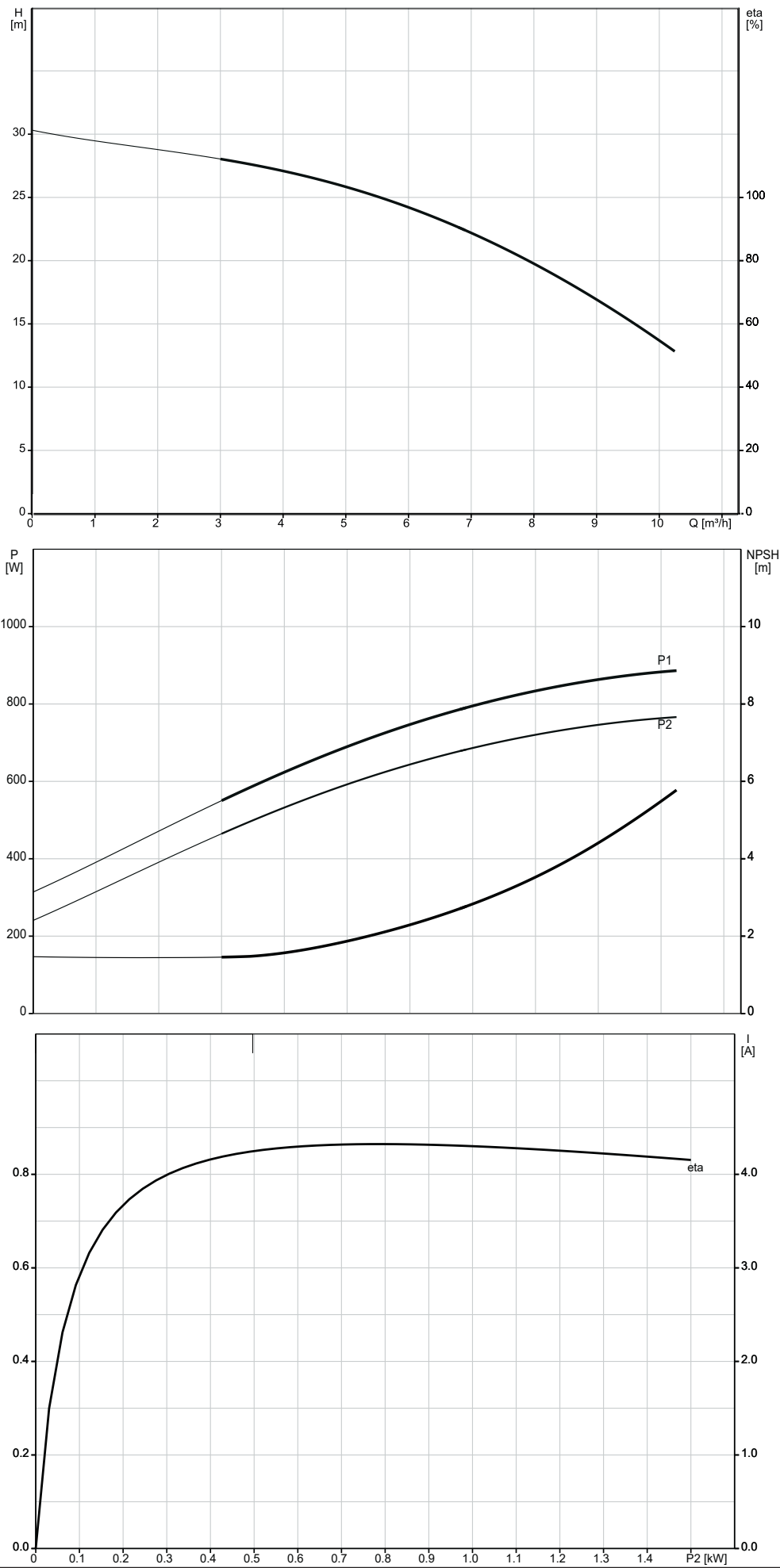
T: Three phase

● Working conditions: -

Temperature range: -20 Up to 104 °C

Ambient temperatue 50 °C

PERFORMANCE CURVE



Model: FCV-5-3VT



Design: Vertical multistage in Line pump where Discharge and suction ports are located on one line to make it easier for connection access, it's either in Oval Victualic threaded to NPT cast iron discharge casing design or Slip on flanged and drilled to Universal standard which suite both DIN and ANSI standards.

Construction of Material:

FCV basic design is made of :

Stainless steel imeller, sleeves and shaft , and cast iron discarge casing.

FCV-N

All wetted parts are made from SUS316 design.

The Shaft Mechanical seal is cartridge pull out type for easier replacement.

Mechanical Seal is EPDM /SUS304 for FCV and VITON / SUS316 for FCV-N Ensuring high reliability.

Coupling:

Shaft Coupling are rigid split type PJE Coupling.

Motor Type:

IE3 / IE2, 3-phase AC squirrel cage synchronous, high efficiency, TEFC fan cooled, IP55 Jetting protected.

General description :

The high efficiency ratings are in conformity to grade 3 efficiency value GB18613-2012 and IE of IEC60034-30 standard, but also the efficiency grade and installation dimensions are totally in conformity to IEC standard.

Motor shape is square motor as standard and round motor for larger sizes.

General specifications:

Operation factor: S1 (Continuous duty)

Insulation class: F

Protection class: IP55

Cooling ref.: IC411

Mount type: B35/B6

Ambient temperature: (-15 Up to 60 °C)

Altitude: ≤1000m

Motor paint finish is made with the most high technology with double painting and epoxy coating.

Motor housing construction are alaminum for motor rated up to 15HP and high grade cast iron for motors above rating.

The colour code: RAL 9005 deep jet black.

A standard split coupling connects the pump and motor shaft.

It's enclosed in the pump head/motor stool by means of two coupling guards.

The pump head and flange for motor mounting bracket is made in one piece (cast iron). the pump head cover is a seperate component (stainless steel).

Pump head cover stage has 1/2" priming plug and vent screw.

The pump is fitted with a balanced O-ring seal unit with a rigid torque-transmission system.

This seal type is assembled in a cartridge unit which makes replacement safe and easy.

In compliance with:

ENERGY EFFICIENCY ASIA JAPAN, IE3 IE2 IE1

MEPS (Minimum energy performance standard)

Current Regulation Energy saving act / top runner program standard

JIS C 4034-30

Power supply system

Three phase

Minimum energy performance IE3

Operated with inverter frequency: ok

Output (KW): 0.75 Up to 55KW

Number of poles: 2,4,6

Voltage (V): Up to 1000V

Frequency (HZ): 50HZ,60HZ,50/60HZ

Service duty: S1 ≥ 100%

Cooling method: tefc

Degree of protection: IP55

Area classification: safe area

Altitude: all

Ambient temperature: from (-20 °C and above)

Pump technical data

Rated flow: 5 m³/h
 Rated head: 23 meter
 Pump speed: 3450 rpm
 Nominal pressure PN: 25 bar
 Maximum pressure rating: 125% than shutoff pressure
 Pump mounting: vertical base mounting pump
 Shaft seal arrangement: single face cartridge type mechanical seal
 Seal code: Sic/Sic/Viton/SS316

Materials:

Wetted parts: Stainless steel EN 1.4408 SUS316
 Bearing: CU or SKF life sealed
 greased bearing
 Insulation: vertical
 Ambient temperature: 60 °C
 Maximum operating pressure: 25 bar
 Maximum pressure at stated temperature: 25 bar @ 0-90 °C

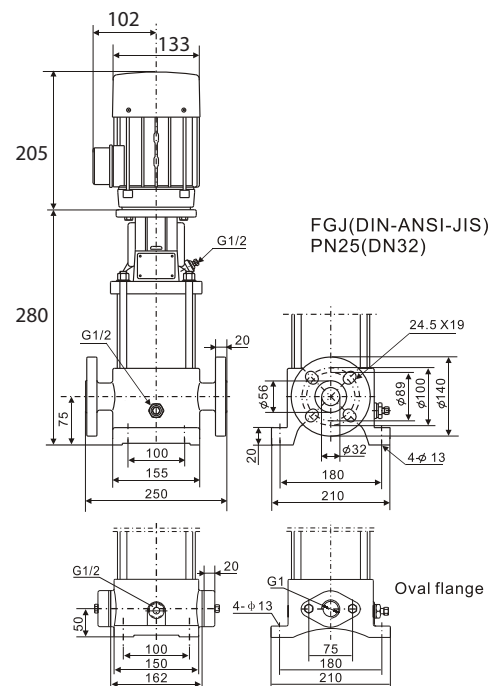
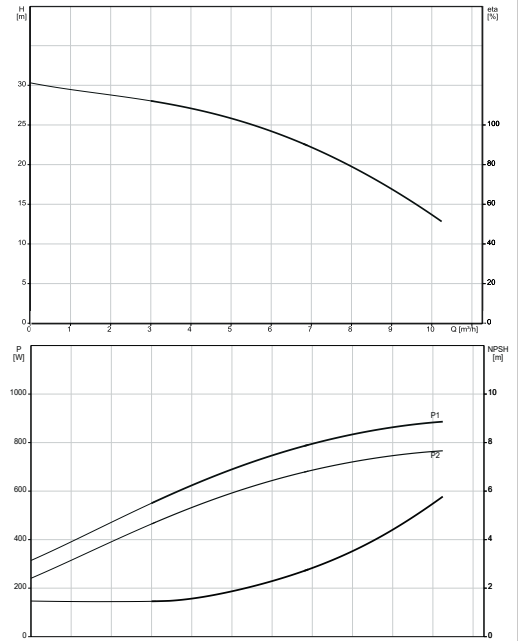
Motor type: by FUJIKA

IE efficiency class: IE3
 Rated power P2: 1.1 KW
 Power (P2) required by pump: 1.1 KW
 Frequency: 60HZ
 Rated voltage: 3 X 220-255V Δ / 380-440V Y
 Rated current: 4.5A / 2.6A
 Starting current: 440-500%
 Cos phi - power: 0.82
 Rotational speed: 3400-3450 rpm
 IE efficiency: IE3 77%
 Minimum efficiency: 76.5%
 Maximum efficiency: 82.9%
 Number of poles: 2
 Enclosure class: IP55
 Insulation class: F (IEC)

Dimensional:

Connection: UNIVERSAL flange drilled to both DIN & ANSI standards.
 Flange size for motor:
 Electrical data: IEC
 Motor standard: IEC DIN
 Approvals: CE
 Certified: IECEE, Coc
 Approvals for drinking water: WRAS
 Performance curve: ISO9906 2012

Gross weight: 28 KG
 Shipping volume: 0.063 m³
 Country of origin: PRC



Material resistivity to chemicals:

Stainless steel 316 material composition Up to :
18% chromium, 10% nickel, and 2% molybdenum.

Allows it to be used in variable applications and
that require superior corrosion resistance and
hygienic.

Applications such as:

Acidic water
Dissolved Chlorine
Sea water
Falls and Fountains
Water treatment
Diesel Fuel
Aggressive oil
Cooking oil
Drinking water
Juices
Water treatment
Food industry
Pharmaceutical Industry
Hygienic process
Fertilization
Fisheries
Irrigation

Characteristics:

The 316 stainless steel is used in salt and other
chlorides applications exposure and lasting
durability
steel over 304.

316 stainless steel's added molybdenum, makes it
much more corrosion resistant than 304 or other
stainless steel alloys. This allows 316 stainless steel
to be used in highly acidic environments that
other steels can't tolerate. 316's corrosion-resist-
ant properties also make it ideal in food process-
ing, pharmaceutical and medical applications due
to its ability to withstand the disinfectants and
high temperatures media.

Exposure to salt and chlorides can cause havoc on
304 stainless steel alloys. Salt can cause 304 stain-
less steel to rust by attacking the protective oxide
layer, and 304 steel is also more easily wear from
the effects of salt and chlorides. However, 316
stainless steel has all of the desirability properties of
304 stainless steel, but with the added benefit of
molybdenum, which allows it to withstand expo-
sure to salt and chlorides for more durability.

Austenitic stainless steels generally show high
corrosion rates when exposed to concentrated,
de-aerated alkaline aqueous environment at tem-
peratures above 60 °C.

Permissible corrosion rates of 0.05–2 mm/year.

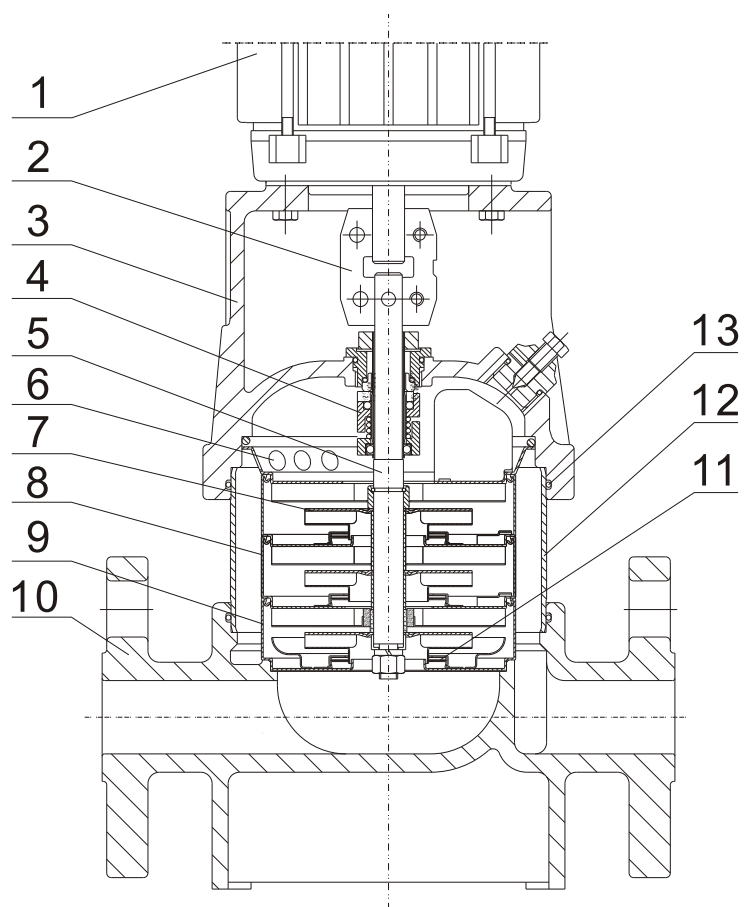
Caution:

The common stainless steel sheets types, 304 and
316 pumps should be considered non-resistant to
hydrochloric acid at any concentration and temper-
ature.

For Sulfuric acid:

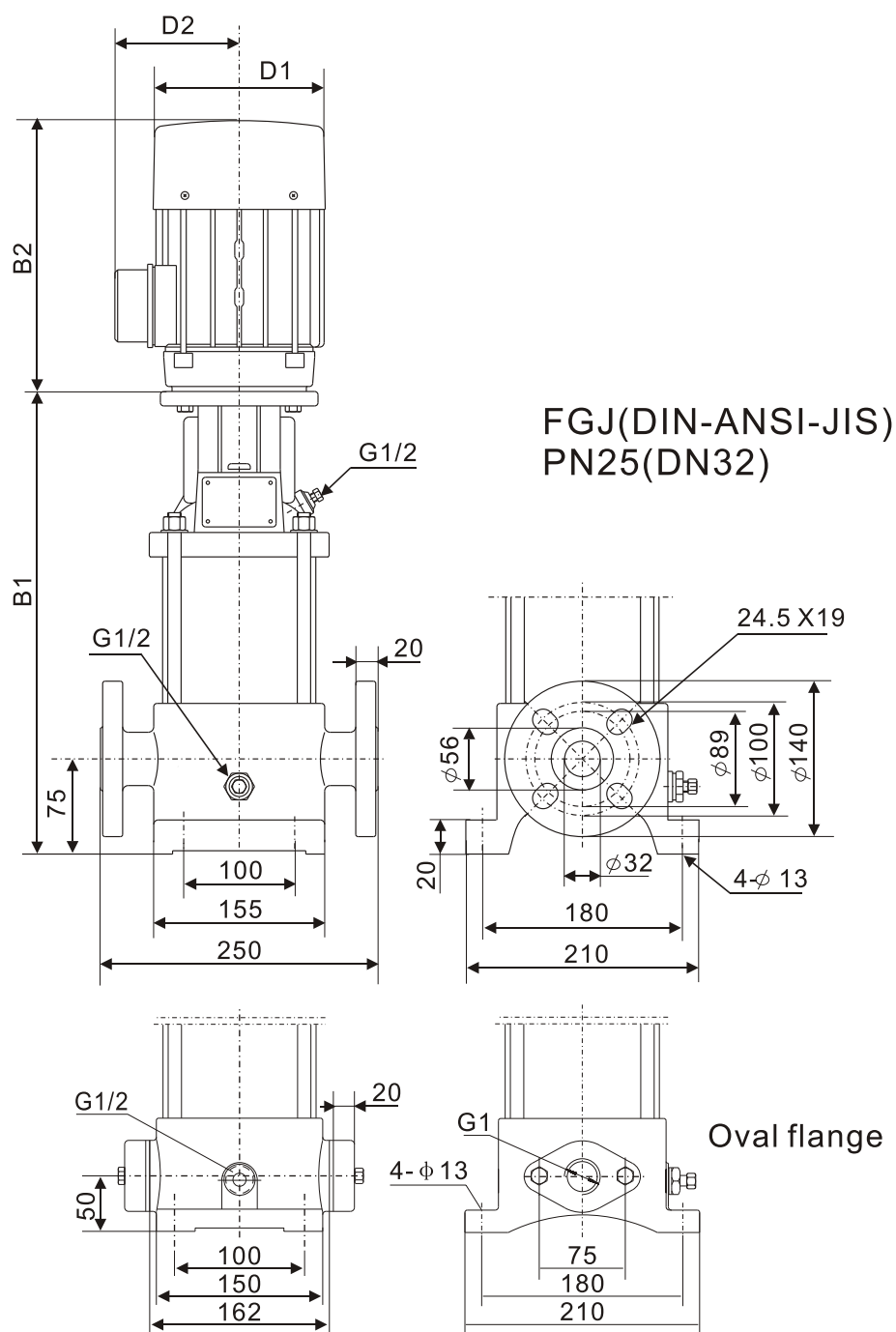
SUS304 is only resistant up to 1% acid while type
316 resistant is up to 10% acid at 50° C temp.

SECTIONAL DRAWING

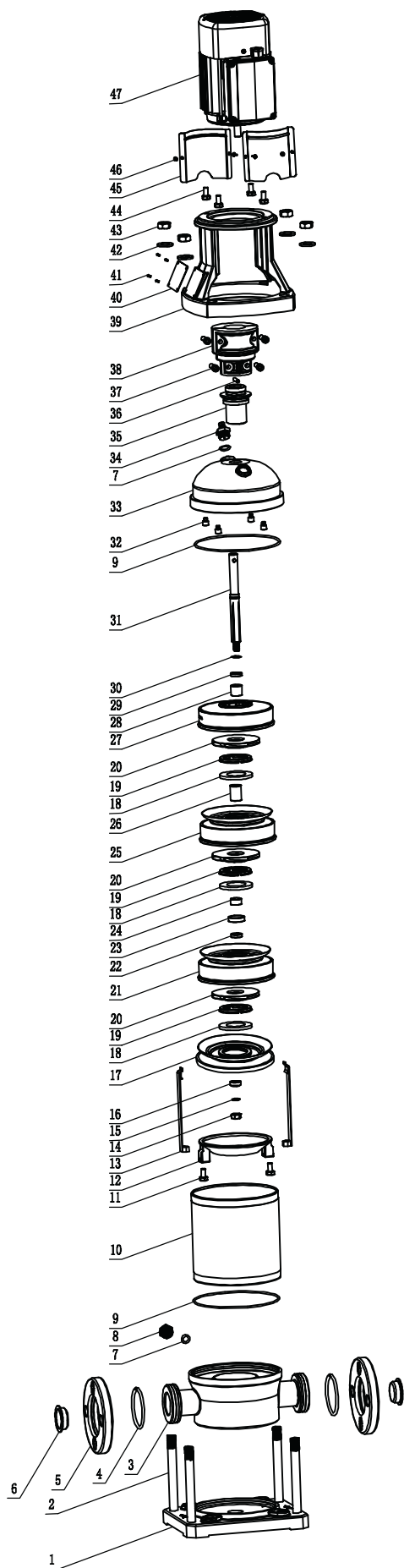


No	Description	Material	EN/DIN	AISI/ASTM
1	Motor	X	X	X
2	Shaft connector	X	X	X
3	Pump head	Cast iron	EN-JL 1030	ASTM25B
4	Mechanical seal	X	X	X
5	Shaft	S.S	X	AISI420
6	Outlet	S.S	1.4301	AISI304
7	Impeller	S.S	1.4301	AISI304
8	Hydraulic stack	S.S	1.4301	AISI304
9	Inlet	S.S	1.4301	AISI304
10	Pump body	Cast iron	EN.JL1030	ASTM25B
11	Neck ring	PTEF	X	X
12	Outer sleeve	S.S	1.4301	AISI316
13	O-ring	EDM/FKM	X	X

DIMENSIONS



Model	Dimensions (mm)					Weight
	B1	B2	B1+B2	D1	D2	KG
FCV-5-3VT	313	241	554	154	111	28



NO	Name	Spec.	Material	QTY
1	Base		HT200	1
2	Two-head bolt stud	M16XL	Chrome	4
3	Pump body		SUS304	1
4	Flange clasp		SUS304	2
5	Flange		SUS304	2
6	Dust cap	G1	PP	2
7	Ring	d18X2.4	NBR	2
8	Drain valve assembly		SUS304	1
9	Ring	d170X3.1	NBR	2
10	Housing		SUS316	1
11	Hexagon bolt	M8X20	SUS304	2
12	Fixed board		SUS316	1
13	Tension rod		SUS316	2
14	Type 1 non-metallic insert hexagon lock nut	M10	SUS316	1
15	Spring washer	ø10	SUS316	1
16	Impeller fix ring		SUS316	1
17	Inlet section		SUS316	1
18	Impeller fix ring		PTFE	N
19	Mouthpiece outer ring		SUS316	N
20	impeller		SUS316	N
21	Diffusor with bearing		SUS316	N
22	Short sleeve	L=4.5	SUS316	N
23	Bearing sleeve		Hard alloy	N
24	Short sleeve	L=10.5	SUS316	N
25	Diffusor		SUS316	N
26	Long sleeve	L=29	SUS316	N
27	Outlet section		SUS316	1
28	Short sleeve	L=19	SUS316	1
29	Snap ring seat		SUS316	1
30	Clasp		SUS316	1
31	Pump shaft		SUS316	1
32	Clasp seat shock colloidal		FI	4
33	Pump cover		SUS316	1
34	Discharge valve screw		SUS316	2
35	Mechanical seal	NJK-16-Q18FF	Carbon silicon carbide-FT	1
36	Cylindrical pin	ø6X25	SUS316	1
37	Hexagon socket head cap screw	M6X20	A3	4
38	Coupling			1
39	Motor base		HT200	1
40	Name plate		AL	1
41	Plate rivets	ø2.5	H62	4
42	Washer	ø16		4
43	Hexagonal nut	M16		4
44	Hexagonal bolt	M10X30	A3	4
45	Protective cover		SUS304	2
46	Cross recessed pan head screw and washer assemblies	M4X8	SUS304	4
47	Motor			1

- FCV-5-4VT
- Vertical Multi-Stage Pump

● Features :-

Stainless Steel 304

3 phase

220-380V

60HZ

IE3

2 pole

3500 rpm

Class 155(F)

Protection: IP55

● Product Photo :-



● Proformance Table :-

Model	Power		Max flow	Max head	Voltage	Inlet	Outlet
	KW	HP	m ³ /h	meters		mm	mm
FCV-5-4VT	1.1	1.5	10	37	220-380V	DN32	DN32

● Design

Vertical multistage in Line pump where Discharge and suction ports are located on one line to make it easier for connection access, it's either in Oval Victualic threaded to NPT cast iron discharge casing design or Slip on flanged and drilled to Universal standard which suite both DIN and ANSI standards.

● Applications: -

Water supply

Industry

Liquid transfer

Water treatment

Irrigation

● FCV Coding:

F: FUJIKA

C: Clean water

V: Vertical pump

● Working conditions: -

Temperature range: -20 Up to 104 °C

Ambient temperatue 50 °C

● Classifications:

FCV: Standard SS304

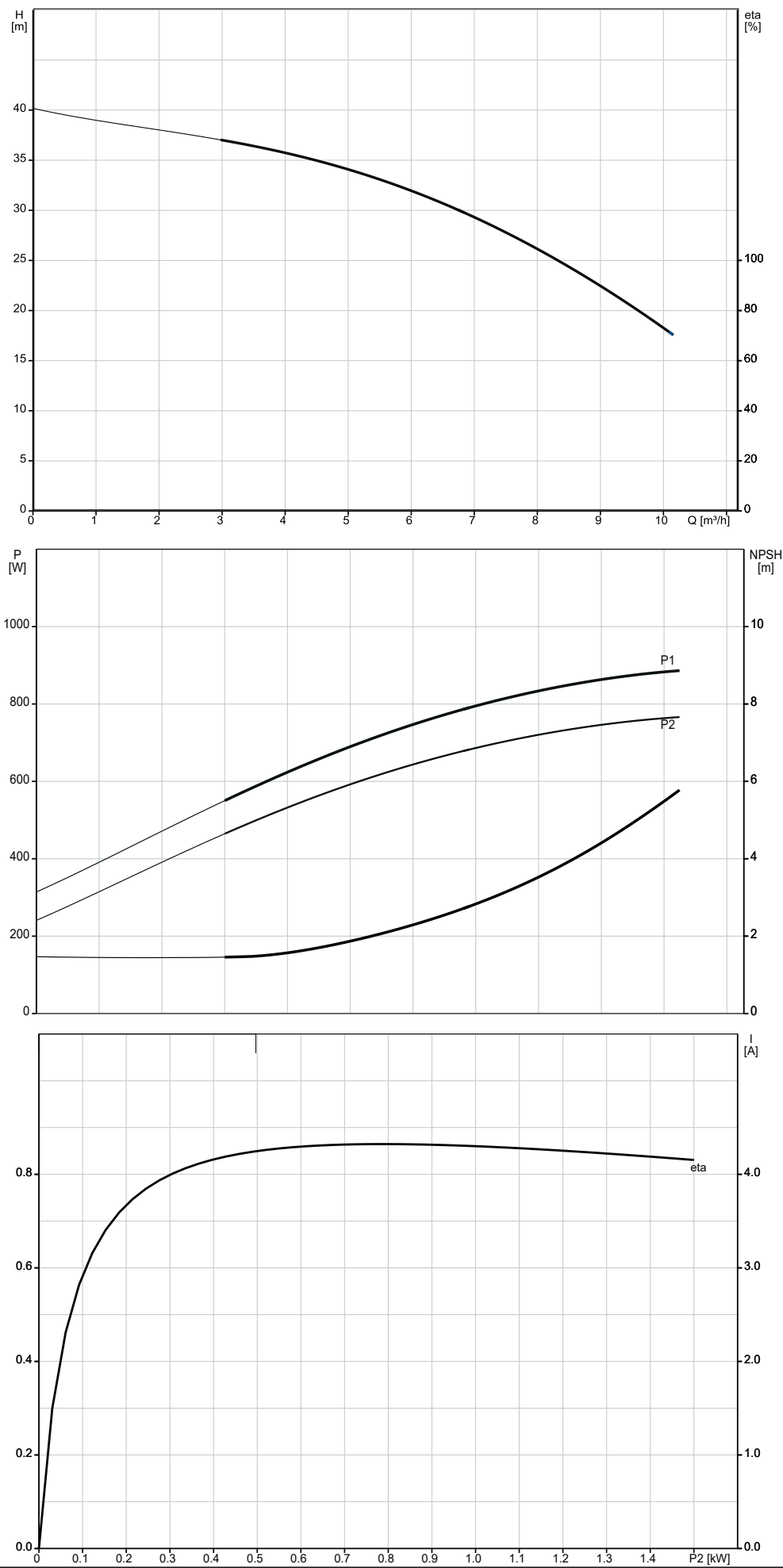
FCV-N: SS316

V: Oval flange

R: Round flange

T: Three phase

PERFORMANCE CURVE



Model: FCV-5-4VT



Design: Vertical multistage in Line pump where Discharge and suction ports are located on one line to make it easier for connection access, it's either in Oval Victualic threaded to NPT cast iron discharge casing design or Slip on flanged and drilled to Universal standard which suite both DIN and ANSI standards.

Construction of Material:

FCV basic design is made of :

Stainless steel imeller, sleeves and shaft , and cast iron discarge casing.

FCV-N

All wetted parts are made from SUS316 design.

The Shaft Mechanical seal is cartridge pull out type for easier replacement.

Mechanical Seal is EPDM /SUS304 for FCV and VITON / SUS316 for FCV-N Ensuring high reliability.

Coupling:

Shaft Coupling are rigid split type PJE Coupling.

Motor Type:

IE3 / IE2, 3-phase AC squirrel cage synchronous, high efficiency, TEFC fan cooled, IP55 Jetting protected.

General description :

The high efficiency ratings are in conformity to grade 3 efficiency value GB18613-2012 and IE of IEC60034-30 standard, but also the efficiency grade and installation dimensions are totally in conformity to IEC standard.

Motor shape is square motor as standard and round motor for larger sizes.

General specifications:

Operation factor: S1 (Continuous duty)

Insulation class: F

Protection class: IP55

Cooling ref.: IC411

Mount type: B35/B6

Ambient temperature: (-15 Up to 60 °C)

Altitude: ≤1000m

Motor paint finish is made with the most high technology with double painting and epoxy coating.

Motor housing construction are alaminum for motor rated up to 15HP and high grade cast iron for motors above rating.

The colour code: RAL 9005 deep jet black.

A standard split coupling connects the pump and motor shaft.

It's enclosed in the pump head/motor stool by means of two coupling guards.

The pump head and flange for motor mounting bracket is made in one piece (cast iron). the pump head cover is a seperate component (stainless steel).

Pump head cover stage has 1/2" priming plug and vent screw.

The pump is fitted with a balanced O-ring seal unit with a rigid torque-transmission system.

This seal type is assembled in a cartridge unit which makes replacement safe and easy.

In compliance with:

ENERGY EFFICIENCY ASIA JAPAN, IE3 IE2 IE1

MEPS (Minimum energy performance standard)

Current Regulation Energy saving act / top runner program standard

JIS C 4034-30

Power supply system

Three phase

Minimum energy performance IE3

Operated with inverter frequency: ok

Output (KW): 0.75 Up to 55KW

Number of poles: 2,4,6

Voltage (V): Up to 1000V

Frequency (HZ): 50HZ,60HZ,50/60HZ

Service duty: S1 ≥ 100%

Cooling method: tefc

Degree of protection: IP55

Area classification: safe area

Altitude: all

Ambient temperature: from (-20 °C and above)

Pump technical data

Rated flow: 5 m³/h
 Rated head: 30 meter
 Pump speed: 3450 rpm
 Nominal pressure PN: 25 bar
 Maximum pressure rating: 125% than shutoff pressure
 Pump mounting: vertical base mounting pump
 Shaft seal arrangement: single face cartridge type mechanical seal
 Seal code: Sic/Sic/Viton/SS316

Materials:

Wetted parts: Stainless steel EN 1.4408 SUS316
 Bearing: CU or SKF life sealed greased bearing
 Insulation: vertical
 Ambient temperature: 60 °C
 Maximum operating pressure: 25 bar
 Maximum pressure at stated temperature: 25 bar @ 0-90 °C

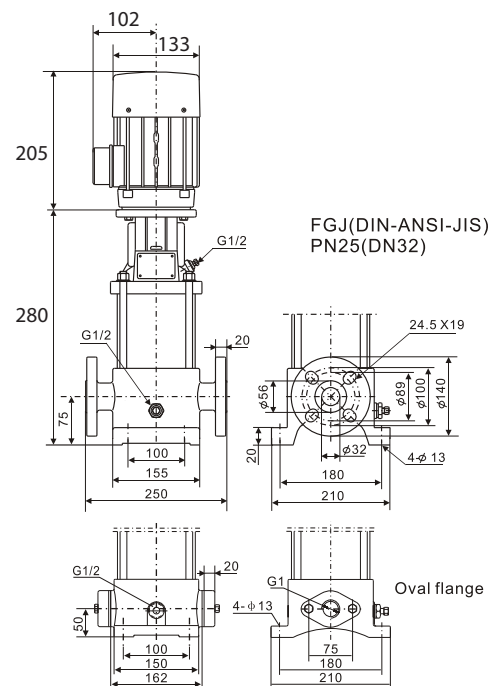
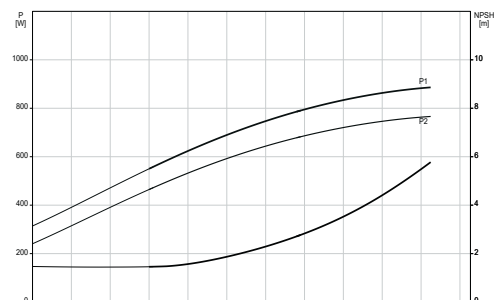
Motor type: by FUJIKA

IE efficiency class: IE3
 Rated power P2: 1.1 KW
 Power (P2) required by pump: 1.1 KW
 Frequency: 60HZ
 Rated voltage: 3 X 220-255V Δ / 380-440V Y
 Rated current: 4.5A / 2.6A
 Starting current: 440-500%
 Cos phi - power: 0.82
 Rotational speed: 3400-3450 rpm
 IE efficiency: IE3 77%
 Minimum efficiency: 76.5%
 Maximum efficiency: 82.9%
 Number of poles: 2
 Enclosure class: IP55
 Insulation class: F (IEC)

Dimensional:

Connection: UNIVERSAL flange drilled to both DIN & ANSI standards.
 Flange size for motor:
 Electrical data: IEC
 Motor standard: IEC DIN
 Approvals: CE
 Certified: IECEE, Coc
 Approvals for drinking water: WRAS
 Performance curve: ISO9906 2012

Gross weight: 29 KG
 Shipping volume: 0.063 m³
 Country of origin: PRC



Material resistivity to chemicals:

Stainless steel 316 material composition Up to :
18% chromium, 10% nickel, and 2% molybdenum.

Allows it to be used in variable applications and
that require superior corrosion resistance and
hygienic.

Applications such as:

Acidic water
Dissolved Chlorine
Sea water
Falls and Fountains
Water treatment
Diesel Fuel
Aggressive oil
Cooking oil
Drinking water
Juices
Water treatment
Food industry
Pharmaceutical Industry
Hygienic process
Fertilization
Fisheries
Irrigation

Characteristics:

The 316 stainless steel is used in salt and other
chlorides applications exposure and lasting
durability
steel over 304.

316 stainless steel's added molybdenum, makes it
much more corrosion resistant than 304 or other
stainless steel alloys. This allows 316 stainless steel
to be used in highly acidic environments that
other steels can't tolerate. 316's corrosion-resist-
ant properties also make it ideal in food process-
ing, pharmaceutical and medical applications due
to its ability to withstand the disinfectants and
high temperatures media.

Exposure to salt and chlorides can cause havoc on
304 stainless steel alloys. Salt can cause 304 stain-
less steel to rust by attacking the protective oxide
layer, and 304 steel is also more easily wear from
the effects of salt and chlorides. However, 316
stainless steel has all of the desirability properties of
304 stainless steel, but with the added benefit of
molybdenum, which allows it to withstand expo-
sure to salt and chlorides for more durability.

Austenitic stainless steels generally show high
corrosion rates when exposed to concentrated,
de-aerated alkaline aqueous environment at tem-
peratures above 60 °C.

Permissible corrosion rates of 0.05–2 mm/year.

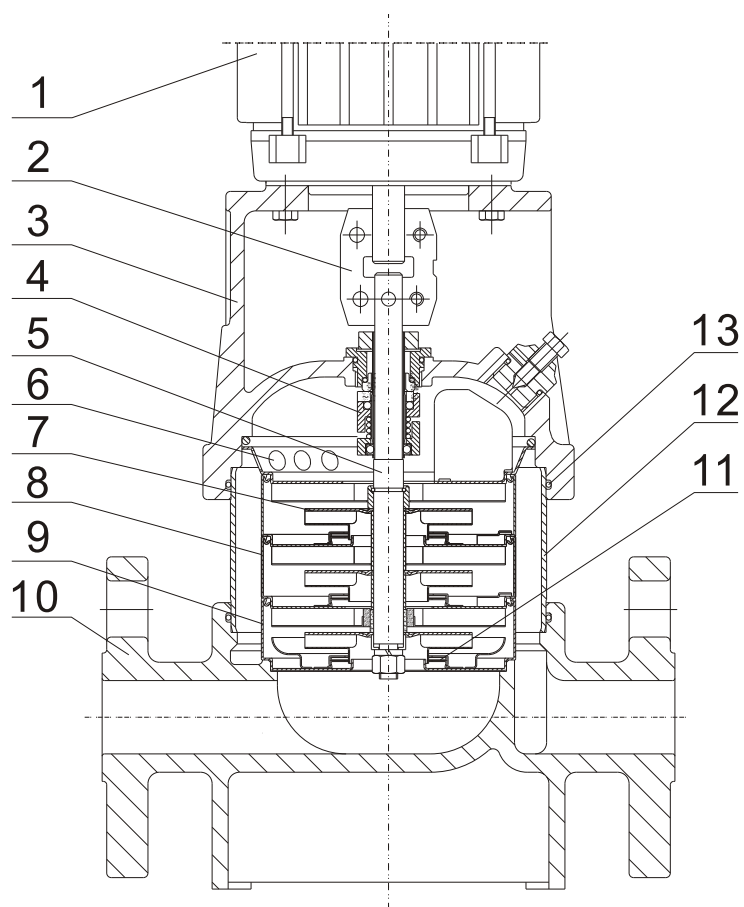
Caution:

The common stainless steel sheets types, 304 and
316 pumps should be considered non-resistant to
hydrochloric acid at any concentration and temper-
ature.

For Sulfuric acid:

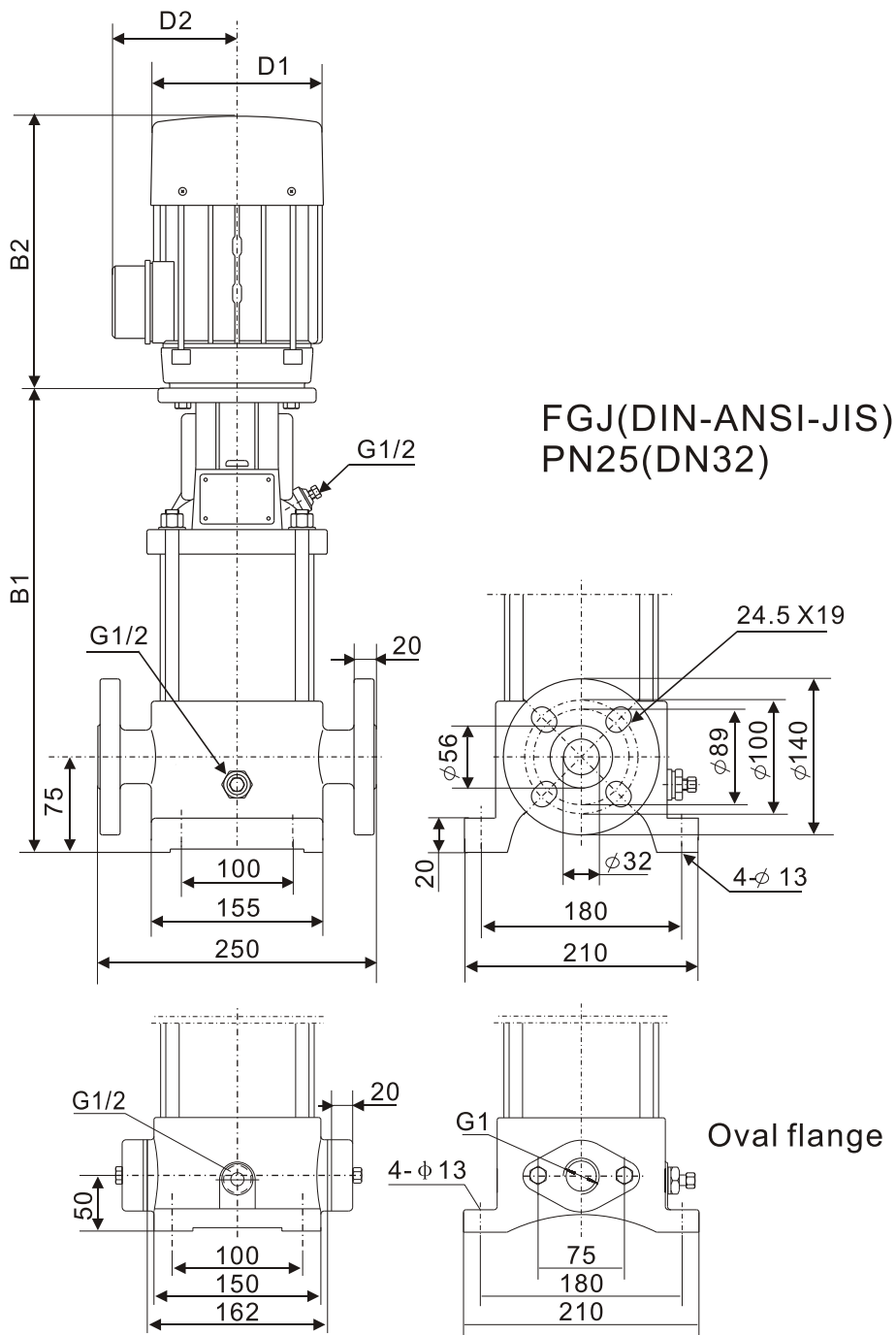
SUS304 is only resistant up to 1% acid while type
316 resistant is up to 10% acid at 50° C temp.

SECTIONAL DRAWING

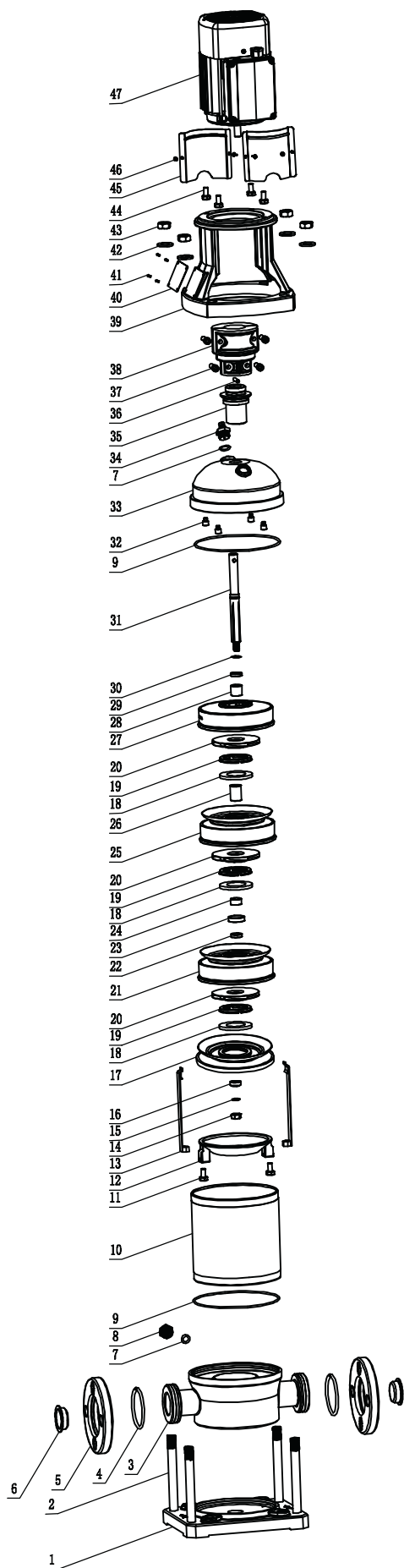


No	Description	Material	EN/DIN	AISI/ASTM
1	Motor	X	X	X
2	Shaft connector	X	X	X
3	Pump head	Cast iron	EN-JL 1030	ASTM25B
4	Mechanical seal	X	X	X
5	Shaft	S.S	X	AISI420
6	Outlet	S.S	1.4301	AISI304
7	Impeller	S.S	1.4301	AISI304
8	Hydraulic stack	S.S	1.4301	AISI304
9	Inlet	S.S	1.4301	AISI304
10	Pump body	Cast iron	EN.JL1030	ASTM25B
11	Neck ring	PTEF	X	X
12	Outer sleeve	S.S	1.4301	AISI316
13	O-ring	EDM/FKM	X	X

DIMENSIONS



Model	Dimensions (mm)					Weight
	B1	B2	B1+B2	D1	D2	KG
FCV-5-4VT	340	241	581	154	111	29



NO	Name	Spec.	Material	QTY
1	Base		HT200	1
2	Two-head bolt stud	M16XL	Chrome	4
3	Pump body		SUS304	1
4	Flange clasp		SUS304	2
5	Flange		SUS304	2
6	Dust cap	G1	PP	2
7	Ring	d18X2.4	NBR	2
8	Drain valve assembly		SUS304	1
9	Ring	d170X3.1	NBR	2
10	Housing		SUS316	1
11	Hexagon bolt	M8X20	SUS304	2
12	Fixed board		SUS316	1
13	Tension rod		SUS316	2
14	Type 1 non-metallic insert hexagon lock nut	M10	SUS316	1
15	Spring washer	ø10	SUS316	1
16	Impeller fix ring		SUS316	1
17	Inlet section		SUS316	1
18	Impeller fix ring		PTFE	N
19	Mouthpiece outer ring		SUS316	N
20	impeller		SUS316	N
21	Diffusor with bearing		SUS316	N
22	Short sleeve	L=4.5	SUS316	N
23	Bearing sleeve		Hard alloy	N
24	Short sleeve	L=10.5	SUS316	N
25	Diffusor		SUS316	N
26	Long sleeve	L=29	SUS316	N
27	Outlet section		SUS316	1
28	Short sleeve	L=19	SUS316	1
29	Snap ring seat		SUS316	1
30	Clasp		SUS316	1
31	Pump shaft		SUS316	1
32	Clasp seat shock colloidal		FI	4
33	Pump cover		SUS316	1
34	Discharge valve screw		SUS316	2
35	Mechanical seal	NJK-16-Q18FF	Carbon silicon carbide-FT	1
36	Cylindrical pin	ø6X25	SUS316	1
37	Hexagon socket head cap screw	M6X20	A3	4
38	Coupling			1
39	Motor base		HT200	1
40	Name plate		AL	1
41	Plate rivets	ø2.5	H62	4
42	Washer	ø16		4
43	Hexagonal nut	M16		4
44	Hexagonal bolt	M10X30	A3	4
45	Protective cover		SUS304	2
46	Cross recessed pan head screw and washer assemblies	M4X8	SUS304	4
47	Motor			1

● FCV-5-5VT

● Vertical Multi-Stage Pump

● Features :-

Stainless Steel 304

3 phase

220-380V

60HZ

IE3

2 pole

3500 rpm

Class 155(F)

Protection: IP55

● Product Photo :-



● Proformance Table :-

Model	Power		Max flow	Max head	Voltage	Inlet	Outlet
	KW	HP	m ³ /h	meters		mm	mm
FCV-5-5VT	1.5	2.0	10	47	220-380V	DN32	DN32

● Design

Vertical multistage in Line pump where Discharge and suction ports are located on one line to make it easier for connection access, it's either in Oval Victualic threaded to NPT cast iron discharge casing design or Slip on flanged and drilled to Universal standard which suite both DIN and ANSI standards.

● Applications: -

Water supply

Industry

Liquid transfer

Water treatment

Irrigation

● FCV Coding:

F: FUJIKA

C: Clean water

V: Vertical pump

● Classifications:

FCV: Standard SS304

FCV-N: SS316

V: Oval flange

R: Round flange

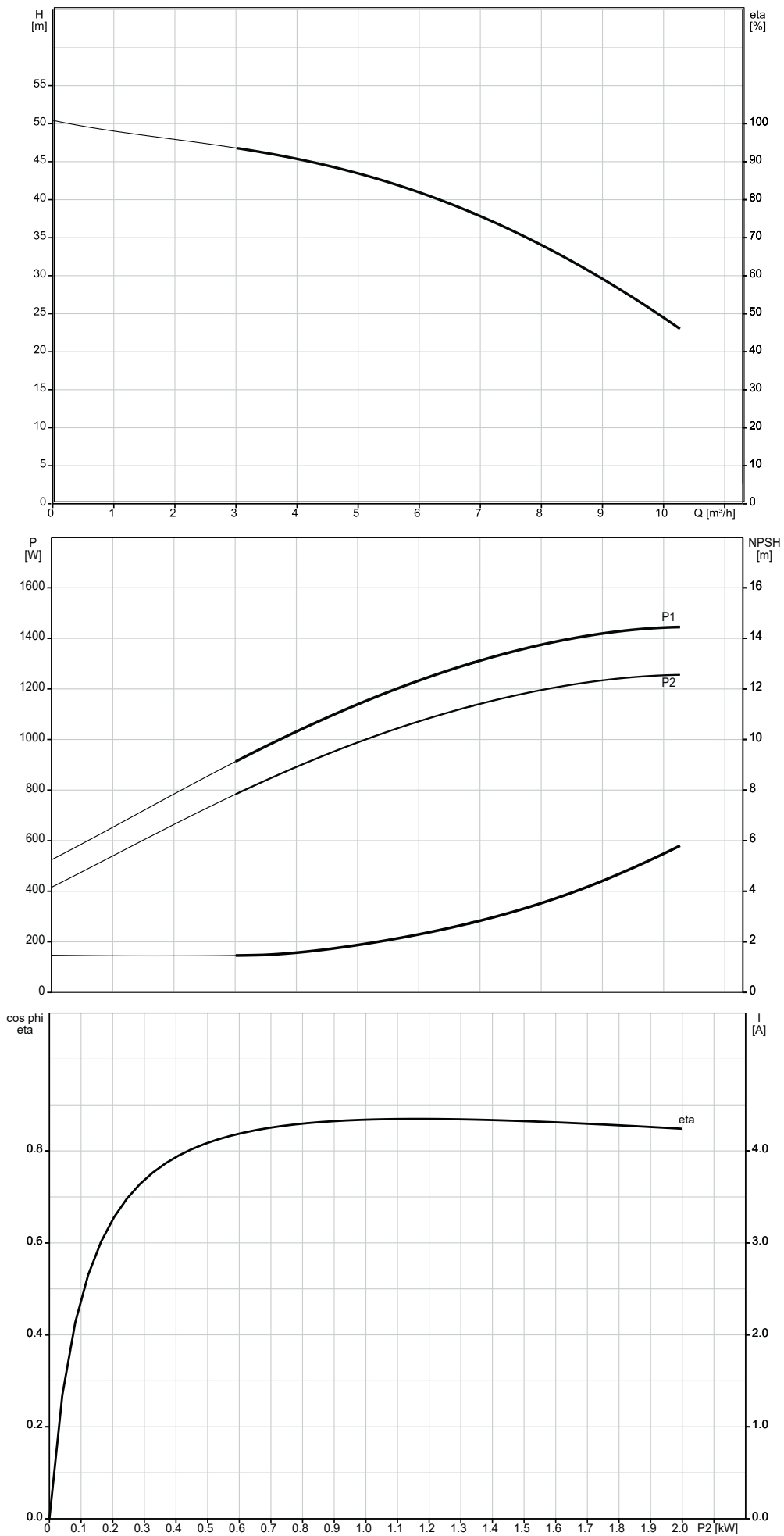
T: Three phase

● Working conditions: -

Temperature range: -20 Up to 104 °C

Ambient temperatue 50 °C

PERFORMANCE CURVE



Model: FCV-5-5VT



Design: Vertical multistage in Line pump where Discharge and suction ports are located on one line to make it easier for connection access, it's either in Oval Victualic threaded to NPT cast iron discharge casing design or Slip on flanged and drilled to Universal standard which suite both DIN and ANSI standards.

Construction of Material:

FCV basic design is made of :

Stainless steel imeller, sleeves and shaft , and cast iron discarge casing.

FCV-N

All wetted parts are made from SUS316 design.

The Shaft Mechanical seal is cartridge pull out type for easier replacement.

Mechanical Seal is EPDM /SUS304 for FCV and VITON / SUS316 for FCV-N Ensuring high reliability.

Coupling:

Shaft Coupling are rigid split type PJE Coupling.

Motor Type:

IE3 / IE2, 3-phase AC squirrel cage synchronous, high efficiency, TEFC fan cooled, IP55 Jetting protected.

General description :

The high efficiency ratings are in conformity to grade 3 efficiency value GB18613-2012 and IE of IEC60034-30 standard, but also the efficiency grade and installation dimensions are totally in conformity to IEC standard.

Motor shape is square motor as standard and round motor for larger sizes.

General specifications:

Operation factor: S1 (Continuous duty)

Insulation class: F

Protection class: IP55

Cooling ref.: IC411

Mount type: B35/B6

Ambient temperature: (-15 Up to 60 °C)

Altitude: ≤1000m

Motor paint finish is made with the most high technology with double painting and epoxy coating.

Motor housing construction are alaminum for motor rated up to 15HP and high grade cast iron for motors above rating.

The colour code: RAL 9005 deep jet black.

A standard split coupling connects the pump and motor shaft.

It's enclosed in the pump head/motor stool by means of two coupling guards.

The pump head and flange for motor mounting bracket is made in one piece (cast iron). the pump head cover is a seperate component (stainless steel).

Pump head cover stage has 1/2" priming plug and vent screw.

The pump is fitted with a balanced O-ring seal unit with a rigid torque-transmission system.

This seal type is assembled in a cartridge unit which makes replacement safe and easy.

In compliance with:

ENERGY EFFICIENCY ASIA JAPAN, IE3 IE2 IE1

MEPS (Minimum energy performance standard)

Current Regulation Energy saving act / top runner program standard

JIS C 4034-30

Power supply system

Three phase

Minimum energy performance IE3

Operated with inverter frequency: ok

Output (KW): 0.75 Up to 55KW

Number of poles: 2,4,6

Voltage (V): Up to 1000V

Frequency (HZ): 50HZ,60HZ,50/60HZ

Service duty: S1 ≥ 100%

Cooling method: tefc

Degree of protection: IP55

Area classification: safe area

Altitude: all

Ambient temperature: from (-20 °C and above)

Pump technical data

Rated flow: 5 m³/h

Rated head: 38 meter

Pump speed: 3450 rpm

Nominal pressure PN: 25 bar

Maximum pressure rating: 125% than shutoff pressure

Pump mounting: vertical base mounting pump

Shaft seal arrangement: single face cartridge type mechanical seal

Seal code: Sic/Sic/Viton/SS316

Materials:

Wetted parts: Stainless steel EN 1.4408 SUS316

Bearing: CU or SKF life sealed

greased bearing

Insulation: vertical

Ambient temperature: 60 °C

Maximum operating pressure: 25 bar

Maximum pressure at stated temperature: 25 bar @ 0-90 °C

Motor type: by FUJIKAWA

IE efficiency class: IE3

Rated power P2: 1.5 KW

Power (P2) required by pump: 1.5 KW

Frequency: 60HZ

Rated voltage: 3 X 220-255V Δ / 380-440V Y

Rated current: 6.0A / 3.5A

Starting current: 780-980%

Cos phi - power: 0.83

Rotational speed: 3400-3450 rpm

IE efficiency: IE3 83.7%

Minimum efficiency: 76.8%

Maximum efficiency: 87.5%

Number of poles: 2

Enclosure class: IP55

Insulation class: F (IEC)

Dimensional:

Connection: UNIVERSAL flange drilled to both DIN & ANSI standards.

Flange size for motor:

Electrical data: IEC

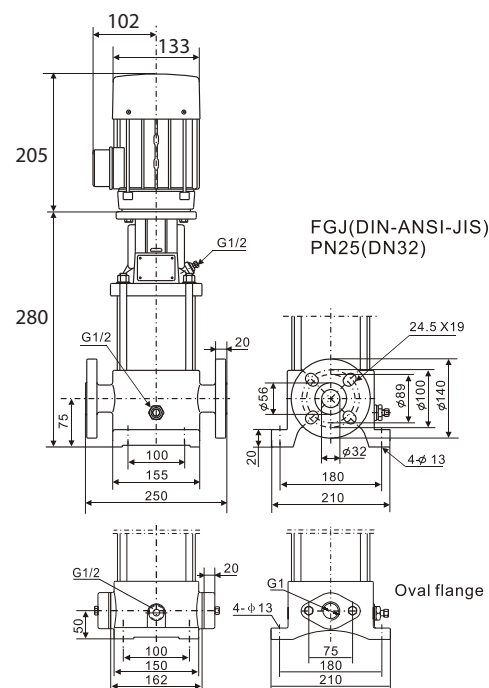
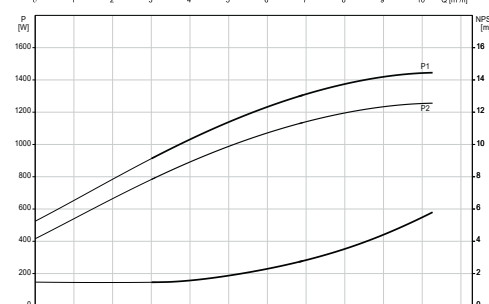
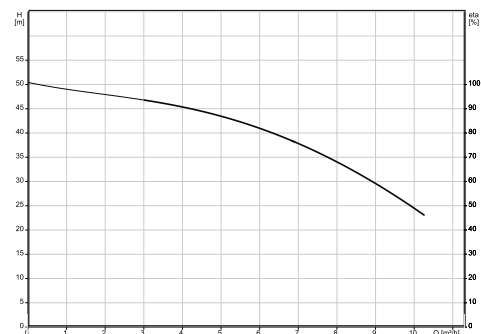
Motor standard: IEC DIN

Approvals: CE

Cerified: IECEE, Coc

Approvals for drinking water: WRAS

Performance curve: ISO9906 2012



FGJ(DIN-ANSI-JIS)
PN25(DN32)

Oval flange

Material resistivity to chemicals:

Stainless steel 316 material composition Up to :
18% chromium, 10% nickel, and 2% molybdenum.

Allows it to be used in variable applications and
that require superior corrosion resistance and
hygienic.

Applications such as:

Acidic water
Dissolved Chlorine
Sea water
Falls and Fountains
Water treatment
Diesel Fuel
Aggressive oil
Cooking oil
Drinking water
Juices
Water treatment
Food industry
Pharmaceutical Industry
Hygienic process
Fertilization
Fisheries
Irrigation

Characteristics:

The 316 stainless steel is used in salt and other
chlorides applications exposure and lasting
durability
steel over 304.

316 stainless steel's added molybdenum, makes it
much more corrosion resistant than 304 or other
stainless steel alloys. This allows 316 stainless steel
to be used in highly acidic environments that
other steels can't tolerate. 316's corrosion-resist-
ant properties also make it ideal in food process-
ing, pharmaceutical and medical applications due
to its ability to withstand the disinfectants and
high temperatures media.

Exposure to salt and chlorides can cause havoc on
304 stainless steel alloys. Salt can cause 304 stain-
less steel to rust by attacking the protective oxide
layer, and 304 steel is also more easily wear from
the effects of salt and chlorides. However, 316
stainless steel has all of the desirability properties of
304 stainless steel, but with the added benefit of
molybdenum, which allows it to withstand expo-
sure to salt and chlorides for more durability.

Austenitic stainless steels generally show high
corrosion rates when exposed to concentrated,
de-aerated alkaline aqueous environment at tem-
peratures above 60 °C.

Permissible corrosion rates of 0.05–2 mm/year.

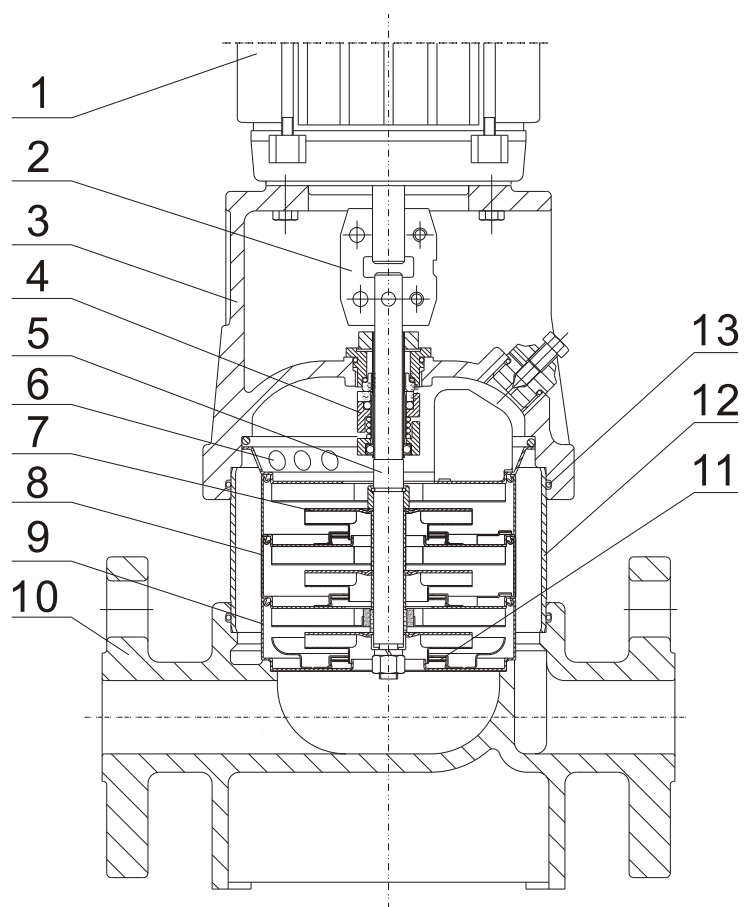
Caution:

The common stainless steel sheets types, 304 and
316 pumps should be considered non-resistant to
hydrochloric acid at any concentration and temper-
ature.

For Sulfuric acid:

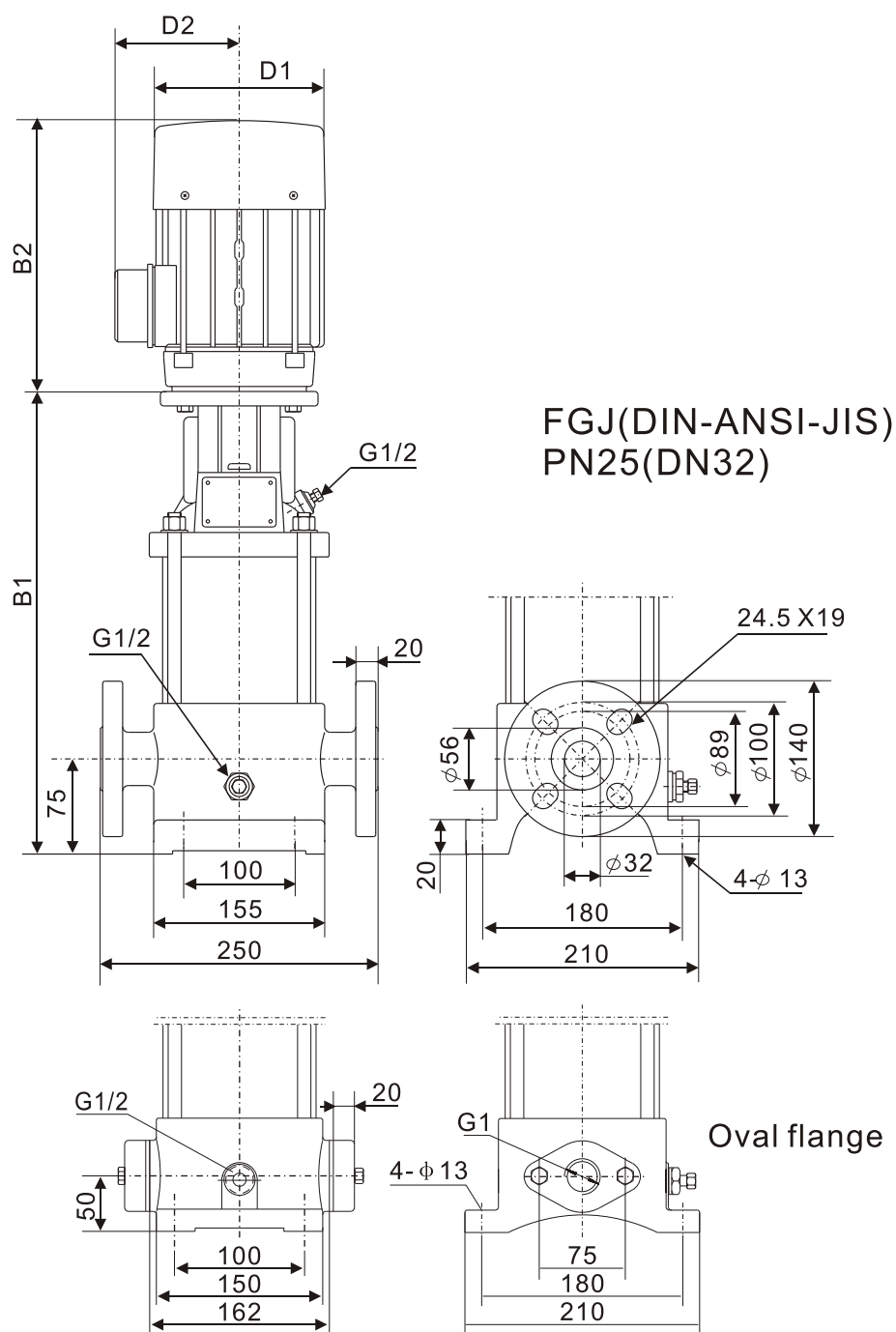
SUS304 is only resistant up to 1% acid while type
316 resistant is up to 10% acid at 50° C temp.

SECTIONAL DRAWING

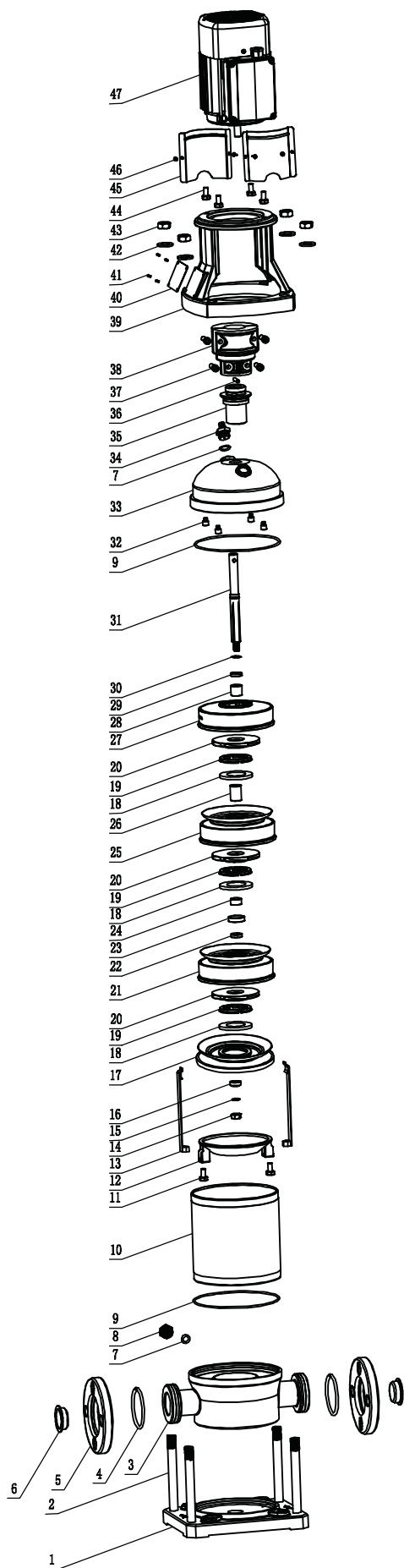


No	Description	Material	EN/DIN	AISI/ASTM
1	Motor	X	X	X
2	Shaft connector	X	X	X
3	Pump head	Cast iron	EN-JL 1030	ASTM25B
4	Mechanical seal	X	X	X
5	Shaft	S.S	X	AISI420
6	Outlet	S.S	1.4301	AISI304
7	Impeller	S.S	1.4301	AISI304
8	Hydraulic stack	S.S	1.4301	AISI304
9	Inlet	S.S	1.4301	AISI304
10	Pump body	Cast iron	EN.JL1030	ASTM25B
11	Neck ring	PTEF	X	X
12	Outer sleeve	S.S	1.4301	AISI316
13	O-ring	EDM/FKM	X	X

DIMENSIONS



Model	Dimensions (mm)					Weight
	B1	B2	B1+B2	D1	D2	KG
FCV-5-5VT	367	241/293	608	154	111	30



NO	Name	Spec.	Material	QTY
1	Base		HT200	1
2	Two-head bolt stud	M16XL	Chrome	4
3	Pump body		SUS304	1
4	Flange clasp		SUS304	2
5	Flange		SUS304	2
6	Dust cap	G1	PP	2
7	Ring	d18X2.4	NBR	2
8	Drain valve assembly		SUS304	1
9	Ring	d170X3.1	NBR	2
10	Housing		SUS316	1
11	Hexagon bolt	M8X20	SUS304	2
12	Fixed board		SUS316	1
13	Tension rod		SUS316	2
14	Type 1 non-metallic insert hexagon lock nut	M10	SUS316	1
15	Spring washer	ø10	SUS316	1
16	Impeller fix ring		SUS316	1
17	Inlet section		SUS316	1
18	Impeller fix ring		PTFE	N
19	Mouthpiece outer ring		SUS316	N
20	impeller		SUS316	N
21	Diffusor with bearing		SUS316	N
22	Short sleeve	L=4.5	SUS316	N
23	Bearing sleeve		Hard alloy	N
24	Short sleeve	L=10.5	SUS316	N
25	Diffusor		SUS316	N
26	Long sleeve	L=29	SUS316	N
27	Outlet section		SUS316	1
28	Short sleeve	L=19	SUS316	1
29	Snap ring seat		SUS316	1
30	Clasp		SUS316	1
31	Pump shaft		SUS316	1
32	Clasp seat shock colloidal		FI	4
33	Pump cover		SUS316	1
34	Discharge valve screw		SUS316	2
35	Mechanical seal	NJK-16-Q18FF	Carbon silicon carbide-FT	1
36	Cylindrical pin	ø6X25	SUS316	1
37	Hexagon socket head cap screw	M6X20	A3	4
38	Coupling			1
39	Motor base		HT200	1
40	Name plate		AL	1
41	Plate rivets	ø2.5	H62	4
42	Washer	ø16		4
43	Hexagonal nut	M16		4
44	Hexagonal bolt	M10X30	A3	4
45	Protective cover		SUS304	2
46	Cross recessed pan head screw and washer assemblies	M4X8	SUS304	4
47	Motor			1

- FCV-5-6VT
- Vertical Multi-Stage Pump

● Features :-

Stainless Steel 304

3 phase

220-380V

60HZ

IE3

2 pole

3500 rpm

Class 155(F)

Protection: IP55

● Product Photo :-



● Proformance Table :-

Model	Power		Max flow	Max head	Voltage	Inlet	Outlet
	KW	HP	m ³ /h	meters		mm	mm
FCV-5-6VT	2.2	3.0	10	65	220-380V	DN32	DN32

● Design

Vertical multistage in Line pump where Discharge and suction ports are located on one line to make it easier for connection access, it's either in Oval Victualic threaded to NPT cast iron discharge casing design or Slip on flanged and drilled to Universal standard which suite both DIN and ANSI standards.

● Applications: -

Water supply

Industry

Liquid transfer

Water treatment

Irrigation

● FCV Coding:

F: FUJIKA

C: Clean water

V: Vertical pump

● Working conditions: -

Temperature range: -20 Up to 104 °C

Ambient temperatue 50 °C

● Classifications:

FCV: Standard SS304

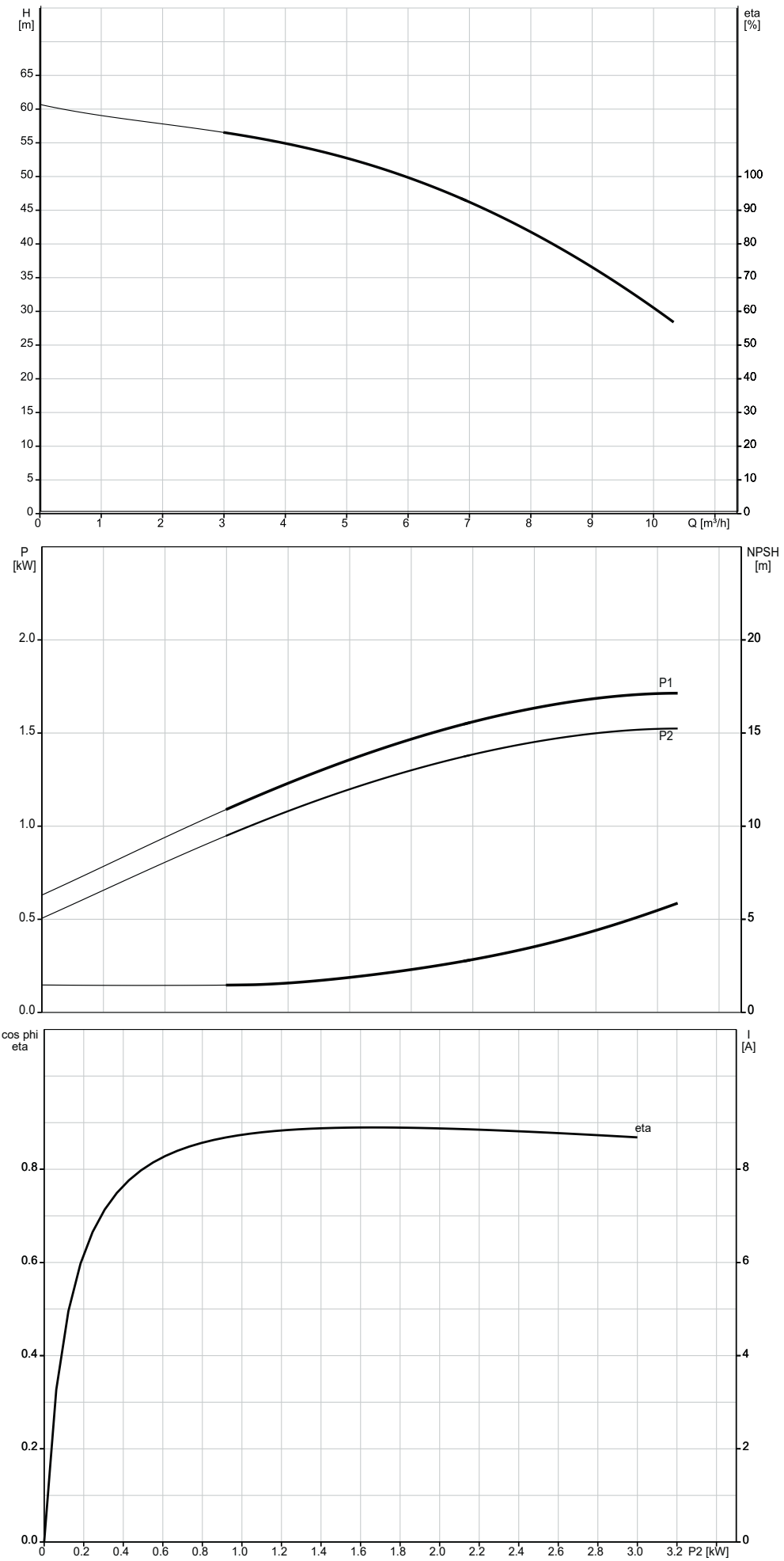
FCV-N: SS316

V: Oval flange

R: Round flange

T: Three phase

PERFORMANCE CURVE



Model: FCV-5-6VT



Design: Vertical multistage in Line pump where Discharge and suction ports are located on one line to make it easier for connection access, it's either in Oval Victualic threaded to NPT cast iron discharge casing design or Slip on flanged and drilled to Universal standard which suite both DIN and ANSI standards.

Construction of Material:

FCV basic design is made of :

Stainless steel imeller, sleeves and shaft , and cast iron discarge casing.

FCV-N

All wetted parts are made from SUS316 design.

The Shaft Mechanical seal is cartridge pull out type for easier replacement.

Mechanical Seal is EPDM /SUS304 for FCV and VITON / SUS316 for FCV-N Ensuring high reliability.

Coupling:

Shaft Coupling are rigid split type PJE Coupling.

Motor Type:

IE3 / IE2, 3-phase AC squirrel cage synchronous, high efficiency, TEFC fan cooled, IP55 Jetting protected.

General description :

The high efficiency ratings are in conformity to grade 3 efficiency value GB18613-2012 and IE of IEC60034-30 standard, but also the efficiency grade and installation dimensions are totally in conformity to IEC standard.

Motor shape is square motor as standard and round motor for larger sizes.

General specifications:

Operation factor: S1 (Continuous duty)

Insulation class: F

Protection class: IP55

Cooling ref.: IC411

Mount type: B35/B6

Ambient temperature: (-15 Up to 60 °C)

Altitude: ≤1000m

Motor paint finish is made with the most high technology with double painting and epoxy coating.

Motor housing construction are alaminum for motor rated up to 15HP and high grade cast iron for motors above rating.

The colour code: RAL 9005 deep jet black.

A standard split coupling connects the pump and motor shaft.

It's enclosed in the pump head/motor stool by means of two coupling guards.

The pump head and flange for motor mounting bracket is made in one piece (cast iron). the pump head cover is a seperate component (stainless steel).

Pump head cover stage has 1/2" priming plug and vent screw.

The pump is fitted with a balanced O-ring seal unit with a rigid torque-transmission system.

This seal type is assembled in a cartridge unit which makes replacement safe and easy.

In compliance with:

ENERGY EFFICIENCY ASIA JAPAN, IE3 IE2 IE1

MEPS (Minimum energy performance standard)

Current Regulation Energy saving act / top runner program standard

JIS C 4034-30

Power supply system

Three phase

Minimum energy performance IE3

Operated with inverter frequency: ok

Output (KW): 0.75 Up to 55KW

Number of poles: 2,4,6

Voltage (V): Up to 1000V

Frequency (HZ): 50HZ,60HZ,50/60HZ

Service duty: S1 ≥ 100%

Cooling method: tefc

Degree of protection: IP55

Area classification: safe area

Altitude: all

Ambient temperature: from (-20 °C and above)

Pump technical data

Rated flow: 5 m³/h
 Rated head: 45 meter
 Pump speed: 3450 rpm
 Nominal pressure PN: 25 bar
 Maximum pressure rating: 125% than shutoff pressure
 Pump mounting: vertical base mounting pump
 Shaft seal arrangement: single face cartridge type mechanical seal
 Seal code: Sic/Sic/Viton/SS316

Materials:

Wetted parts: Stainless steel EN 1.4408 SUS316
 Bearing: CU or SKF life sealed greased bearing
 Insulation: vertical
 Ambient temperature: 60 °C
 Maximum operating pressure: 25 bar
 Maximum pressure at stated temperature: 25 bar @ 0-90 °C

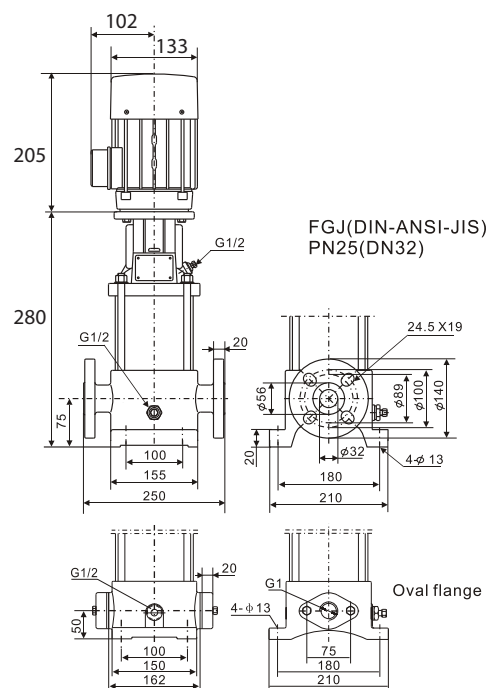
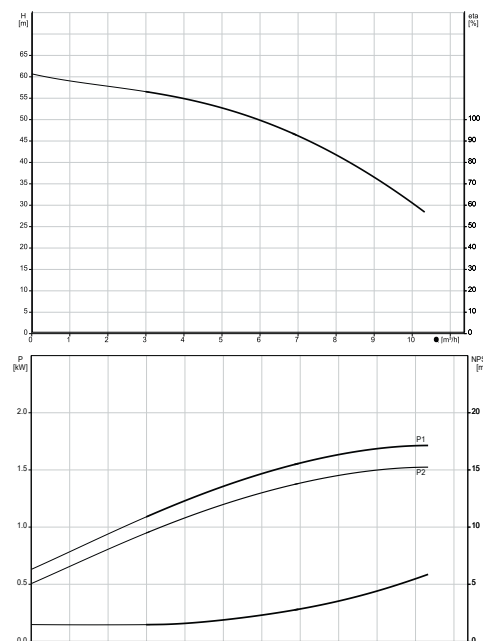
Motor type: by FUJIKA

IE efficiency class: IE3
 Rated power P2: 2.2 KW
 Power (P2) required by pump: 2.2 KW
 Frequency: 60HZ
 Rated voltage: 3 X 220-255V Δ / 380-440V Y
 Rated current: 8.4A / 4.9A
 Starting current: 820-1050%
 Cos phi - power: 0.84
 Rotational speed: 3400-3450 rpm
 IE efficiency: IE3 86.3%
 Minimum efficiency: 81.1%
 Maximum efficiency: 89.5%
 Number of poles: 2
 Enclosure class: IP55
 Insulation class: F (IEC)

Dimensional:

Connection: UNIVERSAL flange drilled to both DIN & ANSI standards.
 Flange size for motor:
 Electrical data: IEC
 Motor standard: IEC DIN
 Approvals: CE
 Certified: IECEE, Coc
 Approvals for drinking water: WRAS
 Performance curve: ISO9906 2012

Gross weight: 33 KG
 Shipping volume: 0.074 m³
 Country of origin: PRC



Material resistivity to chemicals:

Stainless steel 316 material composition Up to :
18% chromium, 10% nickel, and 2% molybdenum.

Allows it to be used in variable applications and
that require superior corrosion resistance and
hygienic.

Applications such as:

Acidic water
Dissolved Chlorine
Sea water
Falls and Fountains
Water treatment
Diesel Fuel
Aggressive oil
Cooking oil
Drinking water
Juices
Water treatment
Food industry
Pharmaceutical Industry
Hygienic process
Fertilization
Fisheries
Irrigation

Characteristics:

The 316 stainless steel is used in salt and other
chlorides applications exposure and lasting
durability
steel over 304.

316 stainless steel's added molybdenum, makes it
much more corrosion resistant than 304 or other
stainless steel alloys. This allows 316 stainless steel
to be used in highly acidic environments that
other steels can't tolerate. 316's corrosion-resist-
ant properties also make it ideal in food process-
ing, pharmaceutical and medical applications due
to its ability to withstand the disinfectants and
high temperatures media.

Exposure to salt and chlorides can cause havoc on
304 stainless steel alloys. Salt can cause 304 stain-
less steel to rust by attacking the protective oxide
layer, and 304 steel is also more easily wear from
the effects of salt and chlorides. However, 316
stainless steel has all of the desirability properties of
304 stainless steel, but with the added benefit of
molybdenum, which allows it to withstand expo-
sure to salt and chlorides for more durability.

Austenitic stainless steels generally show high
corrosion rates when exposed to concentrated,
de-aerated alkaline aqueous environment at tem-
peratures above 60 °C.

Permissible corrosion rates of 0.05–2 mm/year.

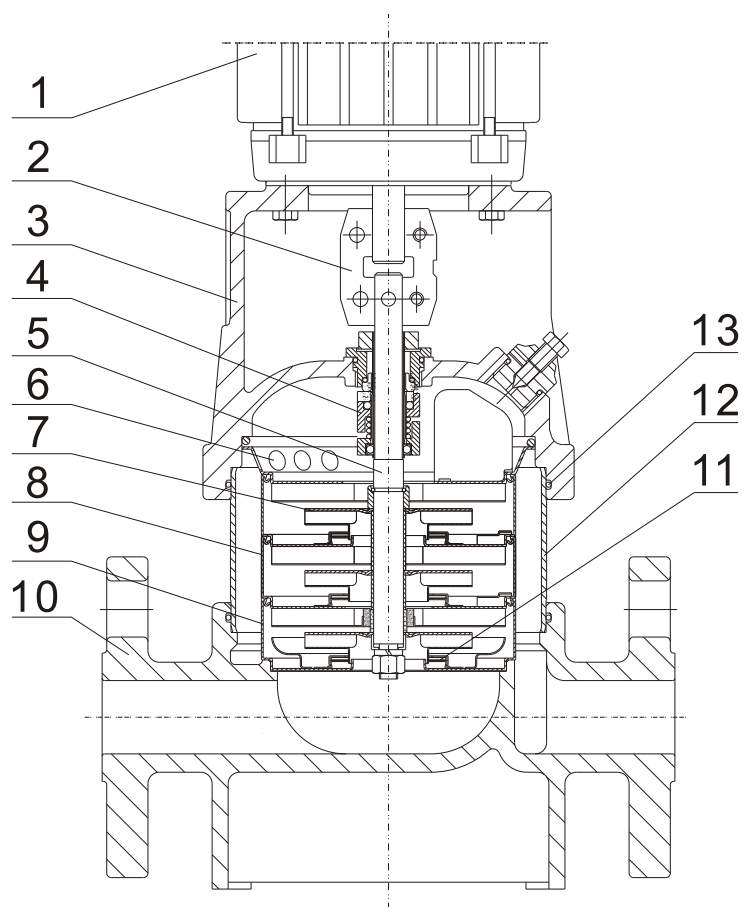
Caution:

The common stainless steel sheets types, 304 and
316 pumps should be considered non-resistant to
hydrochloric acid at any concentration and temper-
ature.

For Sulfuric acid:

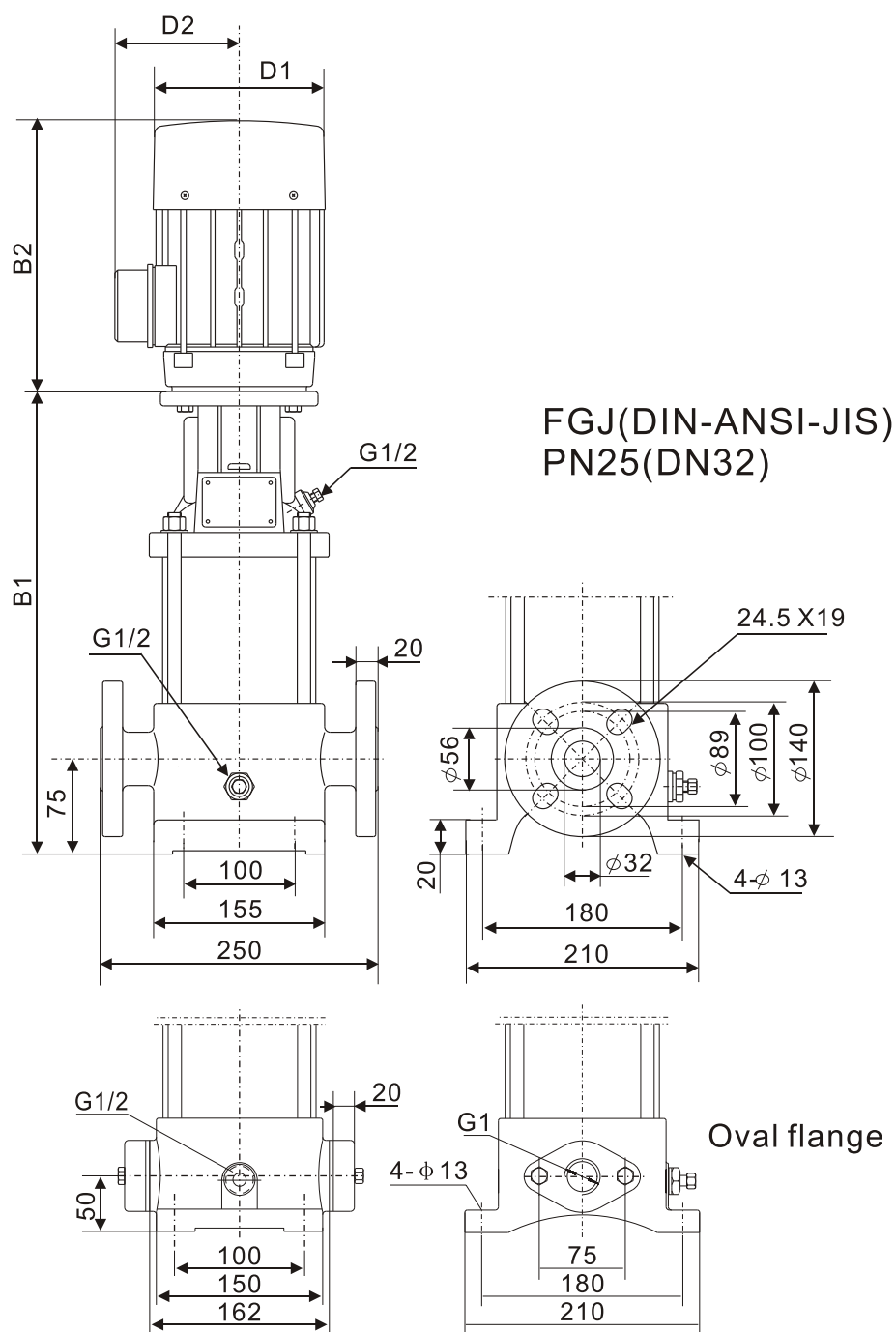
SUS304 is only resistant up to 1% acid while type
316 resistant is up to 10% acid at 50° C temp.

SECTIONAL DRAWING

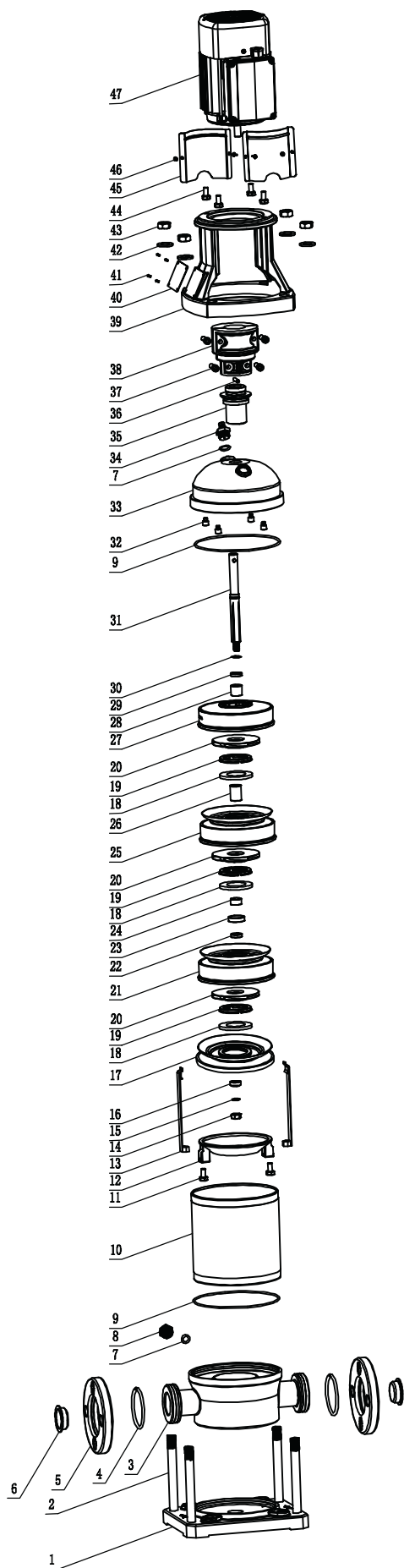


No	Description	Material	EN/DIN	AISI/ASTM
1	Motor	X	X	X
2	Shaft connector	X	X	X
3	Pump head	Cast iron	EN-JL 1030	ASTM25B
4	Mechanical seal	X	X	X
5	Shaft	S.S	X	AISI420
6	Outlet	S.S	1.4301	AISI304
7	Impeller	S.S	1.4301	AISI304
8	Hydraulic stack	S.S	1.4301	AISI304
9	Inlet	S.S	1.4301	AISI304
10	Pump body	Cast iron	EN.JL1030	ASTM25B
11	Neck ring	PTEF	X	X
12	Outer sleeve	S.S	1.4301	AISI316
13	O-ring	EDM/FKM	X	X

DIMENSIONS



Model	Dimensions (mm)					Weight
	B1	B2	B1+B2	D1	D2	KG
FCV-5-6VT	412	275/293	687	177	116	33



NO	Name	Spec.	Material	QTY
1	Base		HT200	1
2	Two-head bolt stud	M16XL	Chrome	4
3	Pump body		SUS304	1
4	Flange clasp		SUS304	2
5	Flange		SUS304	2
6	Dust cap	G1	PP	2
7	Ring	d18X2.4	NBR	2
8	Drain valve assembly		SUS304	1
9	Ring	d170X3.1	NBR	2
10	Housing		SUS316	1
11	Hexagon bolt	M8X20	SUS304	2
12	Fixed board		SUS316	1
13	Tension rod		SUS316	2
14	Type 1 non-metallic insert hexagon lock nut	M10	SUS316	1
15	Spring washer	ø10	SUS316	1
16	Impeller fix ring		SUS316	1
17	Inlet section		SUS316	1
18	Impeller fix ring		PTFE	N
19	Mouthpiece outer ring		SUS316	N
20	impeller		SUS316	N
21	Diffusor with bearing		SUS316	N
22	Short sleeve	L=4.5	SUS316	N
23	Bearing sleeve		Hard alloy	N
24	Short sleeve	L=10.5	SUS316	N
25	Diffusor		SUS316	N
26	Long sleeve	L=29	SUS316	N
27	Outlet section		SUS316	1
28	Short sleeve	L=19	SUS316	1
29	Snap ring seat		SUS316	1
30	Clasp		SUS316	1
31	Pump shaft		SUS316	1
32	Clasp seat shock colloidal		FI	4
33	Pump cover		SUS316	1
34	Discharge valve screw		SUS316	2
35	Mechanical seal	NJK-16-Q18FF	Carbon silicon carbide-FT	1
36	Cylindrical pin	ø6X25	SUS316	1
37	Hexagon socket head cap screw	M6X20	A3	4
38	Coupling			1
39	Motor base		HT200	1
40	Name plate		AL	1
41	Plate rivets	ø2.5	H62	4
42	Washer	ø16		4
43	Hexagonal nut	M16		4
44	Hexagonal bolt	M10X30	A3	4
45	Protective cover		SUS304	2
46	Cross recessed pan head screw and washer assemblies	M4X8	SUS304	4
47	Motor			1

● FCV-5-7VT

● Vertical Multi-Stage Pump

● Features :-

Stainless Steel 304

3 phase

220-380V

60HZ

IE3

2 pole

3500 rpm

Class 155(F)

Protection: IP55

● Product Photo :-



● Proformance Table :-

Model	Power		Max flow	Max head	Voltage	Inlet	Outlet
	KW	HP	m ³ /h	meters		mm	mm
FCV-5-7VT	2.2	3.0	10	65	220-380V	DN32	DN32

● Design

Vertical multistage in Line pump where Discharge and suction ports are located on one line to make it easier for connection access, it's either in Oval Victualic threaded to NPT cast iron discharge casing design or Slip on flanged and drilled to Universal standard which suite both DIN and ANSI standards.

● Applications: -

Water supply

Industry

Liquid transfer

Water treatment

Irrigation

● FCV Coding:

F: FUJIKA

C: Clean water

V: Vertical pump

● Classifications:

FCV: Standard SS304

FCV-N: SS316

V: Oval flange

R: Round flange

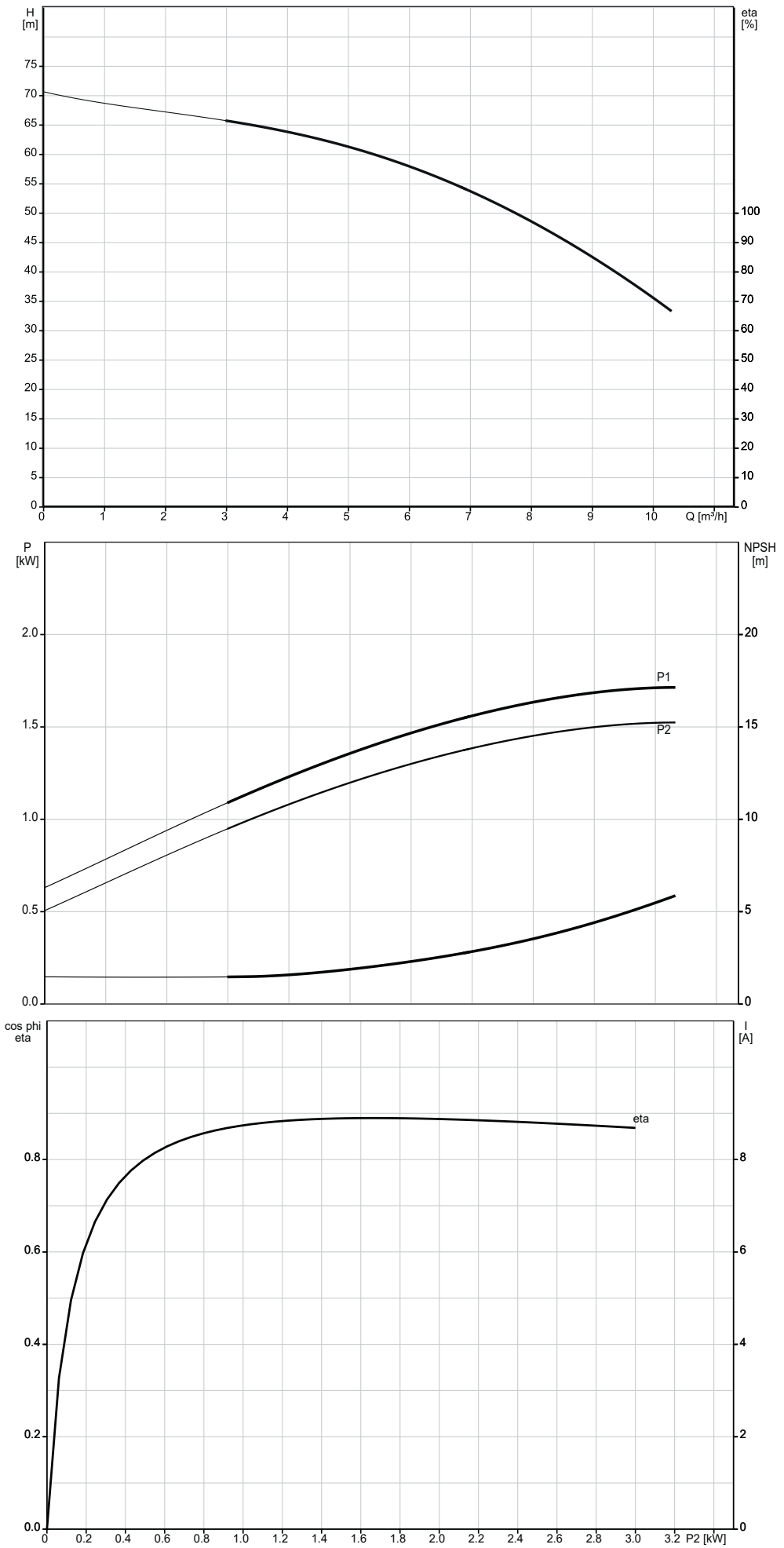
T: Three phase

● Working conditions: -

Temperature range: -20 Up to 104 °C

Ambient temperatue 50 °C

PERFORMANCE CURVE



Model: FCV-5-7VT



Design: Vertical multistage in Line pump where Discharge and suction ports are located on one line to make it easier for connection access, it's either in Oval Victualic threaded to NPT cast iron discharge casing design or Slip on flanged and drilled to Universal standard which suite both DIN and ANSI standards.

Construction of Material:

FCV basic design is made of :

Stainless steel imeller, sleeves and shaft , and cast iron discarge casing.

FCV-N

All wetted parts are made from SUS316 design.

The Shaft Mechanical seal is cartridge pull out type for easier replacement.

Mechanical Seal is EPDM /SUS304 for FCV and VITON / SUS316 for FCV-N Ensuring high reliability.

Coupling:

Shaft Coupling are rigid split type PJE Coupling.

Motor Type:

IE3 / IE2, 3-phase AC squirrel cage synchronous, high efficiency, TEFC fan cooled, IP55 Jetting protected.

General description :

The high efficiency ratings are in conformity to grade 3 efficiency value GB18613-2012 and IE of IEC60034-30 standard, but also the efficiency grade and installation dimensions are totally in conformity to IEC standard.

Motor shape is square motor as standard and round motor for larger sizes.

General specifications:

Operation factor: S1 (Continuous duty)

Insulation class: F

Protection class: IP55

Cooling ref.: IC411

Mount type: B35/B6

Ambient temperature: (-15 Up to 60 °C)

Altitude: ≤1000m

Motor paint finish is made with the most high technology with double painting and epoxy coating.

Motor housing construction are alaminum for motor rated up to 15HP and high grade cast iron for motors above rating.

The colour code: RAL 9005 deep jet black.

A standard split coupling connects the pump and motor shaft.

It's enclosed in the pump head/motor stool by means of two coupling guards.

The pump head and flange for motor mounting bracket is made in one piece (cast iron). the pump head cover is a seperate component (stainless steel).

Pump head cover stage has 1/2" priming plug and vent screw.

The pump is fitted with a balanced O-ring seal unit with a rigid torque-transmission system.

This seal type is assembled in a cartridge unit which makes replacement safe and easy.

In compliance with:

ENERGY EFFICIENCY ASIA JAPAN, IE3 IE2 IE1

MEPS (Minimum energy performance standard)

Current Regulation Energy saving act / top runner program standard

JIS C 4034-30

Power supply system

Three phase

Minimum energy performance IE3

Operated with inverter frequency: ok

Output (KW): 0.75 Up to 55KW

Number of poles: 2,4,6

Voltage (V): Up to 1000V

Frequency (HZ): 50HZ,60HZ,50/60HZ

Service duty: S1 ≥ 100%

Cooling method: tefc

Degree of protection: IP55

Area classification: safe area

Altitude: all

Ambient temperature: from (-20 °C and above)

Pump technical data

Rated flow: 5 m³/h
 Rated head: 25 meter
 Pump speed: 3450 rpm
 Nominal pressure PN: 25 bar
 Maximum pressure rating: 125% than shutoff pressure
 Pump mounting: vertical base mounting pump
 Shaft seal arrangement: single face cartridge type mechanical seal
 Seal code: Sic/Sic/Viton/SS316

Materials:

Wetted parts: Stainless steel EN 1.4408 SUS316
 Bearing: CU or SKF life sealed
 greased bearing
 Insulation: vertical
 Ambient temperature: 60 °C
 Maximum operating pressure: 25 bar
 Maximum pressure at stated temperature: 25 bar @ 0-90 °C

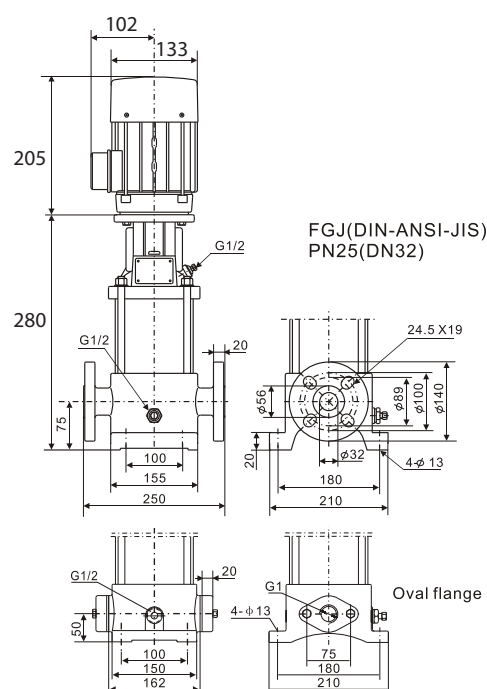
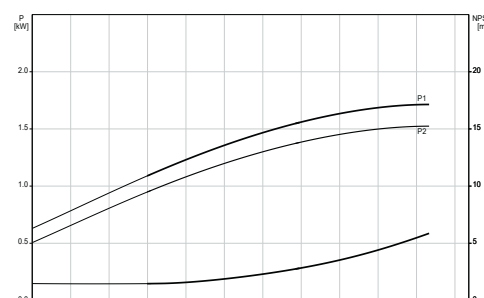
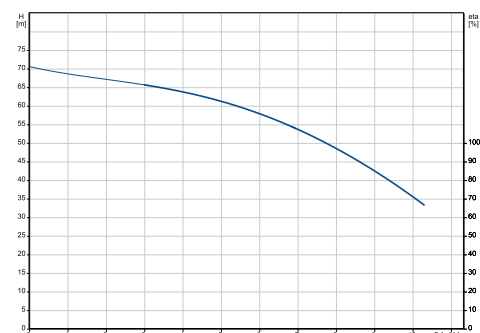
Motor type: by FUJIKA

IE efficiency class: IE3
 Rated power P2: 2.2 KW
 Power (P2) required by pump: 2.2 KW
 Frequency: 60HZ
 Rated voltage: 3 X 220-255V Δ / 380-440V Y
 Rated current: 8.4A / 4.9A
 Starting current: 820-1050%
 Cos phi - power: 0.84
 Rotational speed: 3400-3450 rpm
 IE efficiency: IE3 86.3%
 Minimum efficiency: 81.1%
 Maximum efficiency: 89.5%
 Number of poles: 2
 Enclosure class: IP55
 Insulation class: F (IEC)

Dimensional:

Connection: UNIVERSAL flange drilled to both DIN & ANSI standards.
 Flange size for motor:
 Electrical data: IEC
 Motor standard: IEC DIN
 Approvals: CE
 Certified: IECEE, Coc
 Approvals for drinking water: WRAS
 Performance curve: ISO9906 2012

Gross weight: 34 KG
 Shipping volume: 0.092 m³
 Country of origin: PRC



Material resistivity to chemicals:

Stainless steel 316 material composition Up to :
18% chromium, 10% nickel, and 2% molybdenum.

Allows it to be used in variable applications and
that require superior corrosion resistance and
hygienic.

Applications such as:

Acidic water
Dissolved Chlorine
Sea water
Falls and Fountains
Water treatment
Diesel Fuel
Aggressive oil
Cooking oil
Drinking water
Juices
Water treatment
Food industry
Pharmaceutical Industry
Hygienic process
Fertilization
Fisheries
Irrigation

Characteristics:

The 316 stainless steel is used in salt and other
chlorides applications exposure and lasting
durability
steel over 304.

316 stainless steel's added molybdenum, makes it
much more corrosion resistant than 304 or other
stainless steel alloys. This allows 316 stainless steel
to be used in highly acidic environments that
other steels can't tolerate. 316's corrosion-resist-
ant properties also make it ideal in food process-
ing, pharmaceutical and medical applications due
to its ability to withstand the disinfectants and
high temperatures media.

Exposure to salt and chlorides can cause havoc on
304 stainless steel alloys. Salt can cause 304 stain-
less steel to rust by attacking the protective oxide
layer, and 304 steel is also more easily wear from
the effects of salt and chlorides. However, 316
stainless steel has all of the desirability properties of
304 stainless steel, but with the added benefit of
molybdenum, which allows it to withstand expo-
sure to salt and chlorides for more durability.

Austenitic stainless steels generally show high
corrosion rates when exposed to concentrated,
de-aerated alkaline aqueous environment at tem-
peratures above 60 °C.

Permissible corrosion rates of 0.05–2 mm/year.

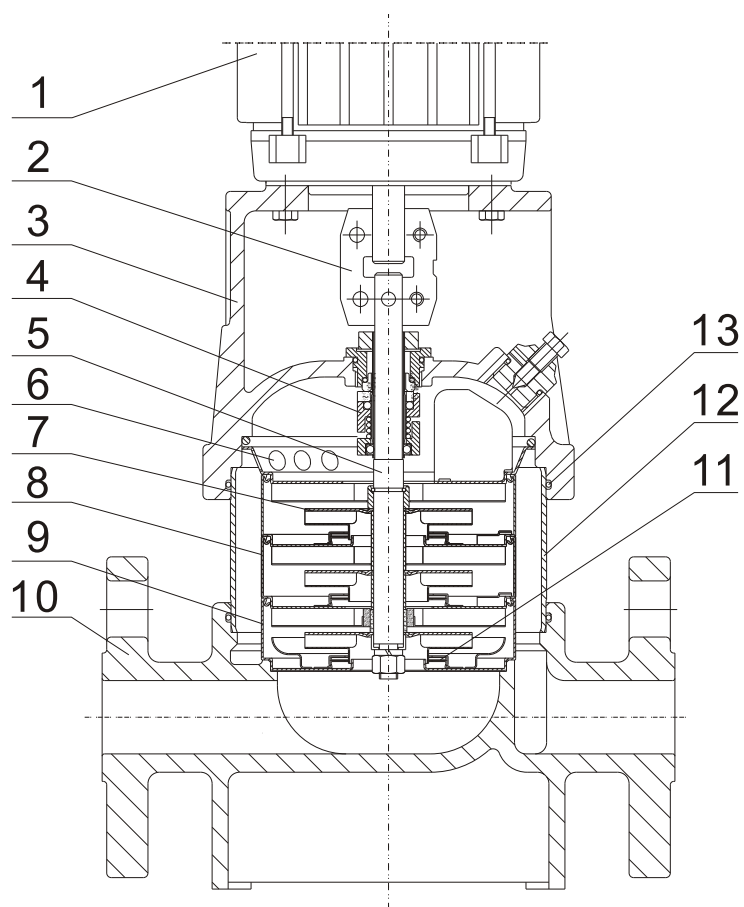
Caution:

The common stainless steel sheets types, 304 and
316 pumps should be considered non-resistant to
hydrochloric acid at any concentration and temper-
ature.

For Sulfuric acid:

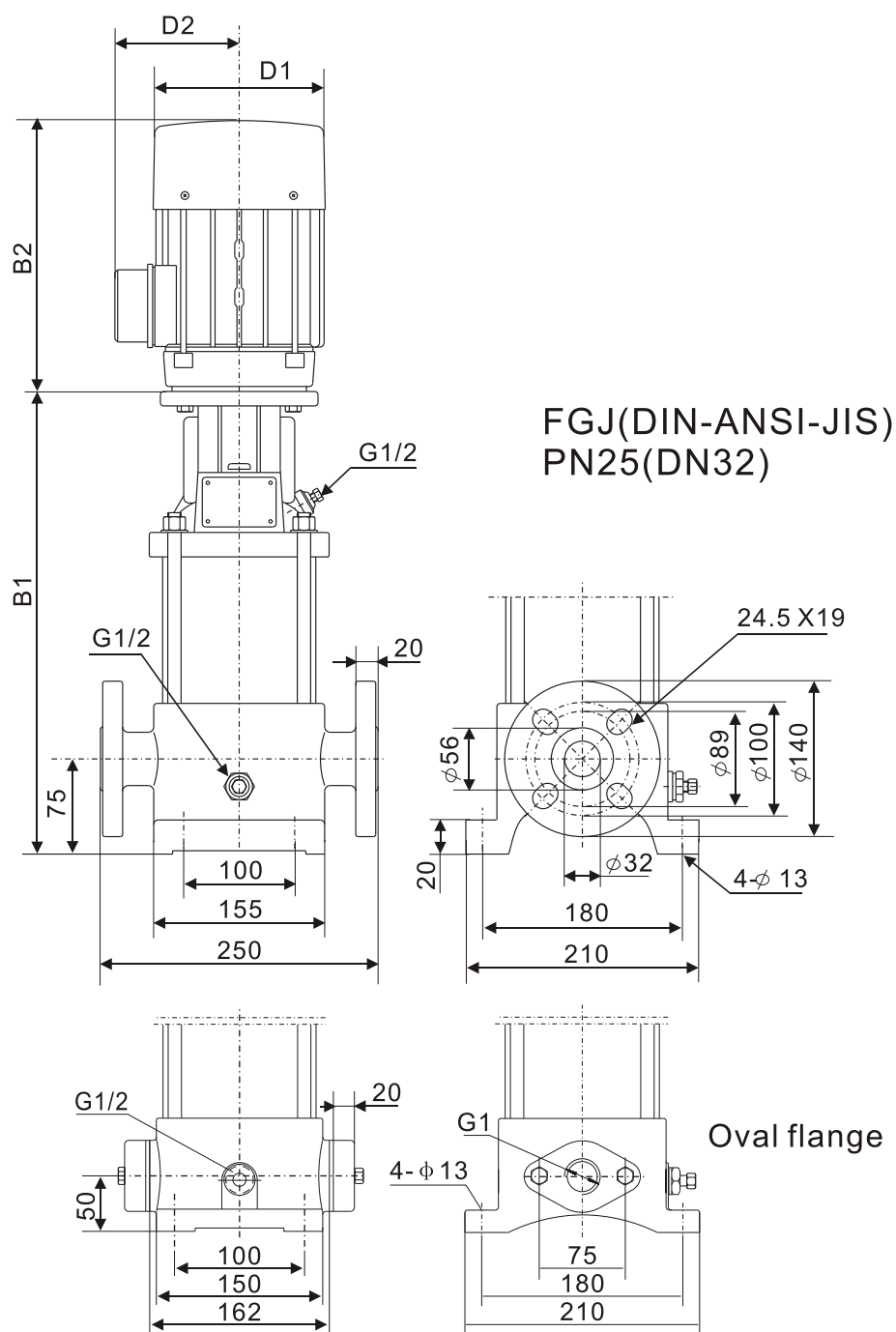
SUS304 is only resistant up to 1% acid while type
316 resistant is up to 10% acid at 50° C temp.

SECTIONAL DRAWING

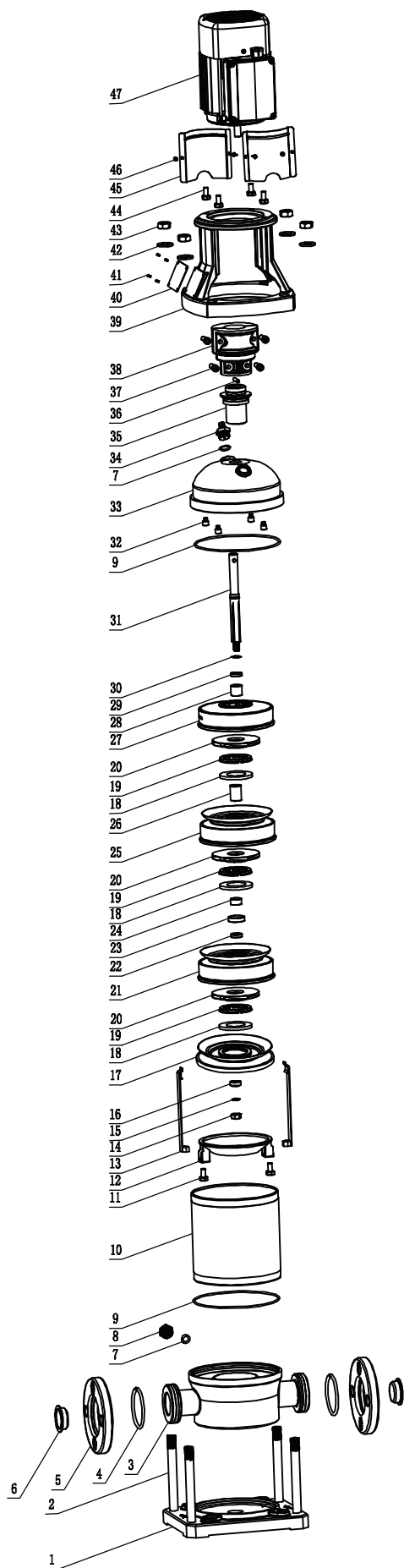


No	Description	Material	EN/DIN	AISI/ASTM
1	Motor	X	X	X
2	Shaft connector	X	X	X
3	Pump head	Cast iron	EN-JL 1030	ASTM25B
4	Mechanical seal	X	X	X
5	Shaft	S.S	X	AISI420
6	Outlet	S.S	1.4301	AISI304
7	Impeller	S.S	1.4301	AISI304
8	Hydraulic stack	S.S	1.4301	AISI304
9	Inlet	S.S	1.4301	AISI304
10	Pump body	Cast iron	EN.JL1030	ASTM25B
11	Neck ring	PTEF	X	X
12	Outer sleeve	S.S	1.4301	AISI316
13	O-ring	EDM/FKM	X	X

DIMENSIONS



Model	Dimensions (mm)					Weight
	B1	B2	B1+B2	D1	D2	KG
FCV-5-7VT	429	275/293	731	177	116	34



NO	Name	Spec.	Material	QTY
1	Base		HT200	1
2	Two-head bolt stud	M16XL	Chrome	4
3	Pump body		SUS304	1
4	Flange clasp		SUS304	2
5	Flange		SUS304	2
6	Dust cap	G1	PP	2
7	Ring	d18X2.4	NBR	2
8	Drain valve assembly		SUS304	1
9	Ring	d170X3.1	NBR	2
10	Housing		SUS316	1
11	Hexagon bolt	M8X20	SUS304	2
12	Fixed board		SUS316	1
13	Tension rod		SUS316	2
14	Type 1 non-metallic insert hexagon lock nut	M10	SUS316	1
15	Spring washer	ø10	SUS316	1
16	Impeller fix ring		SUS316	1
17	Inlet section		SUS316	1
18	Impeller fix ring		PTFE	N
19	Mouthpiece outer ring		SUS316	N
20	impeller		SUS316	N
21	Diffusor with bearing		SUS316	N
22	Short sleeve	L=4.5	SUS316	N
23	Bearing sleeve		Hard alloy	N
24	Short sleeve	L=10.5	SUS316	N
25	Diffusor		SUS316	N
26	Long sleeve	L=29	SUS316	N
27	Outlet section		SUS316	1
28	Short sleeve	L=19	SUS316	1
29	Snap ring seat		SUS316	1
30	Clasp		SUS316	1
31	Pump shaft		SUS316	1
32	Clasp seat shock colloidal		FI	4
33	Pump cover		SUS316	1
34	Discharge valve screw		SUS316	2
35	Mechanical seal	NJK-16-Q18FF	Carbon silicon carbide-FT	1
36	Cylindrical pin	ø6X25	SUS316	1
37	Hexagon socket head cap screw	M6X20	A3	4
38	Coupling			1
39	Motor base		HT200	1
40	Name plate		AL	1
41	Plate rivets	ø2.5	H62	4
42	Washer	ø16		4
43	Hexagonal nut	M16		4
44	Hexagonal bolt	M10X30	A3	4
45	Protective cover		SUS304	2
46	Cross recessed pan head screw and washer assemblies	M4X8	SUS304	4
47	Motor			1

● FCV-5-8VT

● Vertical Multi-Stage Pump

● Features :-

Stainless Steel 304

3 phase

220-380V

60HZ

IE3

2 pole

3500 rpm

Class 155(F)

Protection: IP55

● Product Photo :-



● Proformance Table :-

Model	Power		Max flow	Max head	Voltage	Inlet	Outlet
	KW	HP	m ³ /h	meters		mm	mm
FCV-5-8VT	2.2	3.0	10	75	220-380V	DN32	DN32

● Design

Vertical multistage in Line pump where Discharge and suction ports are located on one line to make it easier for connection access, it's either in Oval Victualic threaded to NPT cast iron discharge casing design or Slip on flanged and drilled to Universal standard which suite both DIN and ANSI standards.

● Applications: -

Water supply

Industry

Liquid transfer

Water treatment

Irrigation

● FCV Coding:

F: FUJIKA

C: Clean water

V: Vertical pump

● Classifications:

FCV: Standard SS304

FCV-N: SS316

V: Oval flange

R: Round flange

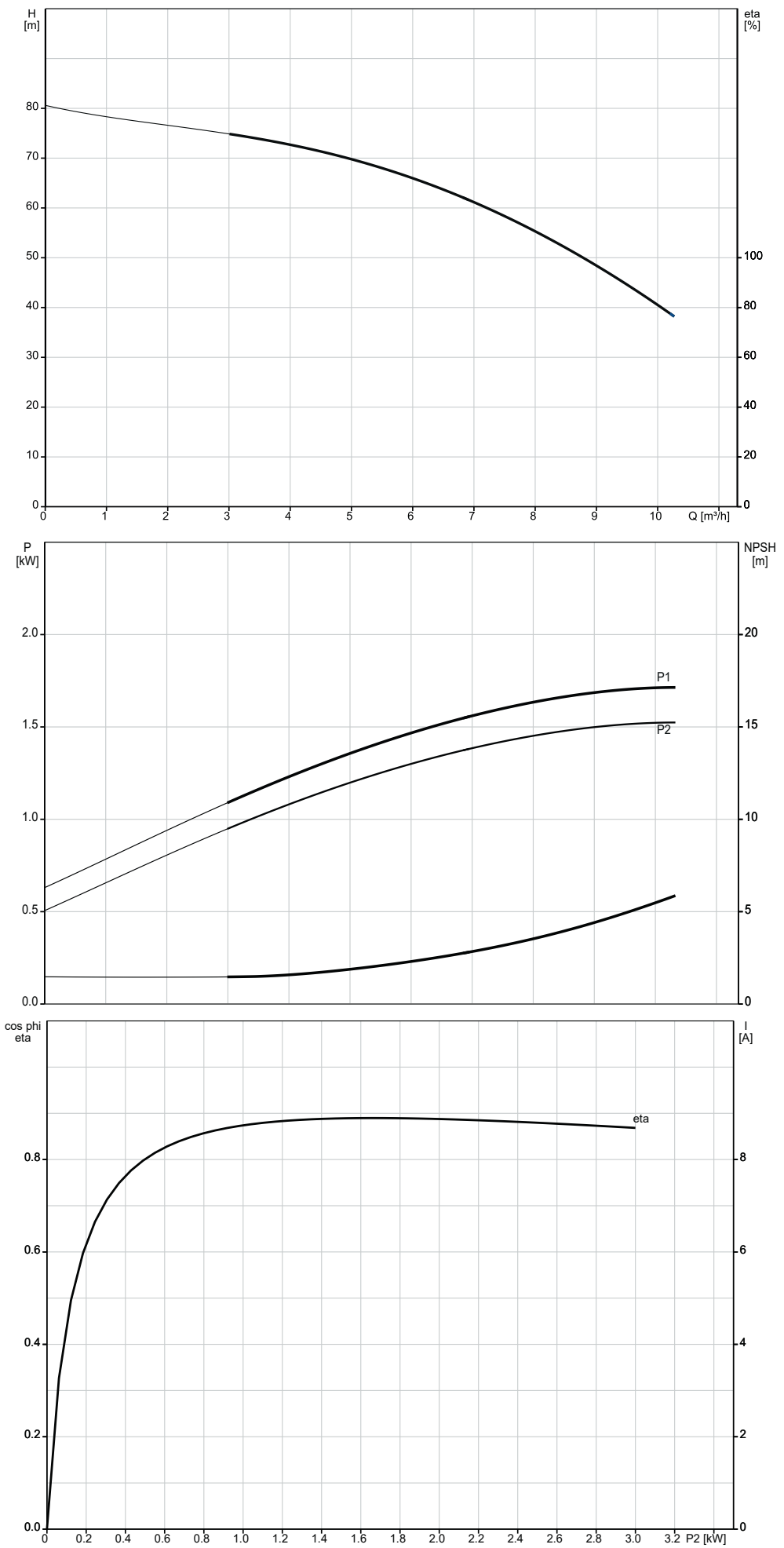
T: Three phase

● Working conditions: -

Temperature range: -20 Up to 104 °C

Ambient temperatue 50 °C

PERFORMANCE CURVE



Model: FCV-5-8VT



Design: Vertical multistage in Line pump where Discharge and suction ports are located on one line to make it easier for connection access, it's either in Oval Victualic threaded to NPT cast iron discharge casing design or Slip on flanged and drilled to Universal standard which suite both DIN and ANSI standards.

Construction of Material:

FCV basic design is made of :

Stainless steel imeller, sleeves and shaft , and cast iron discarge casing.

FCV-N

All wetted parts are made from SUS316 design.

The Shaft Mechanical seal is cartridge pull out type for easier replacement.

Mechanical Seal is EPDM /SUS304 for FCV and VITON / SUS316 for FCV-N Ensuring high reliability.

Coupling:

Shaft Coupling are rigid split type PJE Coupling.

Motor Type:

IE3 / IE2, 3-phase AC squirrel cage synchronous, high efficiency, TEFC fan cooled, IP55 Jetting protected.

General description :

The high efficiency ratings are in conformity to grade 3 efficiency value GB18613-2012 and IE of IEC60034-30 standard, but also the efficiency grade and installation dimensions are totally in conformity to IEC standard.

Motor shape is square motor as standard and round motor for larger sizes.

General specifications:

Operation factor: S1 (Continuous duty)

Insulation class: F

Protection class: IP55

Cooling ref.: IC411

Mount type: B35/B6

Ambient temperature: (-15 Up to 60 °C)

Altitude: ≤1000m

Motor paint finish is made with the most high technology with double painting and epoxy coating.

Motor housing construction are alaminum for motor rated up to 15HP and high grade cast iron for motors above rating.

The colour code: RAL 9005 deep jet black.

A standard split coupling connects the pump and motor shaft.

It's enclosed in the pump head/motor stool by means of two coupling guards.

The pump head and flange for motor mounting bracket is made in one piece (cast iron). the pump head cover is a seperate component (stainless steel).

Pump head cover stage has 1/2" priming plug and vent screw.

The pump is fitted with a balanced O-ring seal unit with a rigid torque-transmission system.

This seal type is assembled in a cartridge unit which makes replacement safe and easy.

In compliance with:

ENERGY EFFICIENCY ASIA JAPAN, IE3 IE2 IE1

MEPS (Minimum energy performance standard)

Current Regulation Energy saving act / top runner program standard

JIS C 4034-30

Power supply system

Three phase

Minimum energy performance IE3

Operated with inverter frequency: ok

Output (KW): 0.75 Up to 55KW

Number of poles: 2,4,6

Voltage (V): Up to 1000V

Frequency (HZ): 50HZ,60HZ,50/60HZ

Service duty: S1 ≥ 100%

Cooling method: tefc

Degree of protection: IP55

Area classification: safe area

Altitude: all

Ambient temperature: from (-20 °C and above)

Pump technical data

Rated flow: 5 m³/h
 Rated head: 60 meter
 Pump speed: 3450 rpm
 Nominal pressure PN: 25 bar
 Maximum pressure rating: 125% than shutoff pressure
 Pump mounting: vertical base mounting pump
 Shaft seal arrangement: single face cartridge type mechanical seal
 Seal code: Sic/Sic/Viton/SS316

Materials:

Wetted parts: Stainless steel EN 1.4408 SUS316
 Bearing: CU or SKF life sealed greased bearing
 Insulation: vertical
 Ambient temperature: 60 °C
 Maximum operating pressure: 25 bar
 Maximum pressure at stated temperature: 25 bar @ 0-90 °C

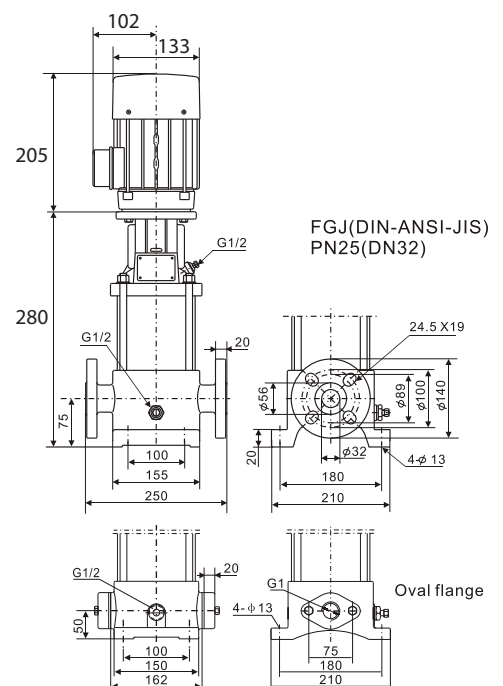
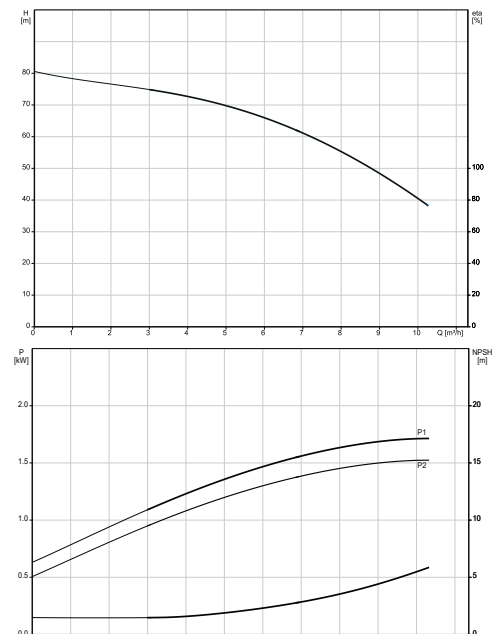
Motor type: by FUJIKA

IE efficiency class: IE3
 Rated power P2: 2.2 KW
 Power (P2) required by pump: 2.2 KW
 Frequency: 60HZ
 Rated voltage: 3 X 220-255V Δ / 380-440V Y
 Rated current: 2.95A / 1.7A
 Starting current: 820-1080%
 Cos phi - power: 0.84
 Rotational speed: 3400-3450 rpm
 IE efficiency: IE3 86.3%
 Minimum efficiency: 81.1%
 Maximum efficiency: 89.5%
 Number of poles: 2
 Enclosure class: IP55
 Insulation class: F (IEC)

Dimensional:

Connection: UNIVERSAL flange drilled to both DIN & ANSI standards.
 Flange size for motor:
 Electrical data: IEC
 Motor standard: IEC DIN
 Approvals: CE
 Certified: IECEE, Coc
 Approvals for drinking water: WRAS
 Performance curve: ISO9906 2012

Gross weight: 38 KG
 Shipping volume: 0.092 m³
 Country of origin: PRC



Material resistivity to chemicals:

Stainless steel 316 material composition Up to :
18% chromium, 10% nickel, and 2% molybdenum.

Allows it to be used in variable applications and
that require superior corrosion resistance and
hygienic.

Applications such as:

Acidic water
Dissolved Chlorine
Sea water
Falls and Fountains
Water treatment
Diesel Fuel
Aggressive oil
Cooking oil
Drinking water
Juices
Water treatment
Food industry
Pharmaceutical Industry
Hygienic process
Fertilization
Fisheries
Irrigation

Characteristics:

The 316 stainless steel is used in salt and other
chlorides applications exposure and lasting
durability
steel over 304.

316 stainless steel's added molybdenum, makes it
much more corrosion resistant than 304 or other
stainless steel alloys. This allows 316 stainless steel
to be used in highly acidic environments that
other steels can't tolerate. 316's corrosion-resist-
ant properties also make it ideal in food process-
ing, pharmaceutical and medical applications due
to its ability to withstand the disinfectants and
high temperatures media.

Exposure to salt and chlorides can cause havoc on
304 stainless steel alloys. Salt can cause 304 stain-
less steel to rust by attacking the protective oxide
layer, and 304 steel is also more easily wear from
the effects of salt and chlorides. However, 316
stainless steel has all of the desirability properties of
304 stainless steel, but with the added benefit of
molybdenum, which allows it to withstand expo-
sure to salt and chlorides for more durability.

Austenitic stainless steels generally show high
corrosion rates when exposed to concentrated,
de-aerated alkaline aqueous environment at tem-
peratures above 60 °C.

Permissible corrosion rates of 0.05–2 mm/year.

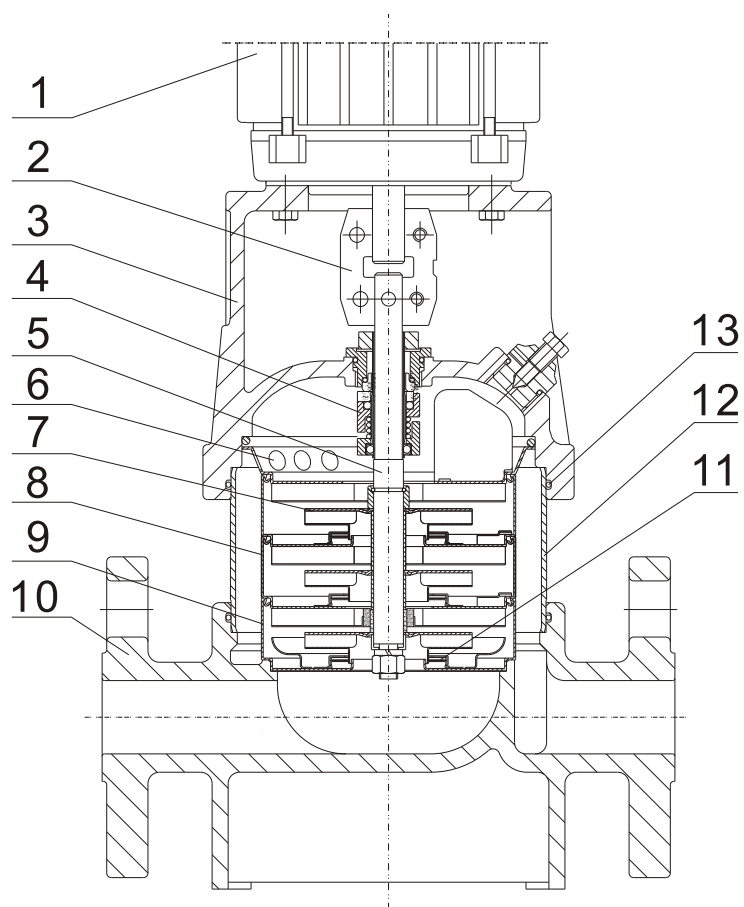
Caution:

The common stainless steel sheets types, 304 and
316 pumps should be considered non-resistant to
hydrochloric acid at any concentration and temper-
ature.

For Sulfuric acid:

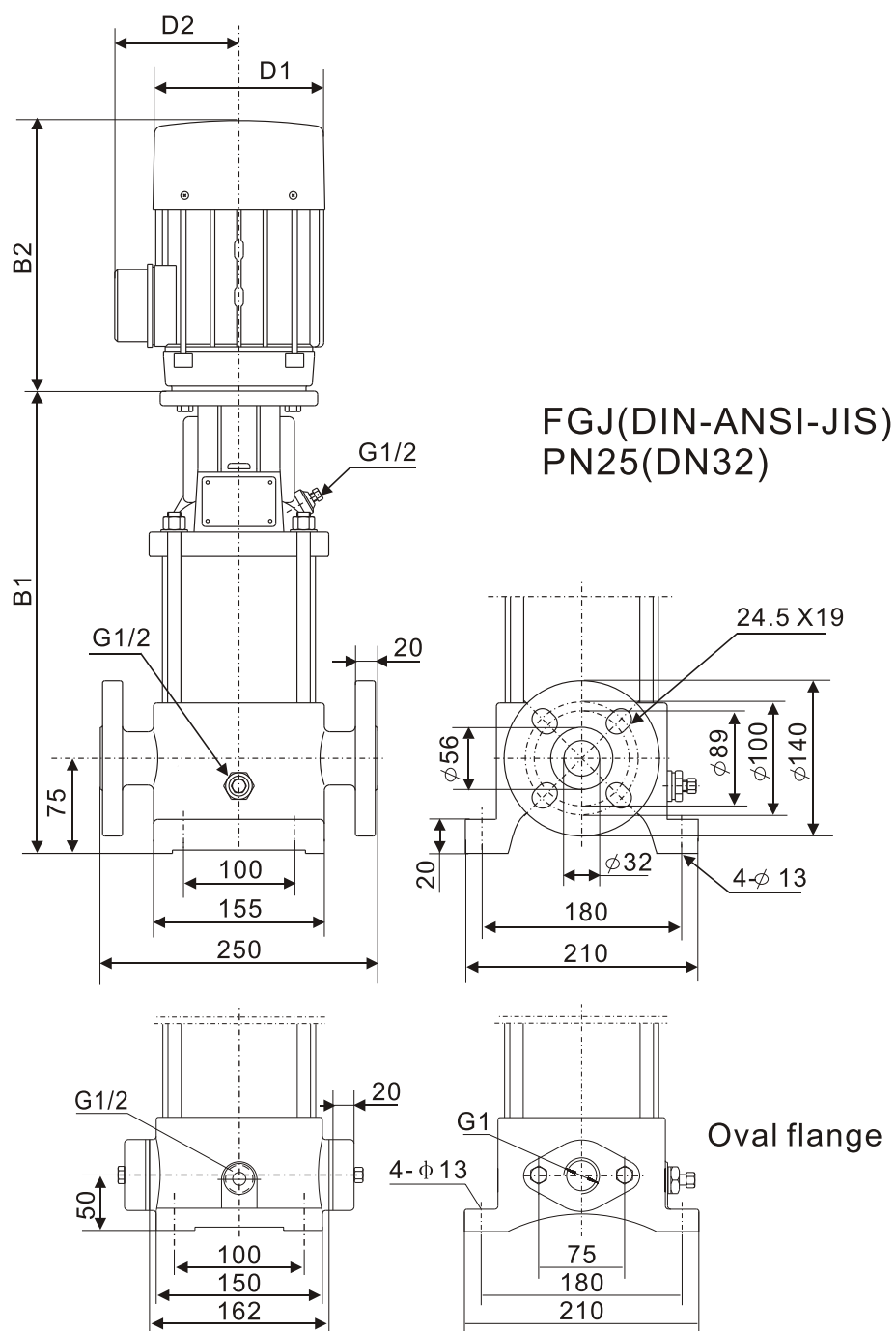
SUS304 is only resistant up to 1% acid while type
316 resistant is up to 10% acid at 50° C temp.

SECTIONAL DRAWING

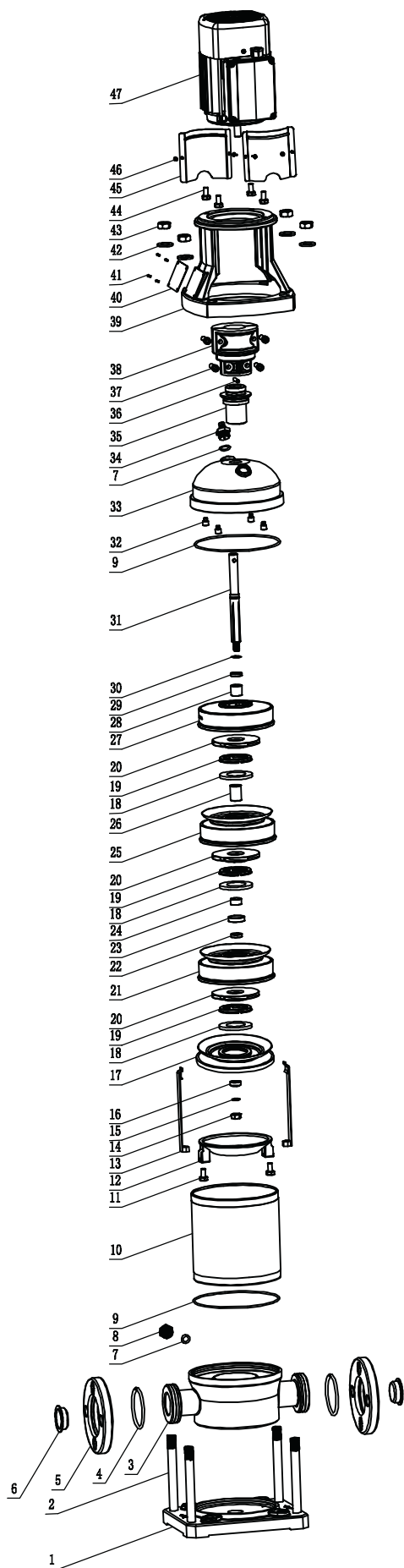


No	Description	Material	EN/DIN	AISI/ASTM
1	Motor	X	X	X
2	Shaft connector	X	X	X
3	Pump head	Cast iron	EN-JL 1030	ASTM25B
4	Mechanical seal	X	X	X
5	Shaft	S.S	X	AISI420
6	Outlet	S.S	1.4301	AISI304
7	Impeller	S.S	1.4301	AISI304
8	Hydraulic stack	S.S	1.4301	AISI304
9	Inlet	S.S	1.4301	AISI304
10	Pump body	Cast iron	EN.JL1030	ASTM25B
11	Neck ring	PTEF	X	X
12	Outer sleeve	S.S	1.4301	AISI316
13	O-ring	EDM/FKM	X	X

DIMENSIONS



Model	Dimensions (mm)					Weight
	B1	B2	B1+B2	D1	D2	KG
FCV-5-8VT	456	275/293	731	177	116	38



NO	Name	Spec.	Material	QTY
1	Base		HT200	1
2	Two-head bolt stud	M16XL	Chrome	4
3	Pump body		SUS304	1
4	Flange clasp		SUS304	2
5	Flange		SUS304	2
6	Dust cap	G1	PP	2
7	Ring	d18X2.4	NBR	2
8	Drain valve assembly		SUS304	1
9	Ring	d170X3.1	NBR	2
10	Housing		SUS316	1
11	Hexagon bolt	M8X20	SUS304	2
12	Fixed board		SUS316	1
13	Tension rod		SUS316	2
14	Type 1 non-metallic insert hexagon lock nut	M10	SUS316	1
15	Spring washer	ø10	SUS316	1
16	Impeller fix ring		SUS316	1
17	Inlet section		SUS316	1
18	Impeller fix ring		PTFE	N
19	Mouthpiece outer ring		SUS316	N
20	impeller		SUS316	N
21	Diffusor with bearing		SUS316	N
22	Short sleeve	L=4.5	SUS316	N
23	Bearing sleeve		Hard alloy	N
24	Short sleeve	L=10.5	SUS316	N
25	Diffusor		SUS316	N
26	Long sleeve	L=29	SUS316	N
27	Outlet section		SUS316	1
28	Short sleeve	L=19	SUS316	1
29	Snap ring seat		SUS316	1
30	Clasp		SUS316	1
31	Pump shaft		SUS316	1
32	Clasp seat shock colloidal		FI	4
33	Pump cover		SUS316	1
34	Discharge valve screw		SUS316	2
35	Mechanical seal	NJK-16-Q18FF	Carbon silicon carbide-FT	1
36	Cylindrical pin	ø6X25	SUS316	1
37	Hexagon socket head cap screw	M6X20	A3	4
38	Coupling			1
39	Motor base		HT200	1
40	Name plate		AL	1
41	Plate rivets	ø2.5	H62	4
42	Washer	ø16		4
43	Hexagonal nut	M16		4
44	Hexagonal bolt	M10X30	A3	4
45	Protective cover		SUS304	2
46	Cross recessed pan head screw and washer assemblies	M4X8	SUS304	4
47	Motor			1

● FCV-5-9VT

● Vertical Multi-Stage Pump

● Features :-

Stainless Steel 304

3 phase

220-380V

60HZ

IE3

2 pole

3500 rpm

Class 155(F)

Protection: IP55

● Product Photo :-



● Proformance Table :-

Model	Power		Max flow	Max head	Voltage	Inlet	Outlet
	KW	HP	m ³ /h	meters		mm	mm
FCV-5-9VT	2.2	3.0	10	84	220-380V	DN32	DN32

● Design

Vertical multistage in Line pump where Discharge and suction ports are located on one line to make it easier for connection access, it's either in Oval Victualic threaded to NPT cast iron discharge casing design or Slip on flanged and drilled to Universal standard which suite both DIN and ANSI standards.

● Applications: -

Water supply

Industry

Liquid transfer

Water treatment

Irrigation

● FCV Coding:

F: FUJIKA

C: Clean water

V: Vertical pump

● Classifications:

FCV: Standard SS304

FCV-N: SS316

V: Oval flange

R: Round flange

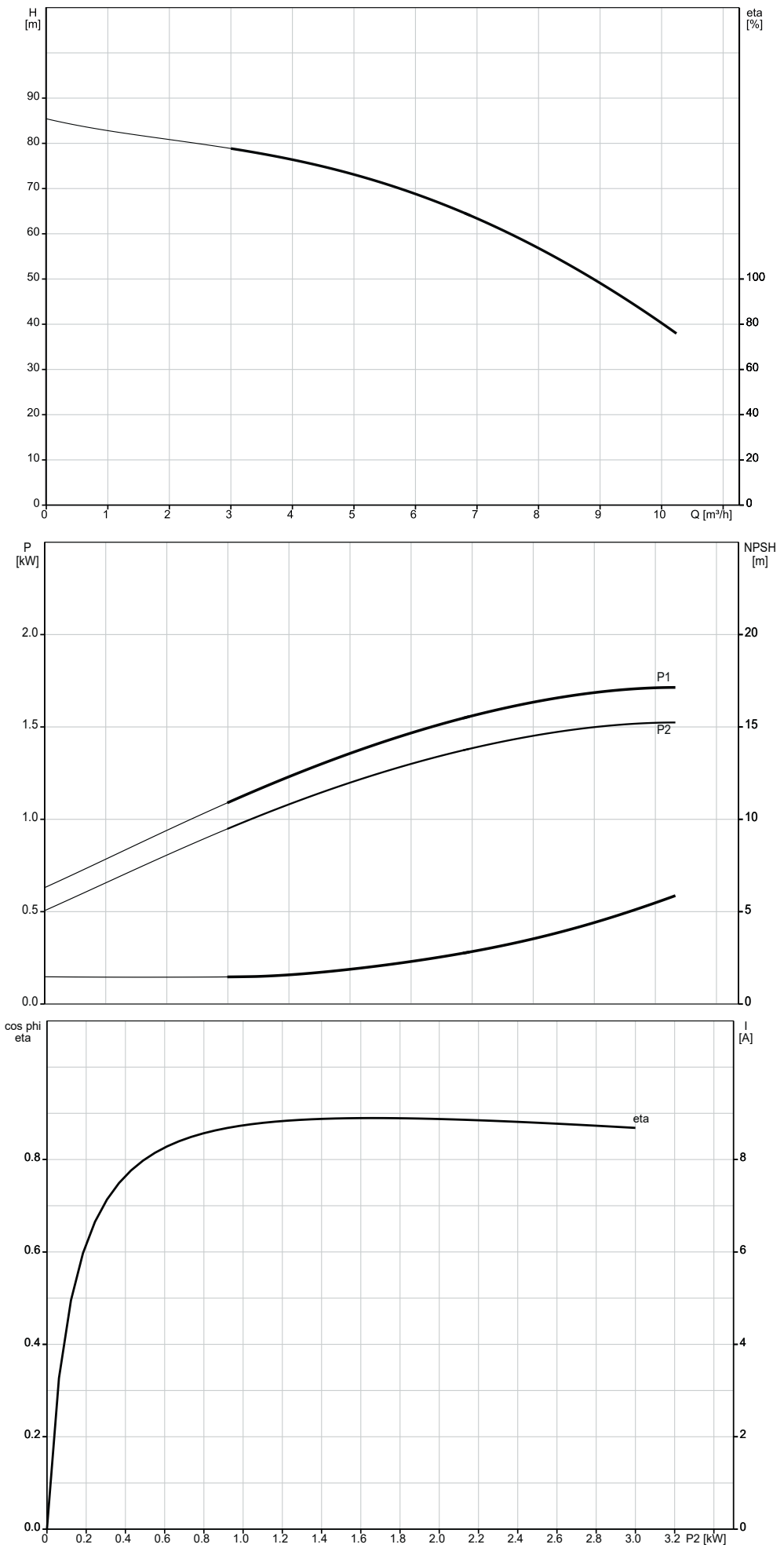
T: Three phase

● Working conditions: -

Temperature range: -20 Up to 104 °C

Ambient temperatue 50 °C

PERFORMANCE CURVE



Model: FCV-5-9VT



Design: Vertical multistage in Line pump where Discharge and suction ports are located on one line to make it easier for connection access, it's either in Oval Victualic threaded to NPT cast iron discharge casing design or Slip on flanged and drilled to Universal standard which suite both DIN and ANSI standards.

Construction of Material:

FCV basic design is made of :

Stainless steel imeller, sleeves and shaft , and cast iron discarge casing.

FCV-N

All wetted parts are made from SUS316 design.

The Shaft Mechanical seal is cartridge pull out type for easier replacement.

Mechanical Seal is EPDM /SUS304 for FCV and VITON / SUS316 for FCV-N Ensuring high reliability.

Coupling:

Shaft Coupling are rigid split type PJE Coupling.

Motor Type:

IE3 / IE2, 3-phase AC squirrel cage synchronous, high efficiency, TEFC fan cooled, IP55 Jetting protected.

General description :

The high efficiency ratings are in conformity to grade 3 efficiency value GB18613-2012 and IE of IEC60034-30 standard, but also the efficiency grade and installation dimensions are totally in conformity to IEC standard.

Motor shape is square motor as standard and round motor for larger sizes.

General specifications:

Operation factor: S1 (Continuous duty)

Insulation class: F

Protection class: IP55

Cooling ref.: IC411

Mount type: B35/B6

Ambient temperature: (-15 Up to 60 °C)

Altitude: ≤1000m

Motor paint finish is made with the most high technology with double painting and epoxy coating.

Motor housing construction are alaminum for motor rated up to 15HP and high grade cast iron for motors above rating.

The colour code: RAL 9005 deep jet black.

A standard split coupling connects the pump and motor shaft.

It's enclosed in the pump head/motor stool by means of two coupling guards.

The pump head and flange for motor mounting bracket is made in one piece (cast iron). the pump head cover is a seperate component (stainless steel).

Pump head cover stage has 1/2" priming plug and vent screw.

The pump is fitted with a balanced O-ring seal unit with a rigid torque-transmission system.

This seal type is assembled in a cartridge unit which makes replacement safe and easy.

In compliance with:

ENERGY EFFICIENCY ASIA JAPAN, IE3 IE2 IE1

MEPS (Minimum energy performance standard)

Current Regulation Energy saving act / top runner program standard

JIS C 4034-30

Power supply system

Three phase

Minimum energy performance IE3

Operated with inverter frequency: ok

Output (KW): 0.75 Up to 55KW

Number of poles: 2,4,6

Voltage (V): Up to 1000V

Frequency (HZ): 50HZ,60HZ,50/60HZ

Service duty: S1 ≥ 100%

Cooling method: tefc

Degree of protection: IP55

Area classification: safe area

Altitude: all

Ambient temperature: from (-20 °C and above)

Pump technical data

Rated flow: 5 m³/h
 Rated head: 68 meter
 Pump speed: 3450 rpm
 Nominal pressure PN: 25 bar
 Maximum pressure rating: 125% than shutoff pressure
 Pump mounting: vertical base mounting pump
 Shaft seal arrangement: single face cartridge type mechanical seal
 Seal code: Sic/Sic/Viton/SS316

Materials:

Wetted parts: Stainless steel EN 1.4408 SUS316
 Bearing: CU or SKF life sealed
 greased bearing
 Insulation: vertical
 Ambient temperature: 60 °C
 Maximum operating pressure: 25 bar
 Maximum pressure at stated temperature: 25 bar @ 0-90 °C

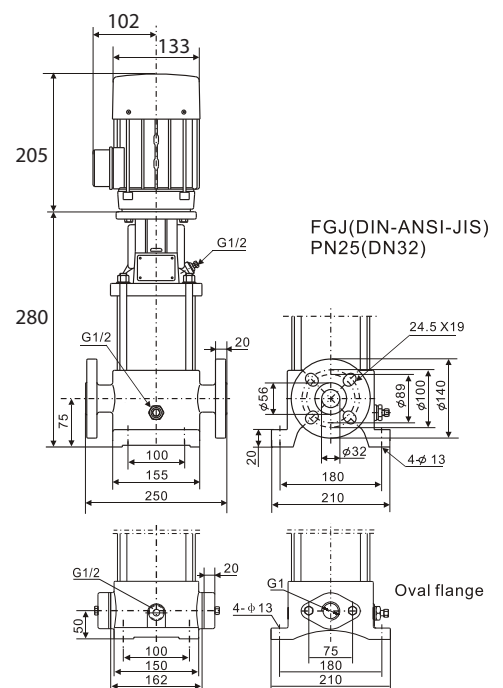
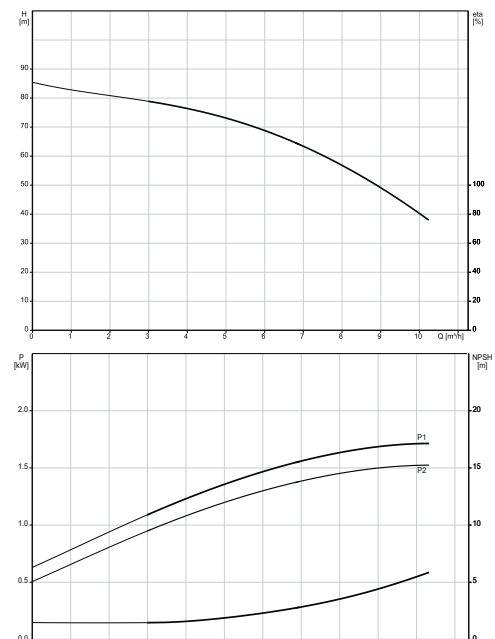
Motor type: by FUJIKA

IE efficiency class: IE3
 Rated power P2: 2.2 KW
 Power (P2) required by pump: 2.2 KW
 Frequency: 60HZ
 Rated voltage: 3 X 220-255V Δ / 380-440V Y
 Rated current: 8.4A / 4.9A
 Starting current: 820-1050%
 Cos phi - power: 0.84
 Rotational speed: 3400-3450 rpm
 IE efficiency: IE3 86.3%
 Minimum efficiency: 81.1%
 Maximum efficiency: 89.5%
 Number of poles: 2
 Enclosure class: IP55
 Insulation class: F (IEC)

Dimensional:

Connection: UNIVERSAL flange drilled to both DIN & ANSI standards.
 Flange size for motor:
 Electrical data: IEC
 Motor standard: IEC DIN
 Approvals: CE
 Certified: IECEE, Coc
 Approvals for drinking water: WRAS
 Performance curve: ISO9906 2012

Gross weight: 40 KG
 Shipping volume: 0.092 m³
 Country of origin: PRC



Material resistivity to chemicals:

Stainless steel 316 material composition Up to : 18% chromium, 10% nickel, and 2% molybdenum. Allows it to be used in variable applications and that require superior corrosion resistance and hygienic.

Applications such as:

- Acidic water
- Dissolved Chlorine
- Sea water
- Falls and Fountains
- Water treatment
- Diesel Fuel
- Aggressive oil
- Cooking oil
- Drinking water
- Juices
- Water treatment
- Food industry
- Pharmaceutical Industry
- Hygienic process
- Fertilization
- Fisheries
- Irrigation

Characteristics:

The 316 stainless steel is used in salt and other chlorides applications exposure and lasting durability steel over 304.

316 stainless steel's added molybdenum, makes it much more corrosion resistant than 304 or other stainless steel alloys. This allows 316 stainless steel to be used in highly acidic environments that other steels can't tolerate. 316's corrosion-resistant properties also make it ideal in food processing, pharmaceutical and medical applications due to its ability to withstand the disinfectants and high temperatures media.

Exposure to salt and chlorides can cause havoc on 304 stainless steel alloys. Salt can cause 304 stainless steel to rust by attacking the protective oxide layer, and 304 steel is also more easily wear from the effects of salt and chlorides. However, 316 stainless steel has all of the desirability properties of 304 stainless steel, but with the added benefit of molybdenum, which allows it to withstand exposure to salt and chlorides for more durability.

Austenitic stainless steels generally show high corrosion rates when exposed to concentrated, de-aerated alkaline aqueous environment at temperatures above 60 °C.

Permissible corrosion rates of 0.05–2 mm/year.

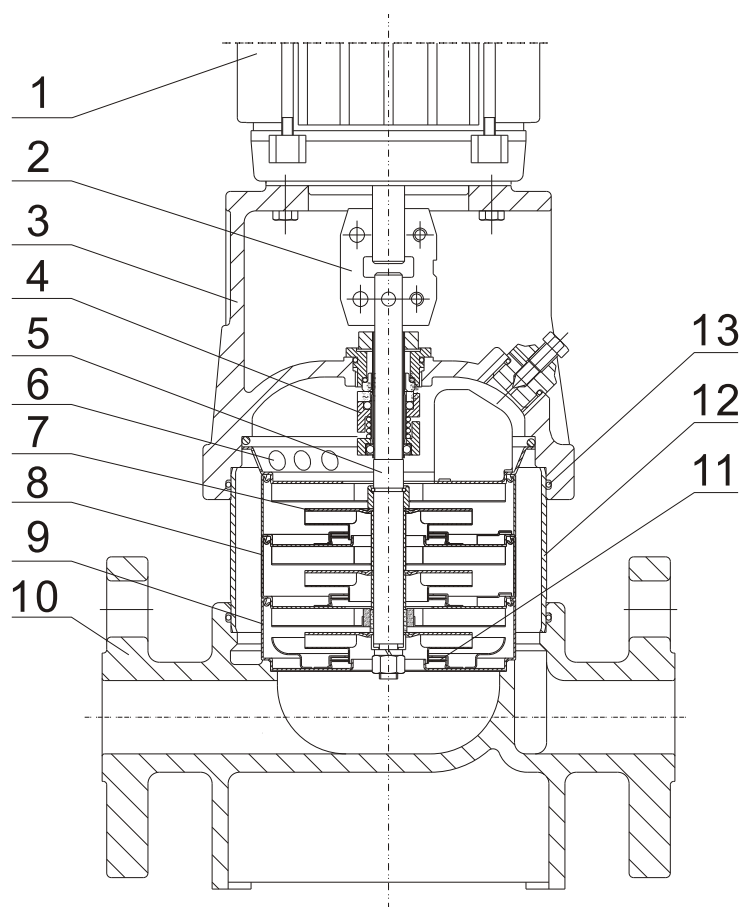
Caution:

The common stainless steel sheets types, 304 and 316 pumps should be considered non-resistant to hydrochloric acid at any concentration and temperature.

For Sulfuric acid:

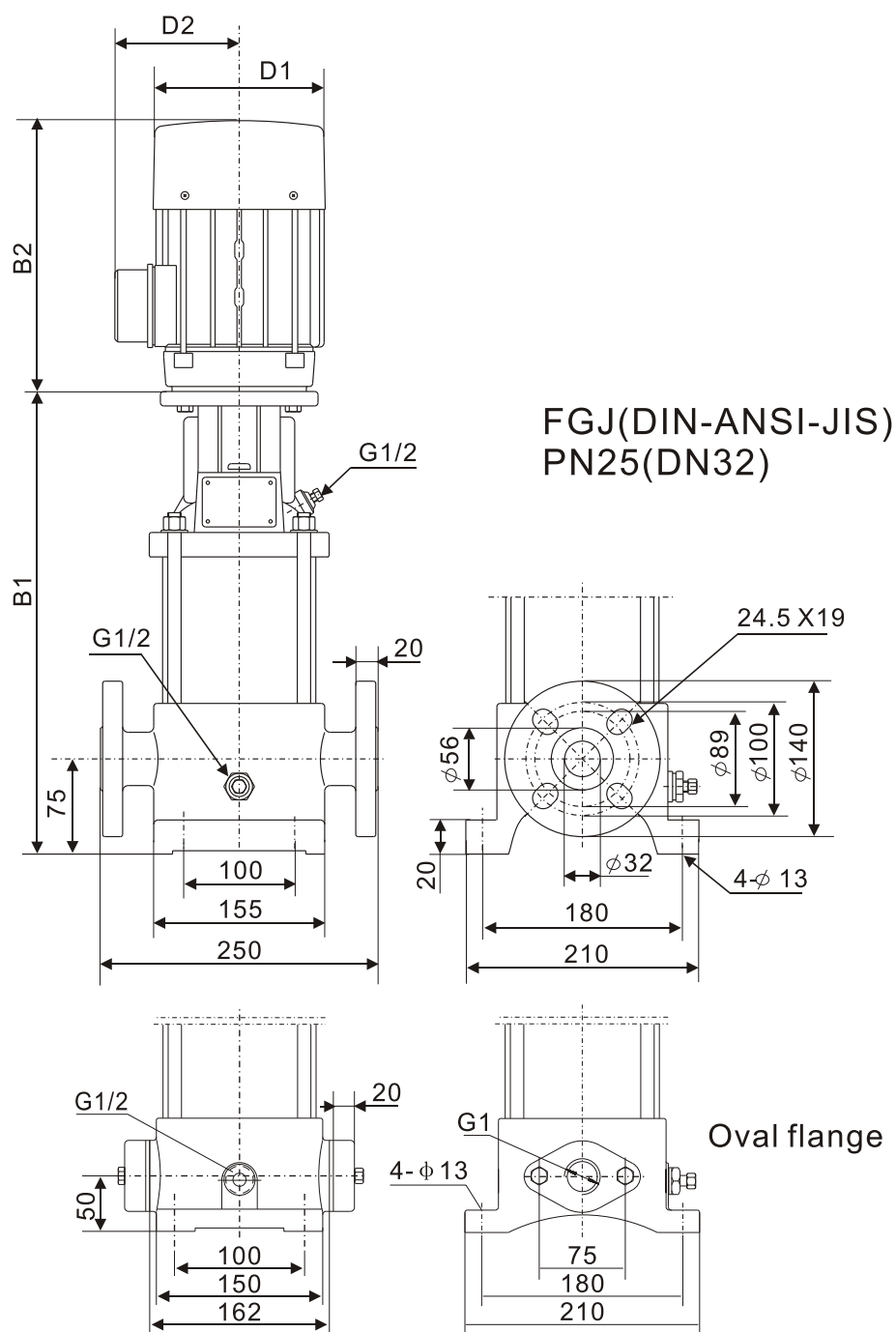
SUS304 is only resistant up to 1% acid while type 316 resistant is up to 10% acid at 50° C temp.

SECTIONAL DRAWING

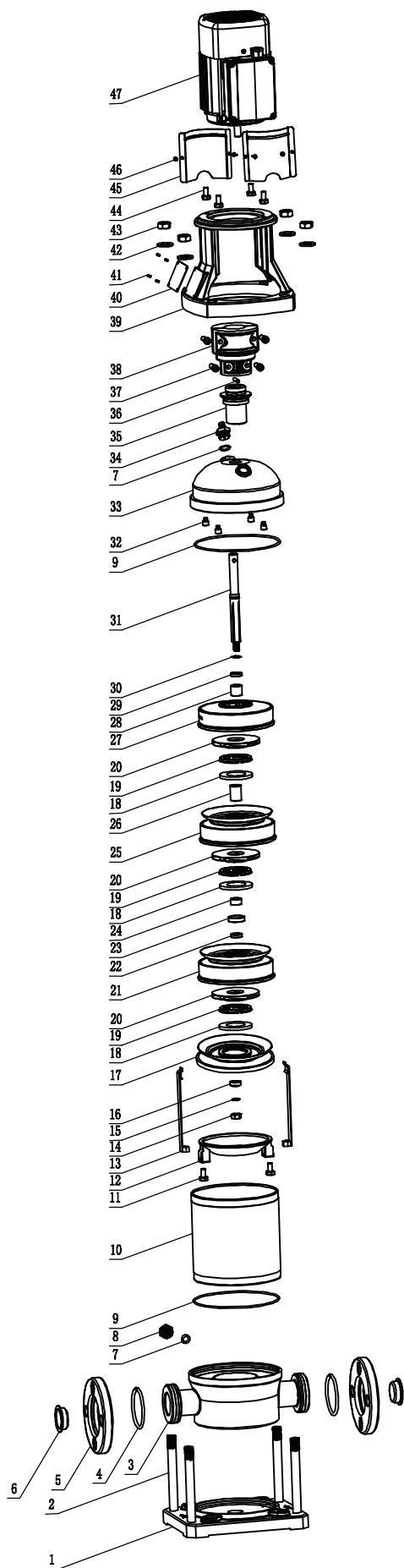


No	Description	Material	EN/DIN	AISI/ASTM
1	Motor	X	X	X
2	Shaft connector	X	X	X
3	Pump head	Cast iron	EN-JL 1030	ASTM25B
4	Mechanical seal	X	X	X
5	Shaft	S.S	X	AISI420
6	Outlet	S.S	1.4301	AISI304
7	Impeller	S.S	1.4301	AISI304
8	Hydraulic stack	S.S	1.4301	AISI304
9	Inlet	S.S	1.4301	AISI304
10	Pump body	Cast iron	EN.JL1030	ASTM25B
11	Neck ring	PTEF	X	X
12	Outer sleeve	S.S	1.4301	AISI316
13	O-ring	EDM/FKM	X	X

DIMENSIONS



Model	Dimensions (mm)					Weight
	B1	B2	B1+B2	D1	D2	KG
FCV-5-9VT	483	275/293	758	177	116	40



NO	Name	Spec.	Material	QTY
1	Base		HT200	1
2	Two-head bolt stud	M16XL	Chrome	4
3	Pump body		SUS304	1
4	Flange clasp		SUS304	2
5	Flange		SUS304	2
6	Dust cap	G1	PP	2
7	Ring	d18X2.4	NBR	2
8	Drain valve assembly		SUS304	1
9	Ring	d170X3.1	NBR	2
10	Housing		SUS316	1
11	Hexagon bolt	M8X20	SUS304	2
12	Fixed board		SUS316	1
13	Tension rod		SUS316	2
14	Type 1 non-metallic insert hexagon lock nut	M10	SUS316	1
15	Spring washer	ø10	SUS316	1
16	Impeller fix ring		SUS316	1
17	Inlet section		SUS316	1
18	Impeller fix ring		PTFE	N
19	Mouthpiece outer ring		SUS316	N
20	impeller		SUS316	N
21	Diffusor with bearing		SUS316	N
22	Short sleeve	L=4.5	SUS316	N
23	Bearing sleeve		Hard alloy	N
24	Short sleeve	L=10.5	SUS316	N
25	Diffusor		SUS316	N
26	Long sleeve	L=29	SUS316	N
27	Outlet section		SUS316	1
28	Short sleeve	L=19	SUS316	1
29	Snap ring seat		SUS316	1
30	Clasp		SUS316	1
31	Pump shaft		SUS316	1
32	Clasp seat shock colloidal		FI	4
33	Pump cover		SUS316	1
34	Discharge valve screw		SUS316	2
35	Mechanical seal	NJK-16-Q18FF	Carbon silicon carbide-FT	1
36	Cylindrical pin	ø6X25	SUS316	1
37	Hexagon socket head cap screw	M6X20	A3	4
38	Coupling			1
39	Motor base		HT200	1
40	Name plate		AL	1
41	Plate rivets	ø2.5	H62	4
42	Washer	ø16		4
43	Hexagonal nut	M16		4
44	Hexagonal bolt	M10X30	A3	4
45	Protective cover		SUS304	2
46	Cross recessed pan head screw and washer assemblies	M4X8	SUS304	4
47	Motor			1

- FCV-5-10VT
- Vertical Multi-Stage Pump

● Features :-

Stainless Steel 304

3 phase

220-380V

60HZ

IE3

2 pole

3500 rpm

Class 155(F)

Protection: IP55

● Product Photo :-



● Proformance Table :-

Model	Power		Max flow	Max head	Voltage	Inlet	Outlet
	KW	HP	m ³ /h	meters		mm	mm
FCV-5-10VT	3.0	4.0	10	93	220-380V	DN32	DN32

● Design

Vertical multistage in Line pump where Discharge and suction ports are located on one line to make it easier for connection access, it's either in Oval Victualic threaded to NPT cast iron discharge casing design or Slip on flanged and drilled to Universal standard which suite both DIN and ANSI standards.

● Applications: -

Water supply

Industry

Liquid transfer

Water treatment

Irrigation

● FCV Coding:

F: FUJIKA

C: Clean water

V: Vertical pump

● Working conditions: -

Temperature range: -20 Up to 104 °C

Ambient temperatue 50 °C

● Classifications:

FCV: Standard SS304

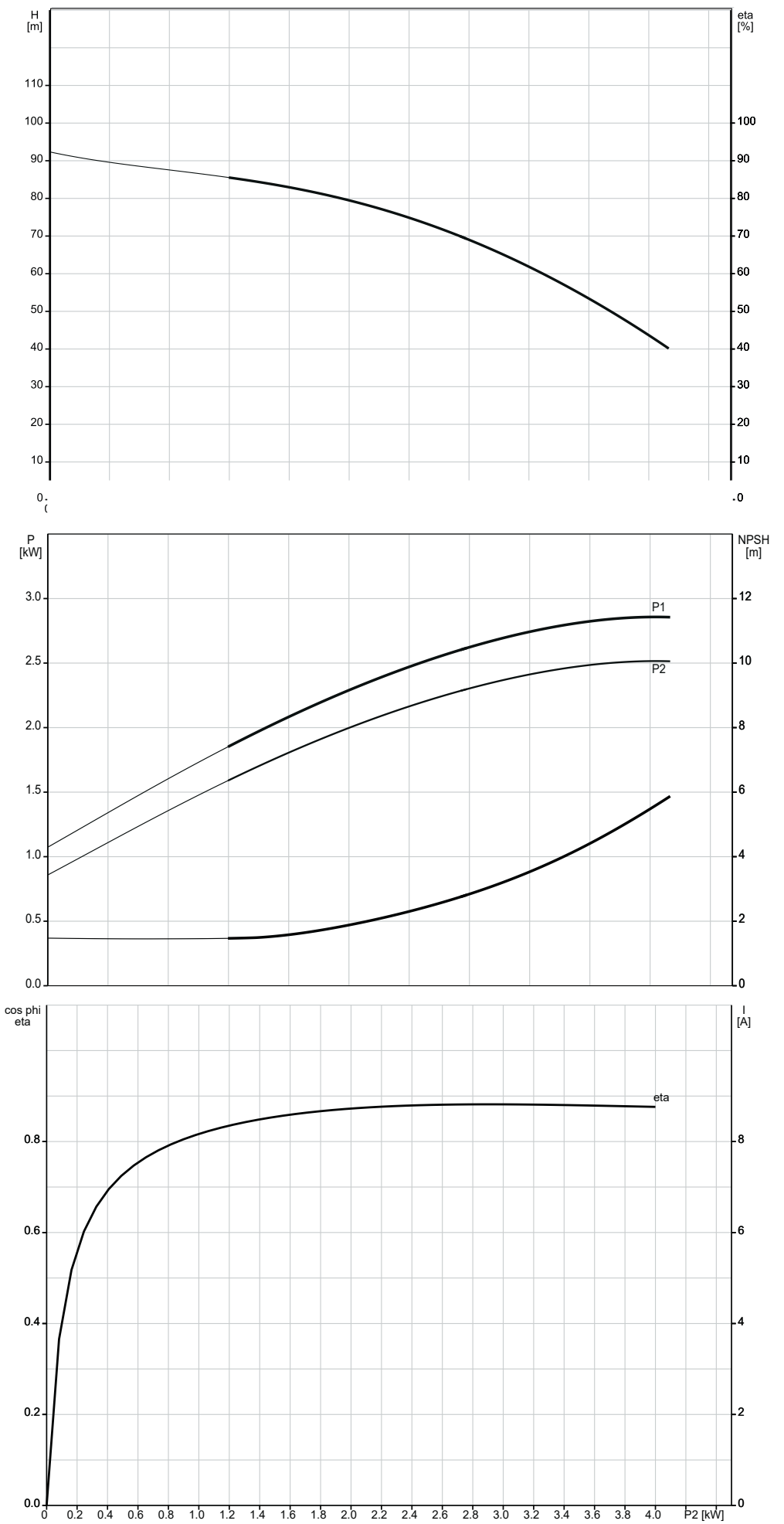
FCV-N: SS316

V: Oval flange

R: Round flange

T: Three phase

PERFORMANCE CURVE



Model: FCV-5-10VT



Design: Vertical multistage in Line pump where Discharge and suction ports are located on one line to make it easier for connection access, it's either in Oval Victualic threaded to NPT cast iron discharge casing design or Slip on flanged and drilled to Universal standard which suite both DIN and ANSI standards.

Construction of Material:

FCV basic design is made of :

Stainless steel imeller, sleeves and shaft , and cast iron discarge casing.

FCV-N

All wetted parts are made from SUS316 design.

The Shaft Mechanical seal is cartridge pull out type for easier replacement.

Mechanical Seal is EPDM /SUS304 for FCV and VITON / SUS316 for FCV-N Ensuring high reliability.

Coupling:

Shaft Coupling are rigid split type PJE Coupling.

Motor Type:

IE3 / IE2, 3-phase AC squirrel cage synchronous, high efficiency, TEFC fan cooled, IP55 Jetting protected.

General description :

The high efficiency ratings are in conformity to grade 3 efficiency value GB18613-2012 and IE of IEC60034-30 standard, but also the efficiency grade and installation dimensions are totally in conformity to IEC standard.

Motor shape is square motor as standard and round motor for larger sizes.

General specifications:

Operation factor: S1 (Continuous duty)

Insulation class: F

Protection class: IP55

Cooling ref.: IC411

Mount type: B35/B6

Ambient temperature: (-15 Up to 60 °C)

Altitude: ≤1000m

Motor paint finish is made with the most high technology with double painting and epoxy coating.

Motor housing construction are alaminum for motor rated up to 15HP and high grade cast iron for motors above rating.

The colour code: RAL 9005 deep jet black.

A standard split coupling connects the pump and motor shaft.

It's enclosed in the pump head/motor stool by means of two coupling guards.

The pump head and flange for motor mounting bracket is made in one piece (cast iron). the pump head cover is a seperate component (stainless steel).

Pump head cover stage has 1/2" priming plug and vent screw.

The pump is fitted with a balanced O-ring seal unit with a rigid torque-transmission system.

This seal type is assembled in a cartridge unit which makes replacement safe and easy.

In compliance with:

ENERGY EFFICIENCY ASIA JAPAN, IE3 IE2 IE1

MEPS (Minimum energy performance standard)

Current Regulation Energy saving act / top runner program standard

JIS C 4034-30

Power supply system

Three phase

Minimum energy performance IE3

Operated with inverter frequency: ok

Output (KW): 0.75 Up to 55KW

Number of poles: 2,4,6

Voltage (V): Up to 1000V

Frequency (HZ): 50HZ,60HZ,50/60HZ

Service duty: S1 ≥ 100%

Cooling method: tefc

Degree of protection: IP55

Area classification: safe area

Altitude: all

Ambient temperature: from (-20 °C and above)

Pump technical data

Rated flow: 5 m³/h
 Rated head: 76 meter
 Pump speed: 3450 rpm
 Nominal pressure PN: 25 bar
 Maximum pressure rating: 125% than shutoff pressure
 Pump mounting: vertical base mounting pump
 Shaft seal arrangement: single face cartridge type mechanical seal
 Seal code: Sic/Sic/Viton/SS316

Materials:

Wetted parts: Stainless steel EN 1.4408 SUS316
 Bearing: CU or SKF life sealed greased bearing
 Insulation: vertical
 Ambient temperature: 60 °C
 Maximum operating pressure: 25 bar
 Maximum pressure at stated temperature: 25 bar @ 0-90 °C

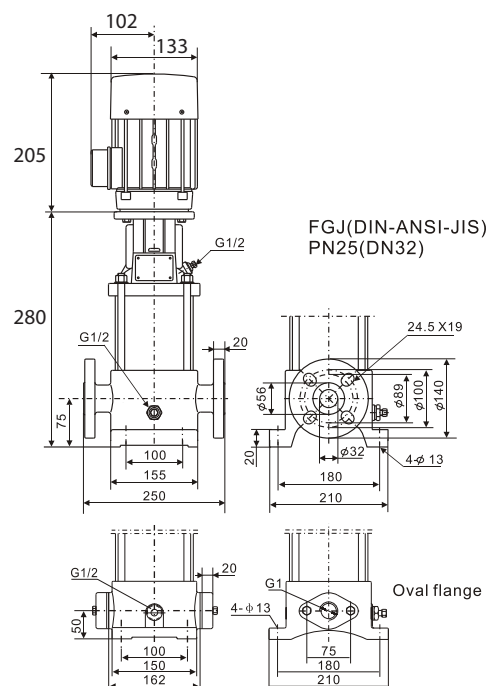
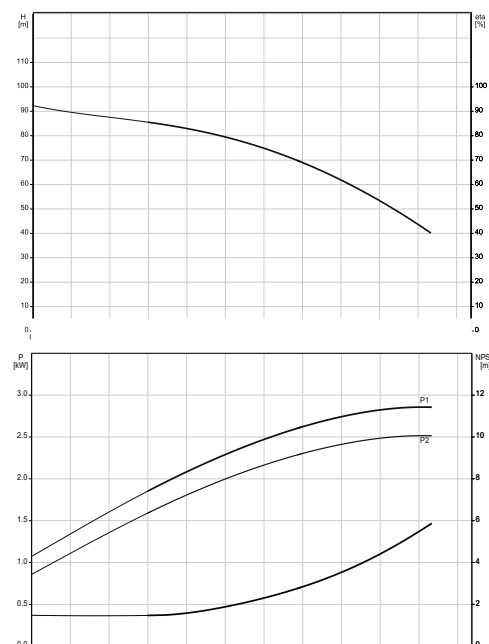
Motor type: by FUJIKA

IE efficiency class: IE3
 Rated power P2: 3 KW
 Power (P2) required by pump: 3 KW
 Frequency: 60HZ
 Rated voltage: 3 X 220-255V Δ / 380-440V Y
 Rated current: 11A / 6.3A
 Starting current: 910-1100%
 Cos phi - power: 0.84
 Rotational speed: 3400-3450 rpm
 IE efficiency: IE3 87.8%
 Minimum efficiency: 81.5%
 Maximum efficiency: 88.5%
 Number of poles: 2
 Enclosure class: IP55
 Insulation class: F (IEC)

Dimensional:

Connection: UNIVERSAL flange drilled to both DIN & ANSI standards.
 Flange size for motor:
 Electrical data: IEC
 Motor standard: IEC DIN
 Approvals: CE
 Certified: IECEE, Coc
 Approvals for drinking water: WRAS
 Performance curve: ISO9906 2012

Gross weight: 41 KG
 Shipping volume: 0.092 m³
 Country of origin: PRC



Material resistivity to chemicals:

Stainless steel 316 material composition Up to :
18% chromium, 10% nickel, and 2% molybdenum.

Allows it to be used in variable applications and
that require superior corrosion resistance and
hygienic.

Applications such as:

Acidic water
Dissolved Chlorine
Sea water
Falls and Fountains
Water treatment
Diesel Fuel
Aggressive oil
Cooking oil
Drinking water
Juices
Water treatment
Food industry
Pharmaceutical Industry
Hygienic process
Fertilization
Fisheries
Irrigation

Characteristics:

The 316 stainless steel is used in salt and other
chlorides applications exposure and lasting
durability
steel over 304.

316 stainless steel's added molybdenum, makes it
much more corrosion resistant than 304 or other
stainless steel alloys. This allows 316 stainless steel
to be used in highly acidic environments that
other steels can't tolerate. 316's corrosion-resist-
ant properties also make it ideal in food process-
ing, pharmaceutical and medical applications due
to its ability to withstand the disinfectants and
high temperatures media.

Exposure to salt and chlorides can cause havoc on
304 stainless steel alloys. Salt can cause 304 stain-
less steel to rust by attacking the protective oxide
layer, and 304 steel is also more easily wear from
the effects of salt and chlorides. However, 316
stainless steel has all of the desirability properties of
304 stainless steel, but with the added benefit of
molybdenum, which allows it to withstand expo-
sure to salt and chlorides for more durability.

Austenitic stainless steels generally show high
corrosion rates when exposed to concentrated,
de-aerated alkaline aqueous environment at tem-
peratures above 60 °C.

Permissible corrosion rates of 0.05–2 mm/year.

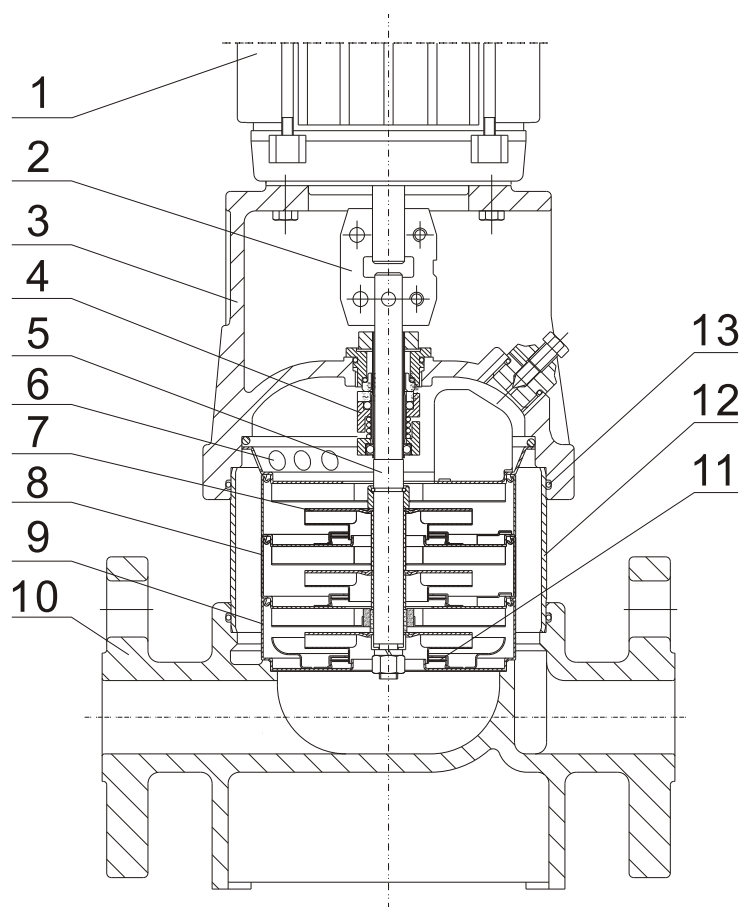
Caution:

The common stainless steel sheets types, 304 and
316 pumps should be considered non-resistant to
hydrochloric acid at any concentration and temper-
ature.

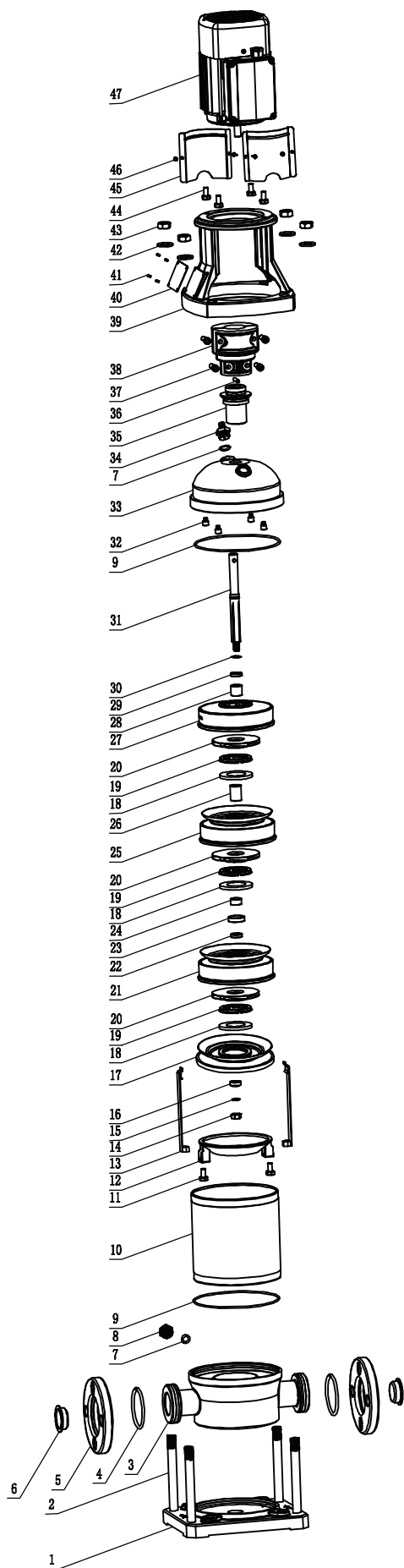
For Sulfuric acid:

SUS304 is only resistant up to 1% acid while type
316 resistant is up to 10% acid at 50° C temp.

SECTIONAL DRAWING



No	Description	Material	EN/DIN	AISI/ASTM
1	Motor	X	X	X
2	Shaft connector	X	X	X
3	Pump head	Cast iron	EN-JL 1030	ASTM25B
4	Mechanical seal	X	X	X
5	Shaft	S.S	X	AISI420
6	Outlet	S.S	1.4301	AISI304
7	Impeller	S.S	1.4301	AISI304
8	Hydraulic stack	S.S	1.4301	AISI304
9	Inlet	S.S	1.4301	AISI304
10	Pump body	Cast iron	EN.JL1030	ASTM25B
11	Neck ring	PTEF	X	X
12	Outer sleeve	S.S	1.4301	AISI316
13	O-ring	EDM/FKM	X	X



NO	Name	Spec.	Material	QTY
1	Base		HT200	1
2	Two-head bolt stud	M16XL	Chrome	4
3	Pump body		SUS304	1
4	Flange clasp		SUS304	2
5	Flange		SUS304	2
6	Dust cap	G1	PP	2
7	Ring	d18X2.4	NBR	2
8	Drain valve assembly		SUS304	1
9	Ring	d170X3.1	NBR	2
10	Housing		SUS316	1
11	Hexagon bolt	M8X20	SUS304	2
12	Fixed board		SUS316	1
13	Tension rod		SUS316	2
14	Type 1 non-metallic insert hexagon lock nut	M10	SUS316	1
15	Spring washer	ø10	SUS316	1
16	Impeller fix ring		SUS316	1
17	Inlet section		SUS316	1
18	Impeller fix ring		PTFE	N
19	Mouthpiece outer ring		SUS316	N
20	impeller		SUS316	N
21	Diffusor with bearing		SUS316	N
22	Short sleeve	L=4.5	SUS316	N
23	Bearing sleeve		Hard alloy	N
24	Short sleeve	L=10.5	SUS316	N
25	Diffusor		SUS316	N
26	Long sleeve	L=29	SUS316	N
27	Outlet section		SUS316	1
28	Short sleeve	L=19	SUS316	1
29	Snap ring seat		SUS316	1
30	Clasp		SUS316	1
31	Pump shaft		SUS316	1
32	Clasp seat shock colloidal		FI	4
33	Pump cover		SUS316	1
34	Discharge valve screw		SUS316	2
35	Mechanical seal	NJK-16-Q18FF	Carbon silicon carbide-FT	1
36	Cylindrical pin	ø6X25	SUS316	1
37	Hexagon socket head cap screw	M6X20	A3	4
38	Coupling			1
39	Motor base		HT200	1
40	Name plate		AL	1
41	Plate rivets	ø2.5	H62	4
42	Washer	ø16		4
43	Hexagonal nut	M16		4
44	Hexagonal bolt	M10X30	A3	4
45	Protective cover		SUS304	2
46	Cross recessed pan head screw and washer assemblies	M4X8	SUS304	4
47	Motor			1

● FCV-5-11VT

● Vertical Multi-Stage Pump

● Features :-

Stainless Steel 304

3 phase

220-380V

60HZ

IE3

2 pole

3500 rpm

Class 155(F)

Protection: IP55

● Product Photo :-



● Proformance Table :-

Model	Power		Max flow	Max head	Voltage	Inlet	Outlet
	KW	HP	m ³ /h	meters		mm	mm
FCV-5-11VT	3.0	4.0	10	103	220-380V	DN32	DN32

● Design

Vertical multistage in Line pump where Discharge and suction ports are located on one line to make it easier for connection access, it's either in Oval Victualic threaded to NPT cast iron discharge casing design or Slip on flanged and drilled to Universal standard which suite both DIN and ANSI standards.

● Applications: -

Water supply

Industry

Liquid transfer

Water treatment

Irrigation

● FCV Coding:

F: FUJIKA

C: Clean water

V: Vertical pump

● Classifications:

FCV: Standard SS304

FCV-N: SS316

V: Oval flange

R: Round flange

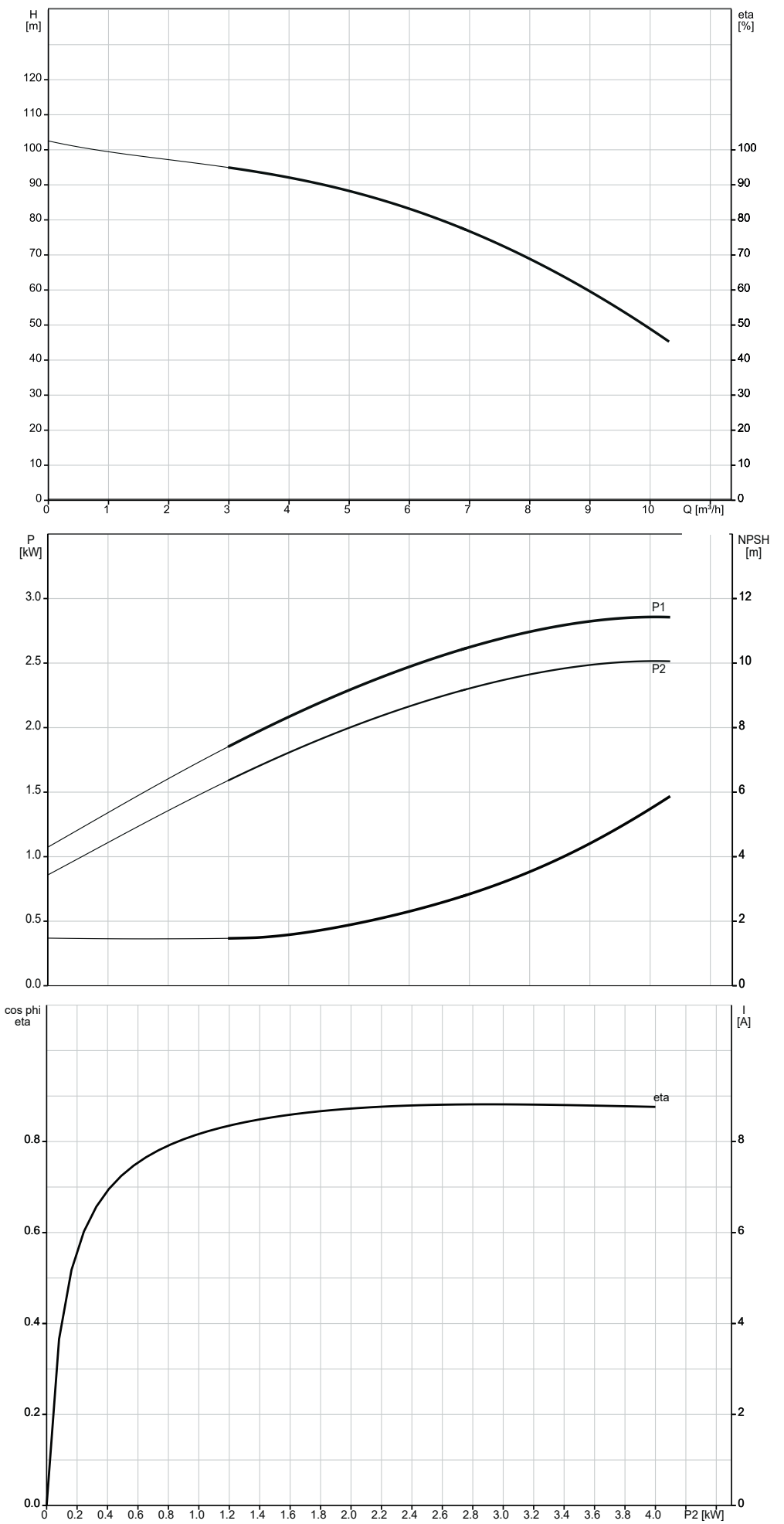
T: Three phase

● Working conditions: -

Temperature range: -20 Up to 104 °C

Ambient temperatue 50 °C

PERFORMANCE CURVE



Model: FCV-5-11VT



Design: Vertical multistage in Line pump where Discharge and suction ports are located on one line to make it easier for connection access, it's either in Oval Victualic threaded to NPT cast iron discharge casing design or Slip on flanged and drilled to Universal standard which suite both DIN and ANSI standards.

Construction of Material:

FCV basic design is made of :

Stainless steel imeller, sleeves and shaft , and cast iron discarge casing.

FCV-N

All wetted parts are made from SUS316 design.

The Shaft Mechanical seal is cartridge pull out type for easier replacement.

Mechanical Seal is EPDM /SUS304 for FCV and VITON / SUS316 for FCV-N Ensuring high reliability.

Coupling:

Shaft Coupling are rigid split type PJE Coupling.

Motor Type:

IE3 / IE2, 3-phase AC squirrel cage synchronous, high efficiency, TEFC fan cooled, IP55 Jetting protected.

General description :

The high efficiency ratings are in conformity to grade 3 efficiency value GB18613-2012 and IE of IEC60034-30 standard, but also the efficiency grade and installation dimensions are totally in conformity to IEC standard.

Motor shape is square motor as standard and round motor for larger sizes.

General specifications:

Operation factor: S1 (Continuous duty)

Insulation class: F

Protection class: IP55

Cooling ref.: IC411

Mount type: B35/B6

Ambient temperature: (-15 Up to 60 °C)

Altitude: ≤1000m

Motor paint finish is made with the most high technology with double painting and epoxy coating.

Motor housing construction are alaminum for motor rated up to 15HP and high grade cast iron for motors above rating.

The colour code: RAL 9005 deep jet black.

A standard split coupling connects the pump and motor shaft.

It's enclosed in the pump head/motor stool by means of two coupling guards.

The pump head and flange for motor mounting bracket is made in one piece (cast iron). the pump head cover is a seperate component (stainless steel).

Pump head cover stage has 1/2" priming plug and vent screw.

The pump is fitted with a balanced O-ring seal unit with a rigid torque-transmission system.

This seal type is assembled in a cartridge unit which makes replacement safe and easy.

In compliance with:

ENERGY EFFICIENCY ASIA JAPAN, IE3 IE2 IE1

MEPS (Minimum energy performance standard)

Current Regulation Energy saving act / top runner program standard

JIS C 4034-30

Power supply system

Three phase

Minimum energy performance IE3

Operated with inverter frequency: ok

Output (KW): 0.75 Up to 55KW

Number of poles: 2,4,6

Voltage (V): Up to 1000V

Frequency (HZ): 50HZ,60HZ,50/60HZ

Service duty: S1 ≥ 100%

Cooling method: tefc

Degree of protection: IP55

Area classification: safe area

Altitude: all

Ambient temperature: from (-20 °C and above)

Pump technical data

Rated flow: 5 m³/h
 Rated head: 84 meter
 Pump speed: 3450 rpm
 Nominal pressure PN: 25 bar
 Maximum pressure rating: 125% than shutoff pressure
 Pump mounting: vertical base mounting pump
 Shaft seal arrangement: single face cartridge type mechanical seal
 Seal code: Sic/Sic/Viton/SS316

Materials:

Wetted parts: Stainless steel EN 1.4408 SUS316
 Bearing: CU or SKF life sealed greased bearing
 Insulation: vertical
 Ambient temperature: 60 °C
 Maximum operating pressure: 25 bar
 Maximum pressure at stated temperature: 25 bar @ 0-90 °C

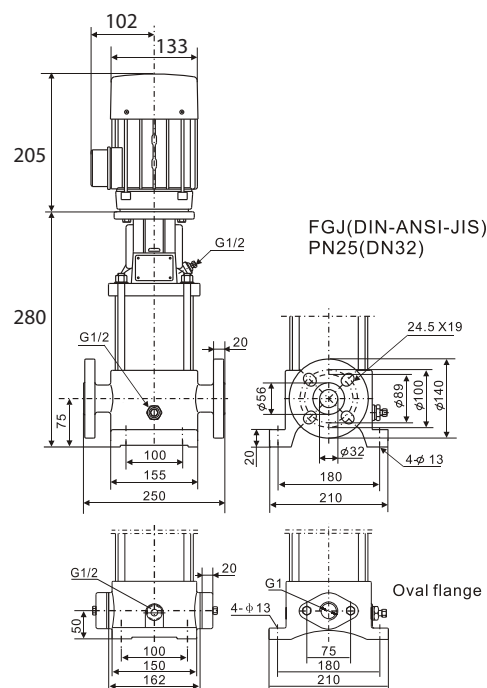
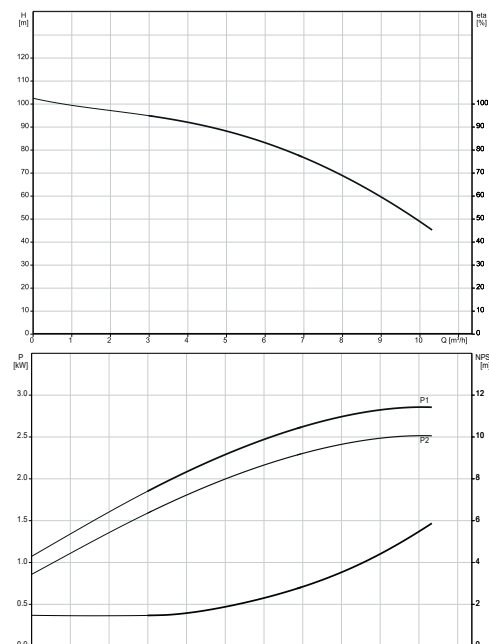
Motor type: by FUJIKA

IE efficiency class: IE3
 Rated power P2: 3 KW
 Power (P2) required by pump: 3 KW
 Frequency: 60HZ
 Rated voltage: 3 X 220-255V Δ / 380-440V Y
 Rated current: 115A / 6.3A
 Starting current: 910-1100%
 Cos phi - power: 0.84
 Rotational speed: 3400-3450 rpm
 IE efficiency: IE3 87.8%
 Minimum efficiency: 81.5%
 Maximum efficiency: 88.5%
 Number of poles: 2
 Enclosure class: IP55
 Insulation class: F (IEC)

Dimensional:

Connection: UNIVERSAL flange drilled to both DIN & ANSI standards.
 Flange size for motor:
 Electrical data: IEC
 Motor standard: IEC DIN
 Approvals: CE
 Certified: IECEE, Coc
 Approvals for drinking water: WRAS
 Performance curve: ISO9906 2012

Gross weight: 42 KG
 Shipping volume: 0.092 m³
 Country of origin: PRC



Material resistivity to chemicals:

Stainless steel 316 material composition Up to : 18% chromium, 10% nickel, and 2% molybdenum. Allows it to be used in variable applications and that require superior corrosion resistance and hygienic.

Applications such as:

- Acidic water
- Dissolved Chlorine
- Sea water
- Falls and Fountains
- Water treatment
- Diesel Fuel
- Aggressive oil
- Cooking oil
- Drinking water
- Juices
- Water treatment
- Food industry
- Pharmaceutical Industry
- Hygienic process
- Fertilization
- Fisheries
- Irrigation

Characteristics:

The 316 stainless steel is used in salt and other chlorides applications exposure and lasting durability steel over 304.

316 stainless steel's added molybdenum, makes it much more corrosion resistant than 304 or other stainless steel alloys. This allows 316 stainless steel to be used in highly acidic environments that other steels can't tolerate. 316's corrosion-resistant properties also make it ideal in food processing, pharmaceutical and medical applications due to its ability to withstand the disinfectants and high temperatures media.

Exposure to salt and chlorides can cause havoc on 304 stainless steel alloys. Salt can cause 304 stainless steel to rust by attacking the protective oxide layer, and 304 steel is also more easily wear from the effects of salt and chlorides. However, 316 stainless steel has all of the desirability properties of 304 stainless steel, but with the added benefit of molybdenum, which allows it to withstand exposure to salt and chlorides for more durability.

Austenitic stainless steels generally show high corrosion rates when exposed to concentrated, de-aerated alkaline aqueous environment at temperatures above 60 °C.

Permissible corrosion rates of 0.05–2 mm/year.

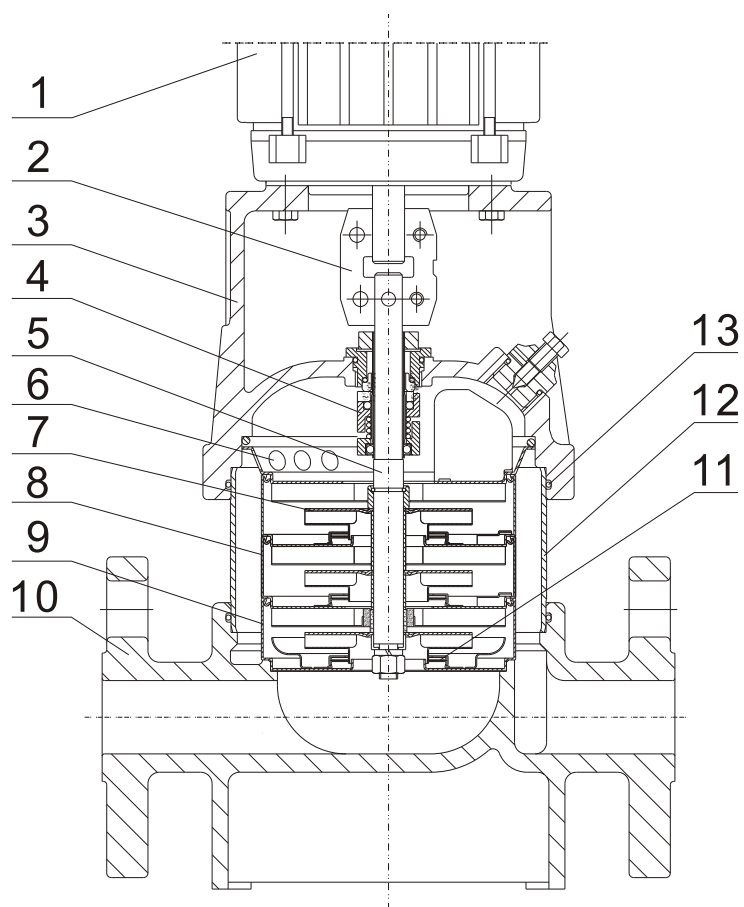
Caution:

The common stainless steel sheets types, 304 and 316 pumps should be considered non-resistant to hydrochloric acid at any concentration and temperature.

For Sulfuric acid:

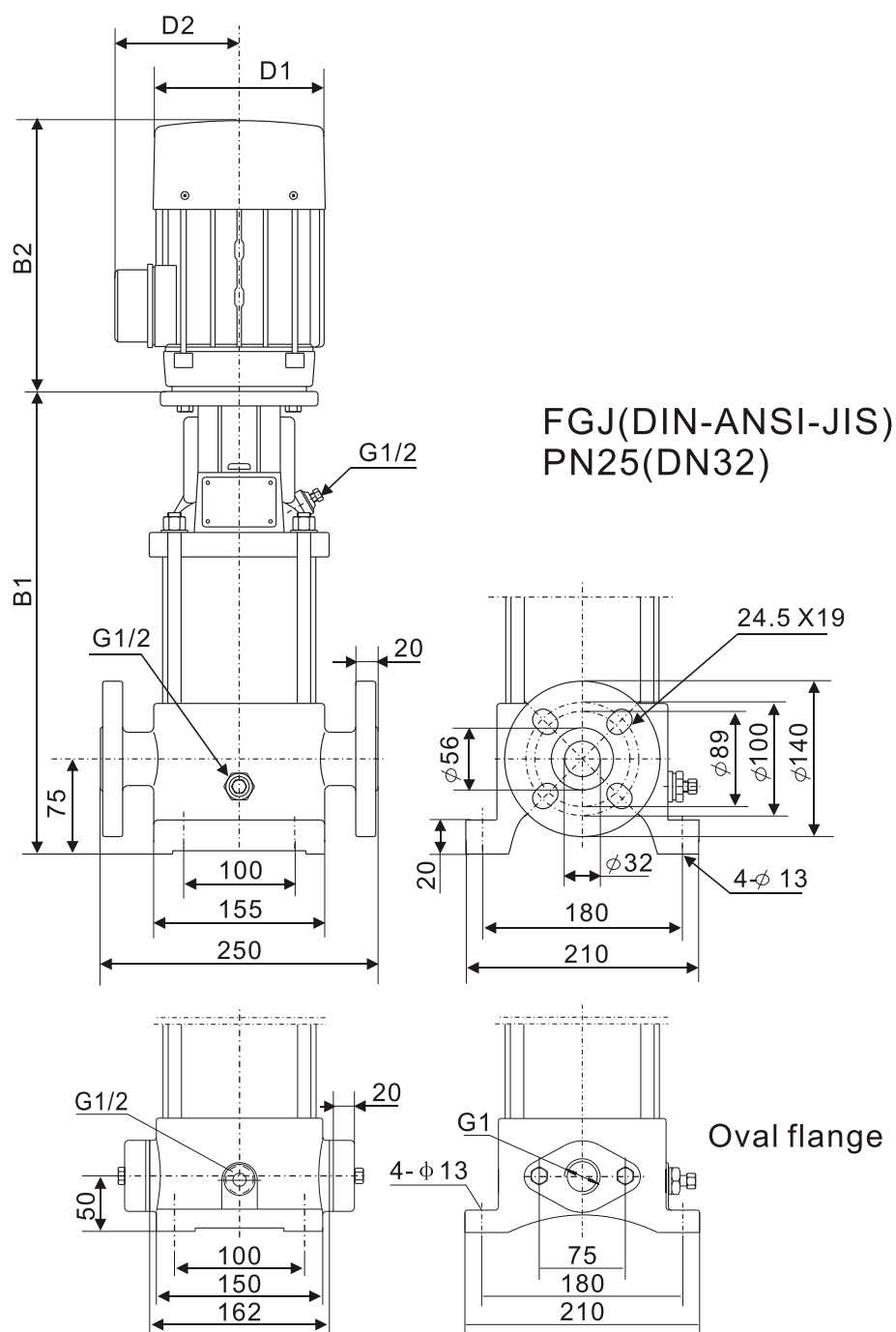
SUS304 is only resistant up to 1% acid while type 316 resistant is up to 10% acid at 50° C temp.

SECTIONAL DRAWING

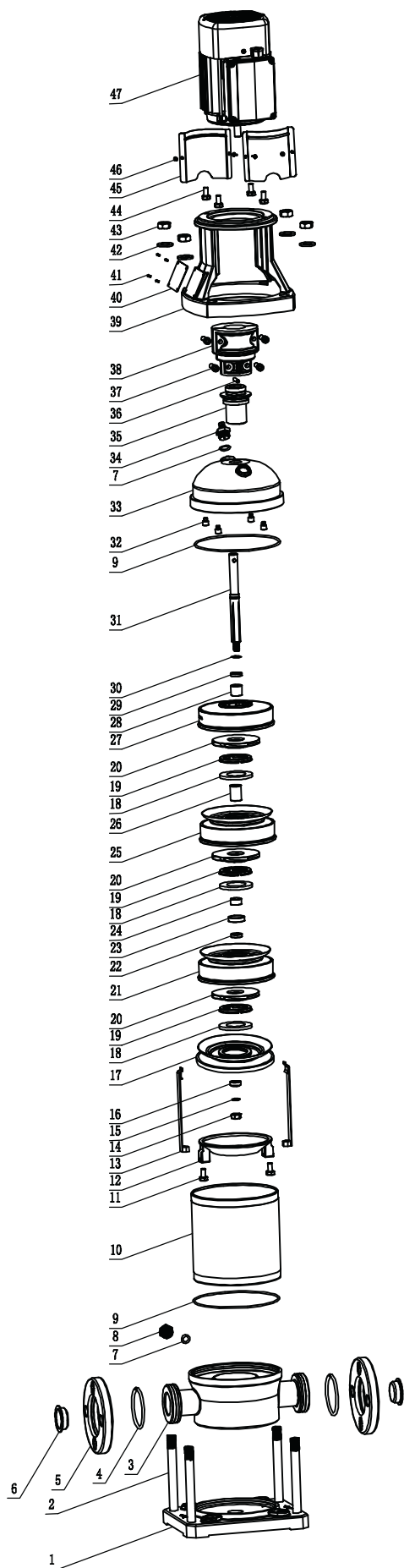


No	Description	Material	EN/DIN	AISI/ASTM
1	Motor	X	X	X
2	Shaft connector	X	X	X
3	Pump head	Cast iron	EN-JL 1030	ASTM25B
4	Mechanical seal	X	X	X
5	Shaft	S.S	X	AISI420
6	Outlet	S.S	1.4301	AISI304
7	Impeller	S.S	1.4301	AISI304
8	Hydraulic stack	S.S	1.4301	AISI304
9	Inlet	S.S	1.4301	AISI304
10	Pump body	Cast iron	EN.JL1030	ASTM25B
11	Neck ring	PTEF	X	X
12	Outer sleeve	S.S	1.4301	AISI316
13	O-ring	EDM/FKM	X	X

DIMENSIONS



Model	Dimensions (mm)					Weight
	B1	B2	B1+B2	D1	D2	KG
FCV-5-11VT	537	293	812	177	116	42



NO	Name	Spec.	Material	QTY
1	Base		HT200	1
2	Two-head bolt stud	M16XL	Chrome	4
3	Pump body		SUS304	1
4	Flange clasp		SUS304	2
5	Flange		SUS304	2
6	Dust cap	G1	PP	2
7	Ring	d18X2.4	NBR	2
8	Drain valve assembly		SUS304	1
9	Ring	d170X3.1	NBR	2
10	Housing		SUS316	1
11	Hexagon bolt	M8X20	SUS304	2
12	Fixed board		SUS316	1
13	Tension rod		SUS316	2
14	Type 1 non-metallic insert hexagon lock nut	M10	SUS316	1
15	Spring washer	ø10	SUS316	1
16	Impeller fix ring		SUS316	1
17	Inlet section		SUS316	1
18	Impeller fix ring		PTFE	N
19	Mouthpiece outer ring		SUS316	N
20	impeller		SUS316	N
21	Diffusor with bearing		SUS316	N
22	Short sleeve	L=4.5	SUS316	N
23	Bearing sleeve		Hard alloy	N
24	Short sleeve	L=10.5	SUS316	N
25	Diffusor		SUS316	N
26	Long sleeve	L=29	SUS316	N
27	Outlet section		SUS316	1
28	Short sleeve	L=19	SUS316	1
29	Snap ring seat		SUS316	1
30	Clasp		SUS316	1
31	Pump shaft		SUS316	1
32	Clasp seat shock colloidal		FI	4
33	Pump cover		SUS316	1
34	Discharge valve screw		SUS316	2
35	Mechanical seal	NJK-16-Q18FF	Carbon silicon carbide-FT	1
36	Cylindrical pin	ø6X25	SUS316	1
37	Hexagon socket head cap screw	M6X20	A3	4
38	Coupling			1
39	Motor base		HT200	1
40	Name plate		AL	1
41	Plate rivets	ø2.5	H62	4
42	Washer	ø16		4
43	Hexagonal nut	M16		4
44	Hexagonal bolt	M10X30	A3	4
45	Protective cover		SUS304	2
46	Cross recessed pan head screw and washer assemblies	M4X8	SUS304	4
47	Motor			1

● FCV-5-12VT

● Vertical Multi-Stage Pump

● Features :-

Stainless Steel 304

3 phase

220-380V

60HZ

IE3

2 pole

3500 rpm

Class 155(F)

Protection: IP55

● Product Photo :-



● Proformance Table :-

Model	Power		Max flow	Max head	Voltage	Inlet	Outlet
	KW	HP	m ³ /h	meters		mm	mm
FCV-5-12VT	3.0	4.0	10	112	220-380V	DN32	DN32

● Design

Vertical multistage in Line pump where Discharge and suction ports are located on one line to make it easier for connection access, it's either in Oval Victualic threaded to NPT cast iron discharge casing design or Slip on flanged and drilled to Universal standard which suite both DIN and ANSI standards.

● Applications: -

Water supply

Industry

Liquid transfer

Water treatment

Irrigation

● FCV Coding:

F: FUJIKA

C: Clean water

V: Vertical pump

● Classifications:

FCV: Standard SS304

FCV-N: SS316

V: Oval flange

R: Round flange

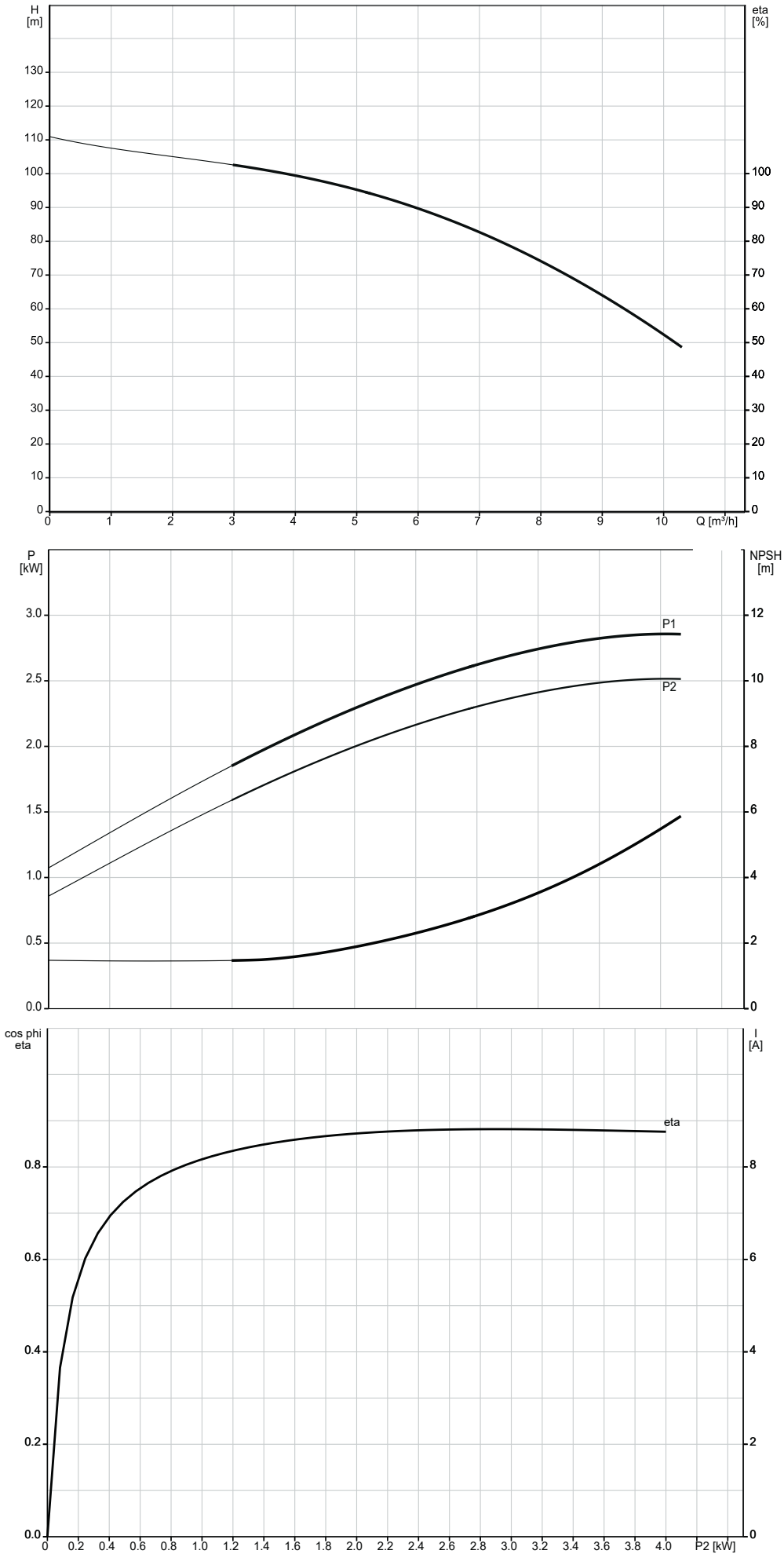
T: Three phase

● Working conditions: -

Temperature range: -20 Up to 104 °C

Ambient temperatue 50 °C

PERFORMANCE CURVE



Model: FCV-5-12VT



Design: Vertical multistage in Line pump where Discharge and suction ports are located on one line to make it easier for connection access, it's either in Oval Victualic threaded to NPT cast iron discharge casing design or Slip on flanged and drilled to Universal standard which suite both DIN and ANSI standards.

Construction of Material:

FCV basic design is made of :

Stainless steel imeller, sleeves and shaft , and cast iron discarge casing.

FCV-N

All wetted parts are made from SUS316 design.

The Shaft Mechanical seal is cartridge pull out type for easier replacement.

Mechanical Seal is EPDM /SUS304 for FCV and VITON / SUS316 for FCV-N Ensuring high reliability.

Coupling:

Shaft Coupling are rigid split type PJE Coupling.

Motor Type:

IE3 / IE2, 3-phase AC squirrel cage synchronous, high efficiency, TEFC fan cooled, IP55 Jetting protected.

General description :

The high efficiency ratings are in conformity to grade 3 efficiency value GB18613-2012 and IE of IEC60034-30 standard, but also the efficiency grade and installation dimensions are totally in conformity to IEC standard.

Motor shape is square motor as standard and round motor for larger sizes.

General specifications:

Operation factor: S1 (Continuous duty)

Insulation class: F

Protection class: IP55

Cooling ref.: IC411

Mount type: B35/B6

Ambient temperature: (-15 Up to 60 °C)

Altitude: ≤1000m

Motor paint finish is made with the most high technology with double painting and epoxy coating.

Motor housing construction are alaminum for motor rated up to 15HP and high grade cast iron for motors above rating.

The colour code: RAL 9005 deep jet black.

A standard split coupling connects the pump and motor shaft.

It's enclosed in the pump head/motor stool by means of two coupling guards.

The pump head and flange for motor mounting bracket is made in one piece (cast iron). the pump head cover is a seperate component (stainless steel).

Pump head cover stage has 1/2" priming plug and vent screw.

The pump is fitted with a balanced O-ring seal unit with a rigid torque-transmission system.

This seal type is assembled in a cartridge unit which makes replacement safe and easy.

In compliance with:

ENERGY EFFICIENCY ASIA JAPAN, IE3 IE2 IE1

MEPS (Minimum energy performance standard)

Current Regulation Energy saving act / top runner program standard

JIS C 4034-30

Power supply system

Three phase

Minimum energy performance IE3

Operated with inverter frequency: ok

Output (KW): 0.75 Up to 55KW

Number of poles: 2,4,6

Voltage (V): Up to 1000V

Frequency (HZ): 50HZ,60HZ,50/60HZ

Service duty: S1 ≥ 100%

Cooling method: tefc

Degree of protection: IP55

Area classification: safe area

Altitude: all

Ambient temperature: from (-20 °C and above)

Material resistivity to chemicals:

Stainless steel 316 material composition Up to :
18% chromium, 10% nickel, and 2% molybdenum.

Allows it to be used in variable applications and
that require superior corrosion resistance and
hygienic.

Applications such as:

Acidic water
Dissolved Chlorine
Sea water
Falls and Fountains
Water treatment
Diesel Fuel
Aggressive oil
Cooking oil
Drinking water
Juices
Water treatment
Food industry
Pharmaceutical Industry
Hygienic process
Fertilization
Fisheries
Irrigation

Characteristics:

The 316 stainless steel is used in salt and other
chlorides applications exposure and lasting
durability
steel over 304.

316 stainless steel's added molybdenum, makes it
much more corrosion resistant than 304 or other
stainless steel alloys. This allows 316 stainless steel
to be used in highly acidic environments that
other steels can't tolerate. 316's corrosion-resist-
ant properties also make it ideal in food process-
ing, pharmaceutical and medical applications due
to its ability to withstand the disinfectants and
high temperatures media.

Exposure to salt and chlorides can cause havoc on
304 stainless steel alloys. Salt can cause 304 stain-
less steel to rust by attacking the protective oxide
layer, and 304 steel is also more easily wear from
the effects of salt and chlorides. However, 316
stainless steel has all of the desirability properties of
304 stainless steel, but with the added benefit of
molybdenum, which allows it to withstand expo-
sure to salt and chlorides for more durability.

Austenitic stainless steels generally show high
corrosion rates when exposed to concentrated,
de-aerated alkaline aqueous environment at tem-
peratures above 60 °C.

Permissible corrosion rates of 0.05–2 mm/year.

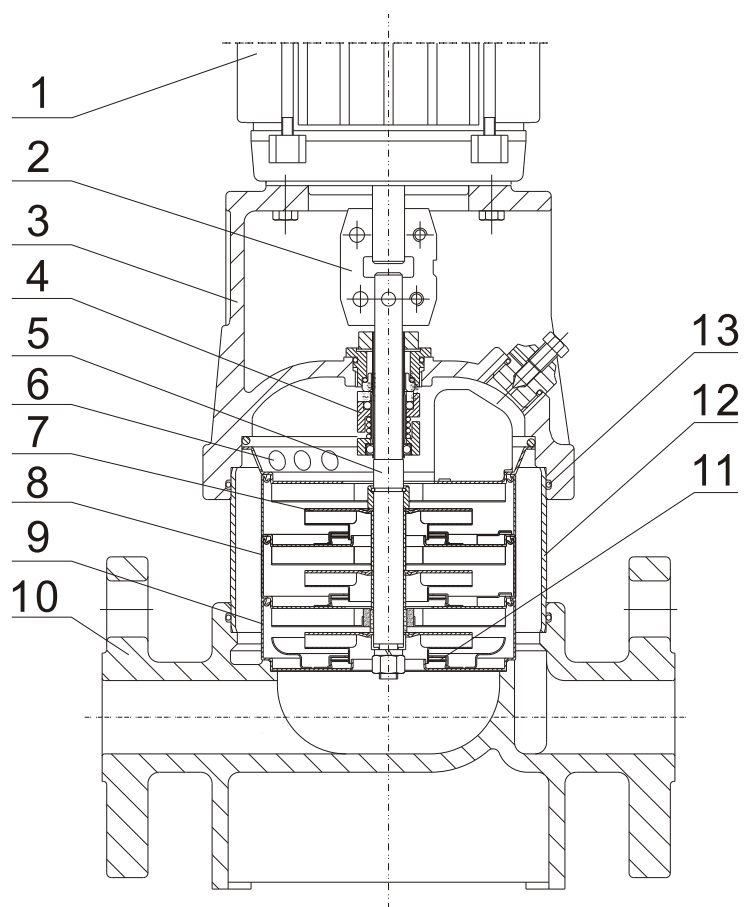
Caution:

The common stainless steel sheets types, 304 and
316 pumps should be considered non-resistant to
hydrochloric acid at any concentration and temper-
ature.

For Sulfuric acid:

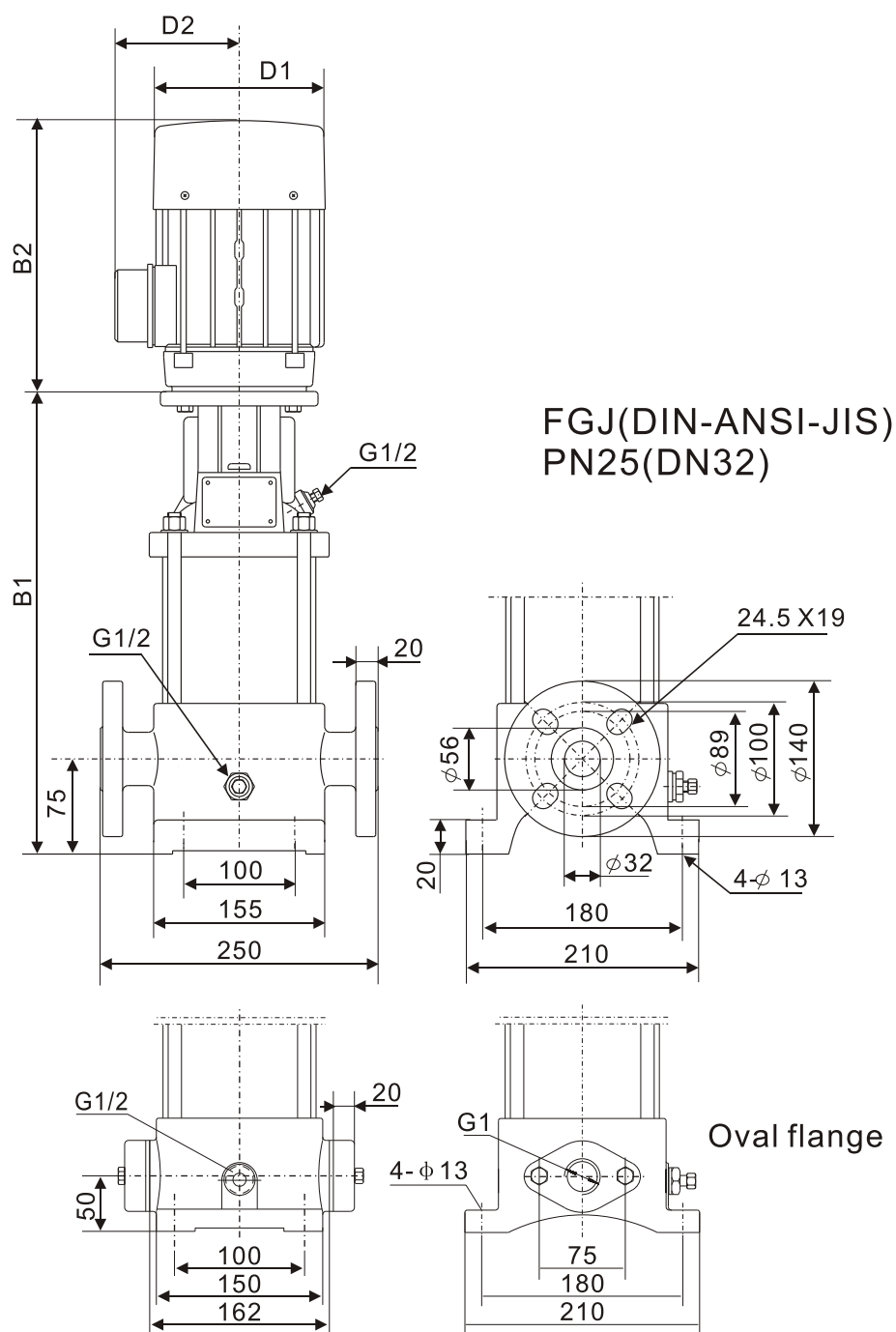
SUS304 is only resistant up to 1% acid while type
316 resistant is up to 10% acid at 50° C temp.

SECTIONAL DRAWING

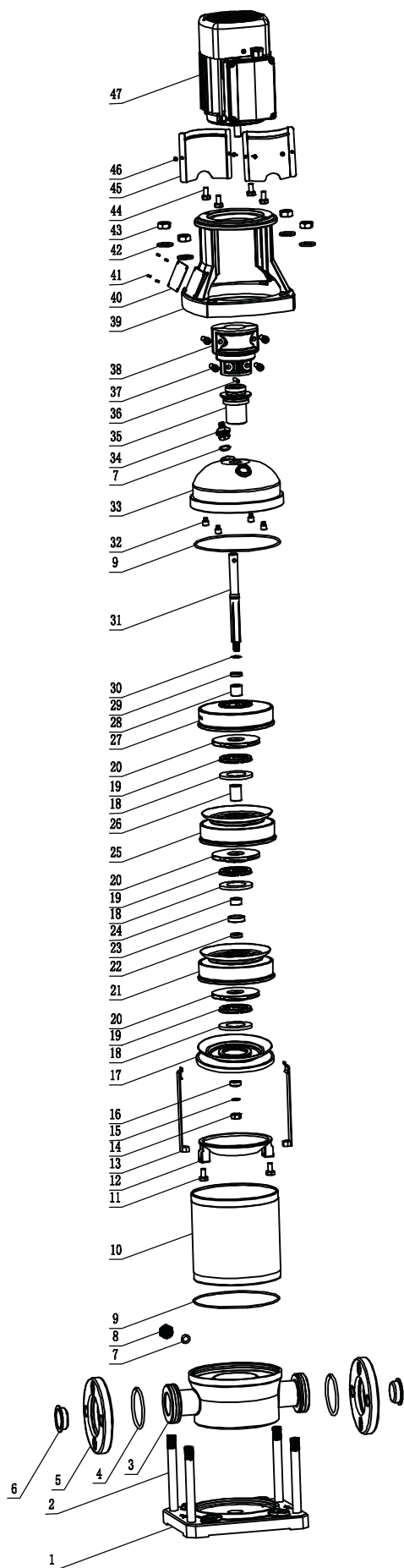


No	Description	Material	EN/DIN	AISI/ASTM
1	Motor	X	X	X
2	Shaft connector	X	X	X
3	Pump head	Cast iron	EN-JL 1030	ASTM25B
4	Mechanical seal	X	X	X
5	Shaft	S.S	X	AISI420
6	Outlet	S.S	1.4301	AISI304
7	Impeller	S.S	1.4301	AISI304
8	Hydraulic stack	S.S	1.4301	AISI304
9	Inlet	S.S	1.4301	AISI304
10	Pump body	Cast iron	EN.JL1030	ASTM25B
11	Neck ring	PTEF	X	X
12	Outer sleeve	S.S	1.4301	AISI316
13	O-ring	EDM/FKM	X	X

DIMENSIONS



Model	Dimensions (mm)					Weight
	B1	B2	B1+B2	D1	D2	KG
FCV-5-12VT	564	293	839	177	116	43



NO	Name	Spec.	Material	QTY
1	Base		HT200	1
2	Two-head bolt stud	M16XL	Chrome	4
3	Pump body		SUS304	1
4	Flange clasp		SUS304	2
5	Flange		SUS304	2
6	Dust cap	G1	PP	2
7	Ring	d18X2.4	NBR	2
8	Drain valve assembly		SUS304	1
9	Ring	d170X3.1	NBR	2
10	Housing		SUS316	1
11	Hexagon bolt	M8X20	SUS304	2
12	Fixed board		SUS316	1
13	Tension rod		SUS316	2
14	Type 1 non-metallic insert hexagon lock nut	M10	SUS316	1
15	Spring washer	ø10	SUS316	1
16	Impeller fix ring		SUS316	1
17	Inlet section		SUS316	1
18	Impeller fix ring		PTFE	N
19	Mouthpiece outer ring		SUS316	N
20	impeller		SUS316	N
21	Diffusor with bearing		SUS316	N
22	Short sleeve	L=4.5	SUS316	N
23	Bearing sleeve		Hard alloy	N
24	Short sleeve	L=10.5	SUS316	N
25	Diffusor		SUS316	N
26	Long sleeve	L=29	SUS316	N
27	Outlet section		SUS316	1
28	Short sleeve	L=19	SUS316	1
29	Snap ring seat		SUS316	1
30	Clasp		SUS316	1
31	Pump shaft		SUS316	1
32	Clasp seat shock colloidal		FI	4
33	Pump cover		SUS316	1
34	Discharge valve screw		SUS316	2
35	Mechanical seal	NJK-16-Q18FF	Carbon silicon carbide-FT	1
36	Cylindrical pin	ø6X25	SUS316	1
37	Hexagon socket head cap screw	M6X20	A3	4
38	Coupling			1
39	Motor base		HT200	1
40	Name plate		AL	1
41	Plate rivets	ø2.5	H62	4
42	Washer	ø16		4
43	Hexagonal nut	M16		4
44	Hexagonal bolt	M10X30	A3	4
45	Protective cover		SUS304	2
46	Cross recessed pan head screw and washer assemblies	M4X8	SUS304	4
47	Motor			1

● FCV-5-13VT

● Vertical Multi-Stage Pump

● Features :-

Stainless Steel 304

3 phase

220-380V

60HZ

IE3

2 pole

3500 rpm

Class 155(F)

Protection: IP55

● Product Photo :-



● Proformance Table :-

Model	Power		Max flow	Max head	Voltage	Inlet	Outlet
	KW	HP	m ³ /h	meters		mm	mm
FCV-5-13VT	4.0	5.5	10	122	220-380V	DN32	DN32

● Design

Vertical multistage in Line pump where Discharge and suction ports are located on one line to make it easier for connection access, it's either in Oval Victualic threaded to NPT cast iron discharge casing design or Slip on flanged and drilled to Universal standard which suite both DIN and ANSI standards.

● Applications: -

Water supply

Industry

Liquid transfer

Water treatment

Irrigation

● FCV Coding:

F: FUJIKA

C: Clean water

V: Vertical pump

● Classifications:

FCV: Standard SS304

FCV-N: SS316

V: Oval flange

R: Round flange

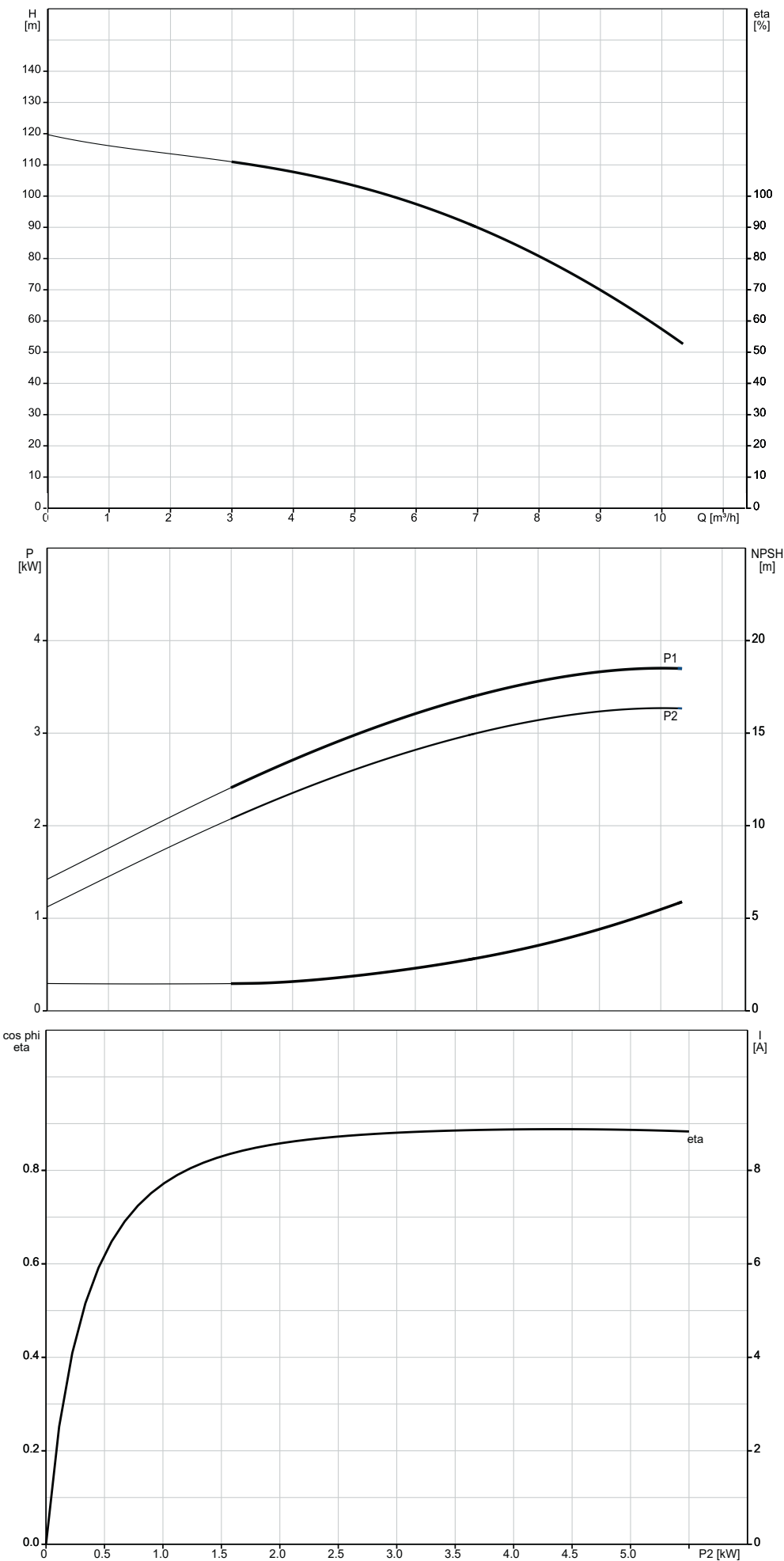
T: Three phase

● Working conditions: -

Temperature range: -20 Up to 104 °C

Ambient temperatue 50 °C

PERFORMANCE CURVE



Model: FCV-5-13VT



Design: Vertical multistage in Line pump where Discharge and suction ports are located on one line to make it easier for connection access, it's either in Oval Victualic threaded to NPT cast iron discharge casing design or Slip on flanged and drilled to Universal standard which suite both DIN and ANSI standards.

Construction of Material:

FCV basic design is made of :

Stainless steel imeller, sleeves and shaft , and cast iron discarge casing.

FCV-N

All wetted parts are made from SUS316 design.

The Shaft Mechanical seal is cartridge pull out type for easier replacement.

Mechanical Seal is EPDM /SUS304 for FCV and VITON / SUS316 for FCV-N Ensuring high reliability.

Coupling:

Shaft Coupling are rigid split type PJE Coupling.

Motor Type:

IE3 / IE2, 3-phase AC squirrel cage synchronous, high efficiency, TEFC fan cooled, IP55 Jetting protected.

General description :

The high efficiency ratings are in conformity to grade 3 efficiency value GB18613-2012 and IE of IEC60034-30 standard, but also the efficiency grade and installation dimensions are totally in conformity to IEC standard.

Motor shape is square motor as standard and round motor for larger sizes.

General specifications:

Operation factor: S1 (Continuous duty)

Insulation class: F

Protection class: IP55

Cooling ref.: IC411

Mount type: B35/B6

Ambient temperature: (-15 Up to 60 °C)

Altitude: ≤1000m

Motor paint finish is made with the most high technology with double painting and epoxy coating.

Motor housing construction are alaminum for motor rated up to 15HP and high grade cast iron for motors above rating.

The colour code: RAL 9005 deep jet black.

A standard split coupling connects the pump and motor shaft.

It's enclosed in the pump head/motor stool by means of two coupling guards.

The pump head and flange for motor mounting bracket is made in one piece (cast iron). the pump head cover is a seperate component (stainless steel).

Pump head cover stage has 1/2" priming plug and vent screw.

The pump is fitted with a balanced O-ring seal unit with a rigid torque-transmission system.

This seal type is assembled in a cartridge unit which makes replacement safe and easy.

In compliance with:

ENERGY EFFICIENCY ASIA JAPAN, IE3 IE2 IE1

MEPS (Minimum energy performance standard)

Current Regulation Energy saving act / top runner program standard

JIS C 4034-30

Power supply system

Three phase

Minimum energy performance IE3

Operated with inverter frequency: ok

Output (KW): 0.75 Up to 55KW

Number of poles: 2,4,6

Voltage (V): Up to 1000V

Frequency (HZ): 50HZ,60HZ,50/60HZ

Service duty: S1 ≥ 100%

Cooling method: tefc

Degree of protection: IP55

Area classification: safe area

Altitude: all

Ambient temperature: from (-20 °C and above)

Pump technical data

Rated flow: 5 m³/h
 Rated head: 99 meter
 Pump speed: 3450 rpm
 Nominal pressure PN: 25 bar
 Maximum pressure rating: 125% than shutoff pressure
 Pump mounting: vertical base mounting pump
 Shaft seal arrangement: single face cartridge type mechanical seal
 Seal code: Sic/Sic/Viton/SS316

Materials:

Wetted parts: Stainless steel EN 1.4408 SUS316
 Bearing: CU or SKF life sealed greased bearing
 Insulation: vertical
 Ambient temperature: 60 °C
 Maximum operating pressure: 25 bar
 Maximum pressure at stated temperature: 25 bar @ 0-90 °C

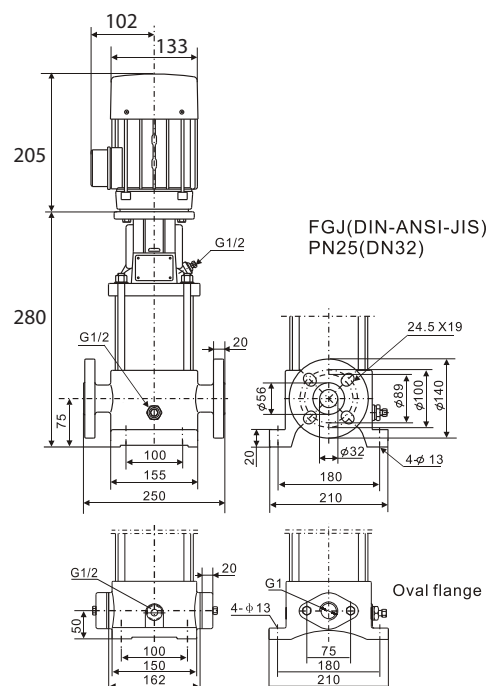
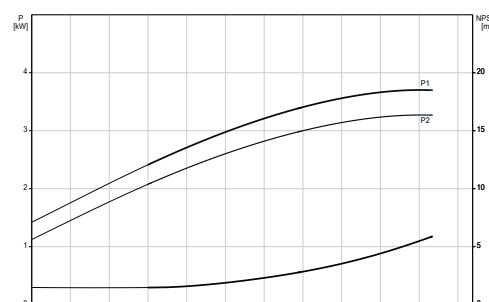
Motor type: by FUJIKA

IE efficiency class: IE3
 Rated power P2: 4 KW
 Power (P2) required by pump: 4 KW
 Frequency: 60HZ
 Rated voltage: 3 X 220-255V Δ / 380-440V Y
 Rated current: 13.6A / 7.8A
 Starting current: 1000-1470%
 Cos phi - power: 0.82-091
 Rotational speed: 3400-3450 rpm
 IE efficiency: IE3 87.3%
 Minimum efficiency: 86.9%
 Maximum efficiency: 88.5%
 Number of poles: 2
 Enclosure class: IP55
 Insulation class: F (IEC)

Dimensional:

Connection: UNIVERSAL flange drilled to both DIN & ANSI standards.
 Flange size for motor:
 Electrical data: IEC
 Motor standard: IEC DIN
 Approvals: CE
 Certified: IECEE, Coc
 Approvals for drinking water: WRAS
 Performance curve: ISO9906 2012

Gross weight: 53 KG
 Shipping volume: 0.092 m³
 Country of origin: PRC



Material resistivity to chemicals:

Stainless steel 316 material composition Up to :
18% chromium, 10% nickel, and 2% molybdenum.

Allows it to be used in variable applications and
that require superior corrosion resistance and
hygienic.

Applications such as:

Acidic water
Dissolved Chlorine
Sea water
Falls and Fountains
Water treatment
Diesel Fuel
Aggressive oil
Cooking oil
Drinking water
Juices
Water treatment
Food industry
Pharmaceutical Industry
Hygienic process
Fertilization
Fisheries
Irrigation

Characteristics:

The 316 stainless steel is used in salt and other
chlorides applications exposure and lasting
durability
steel over 304.

316 stainless steel's added molybdenum, makes it
much more corrosion resistant than 304 or other
stainless steel alloys. This allows 316 stainless steel
to be used in highly acidic environments that
other steels can't tolerate. 316's corrosion-resist-
ant properties also make it ideal in food process-
ing, pharmaceutical and medical applications due
to its ability to withstand the disinfectants and
high temperatures media.

Exposure to salt and chlorides can cause havoc on
304 stainless steel alloys. Salt can cause 304 stain-
less steel to rust by attacking the protective oxide
layer, and 304 steel is also more easily wear from
the effects of salt and chlorides. However, 316
stainless steel has all of the desirability properties of
304 stainless steel, but with the added benefit of
molybdenum, which allows it to withstand expo-
sure to salt and chlorides for more durability.

Austenitic stainless steels generally show high
corrosion rates when exposed to concentrated,
de-aerated alkaline aqueous environment at tem-
peratures above 60 °C.

Permissible corrosion rates of 0.05–2 mm/year.

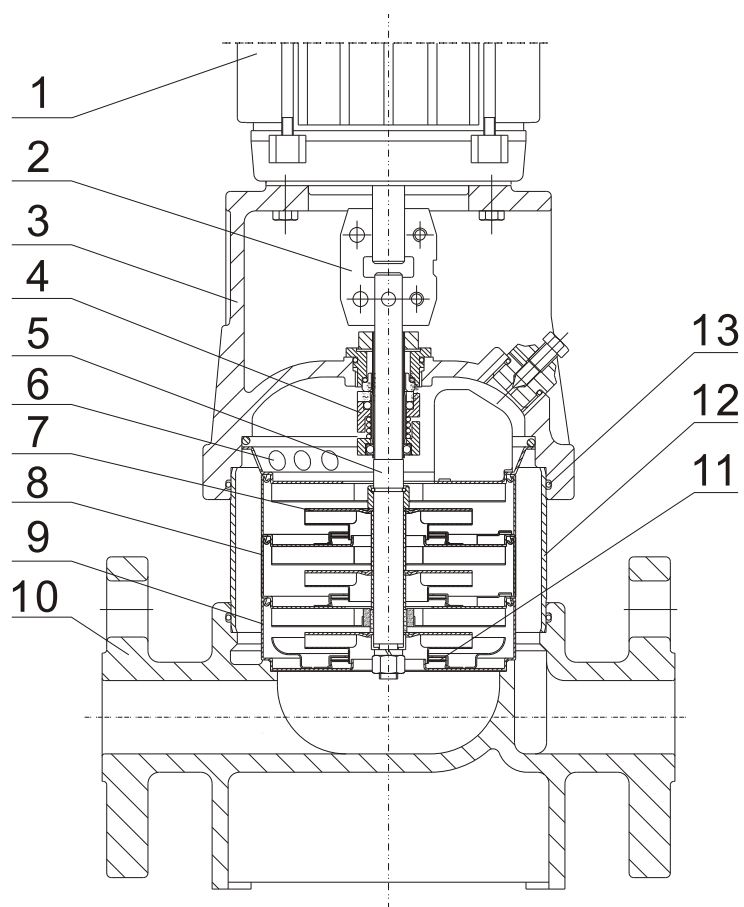
Caution:

The common stainless steel sheets types, 304 and
316 pumps should be considered non-resistant to
hydrochloric acid at any concentration and temper-
ature.

For Sulfuric acid:

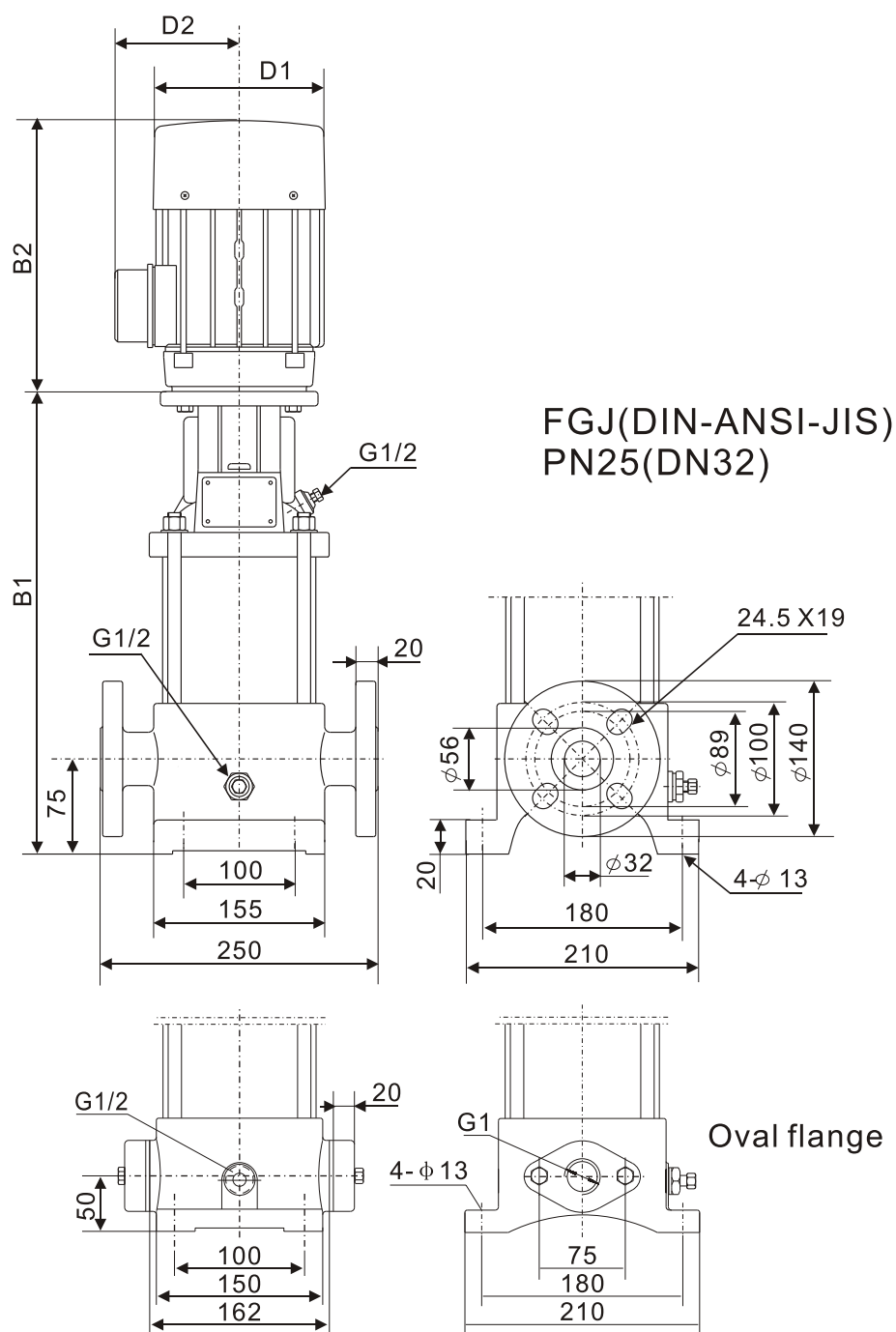
SUS304 is only resistant up to 1% acid while type
316 resistant is up to 10% acid at 50° C temp.

SECTIONAL DRAWING

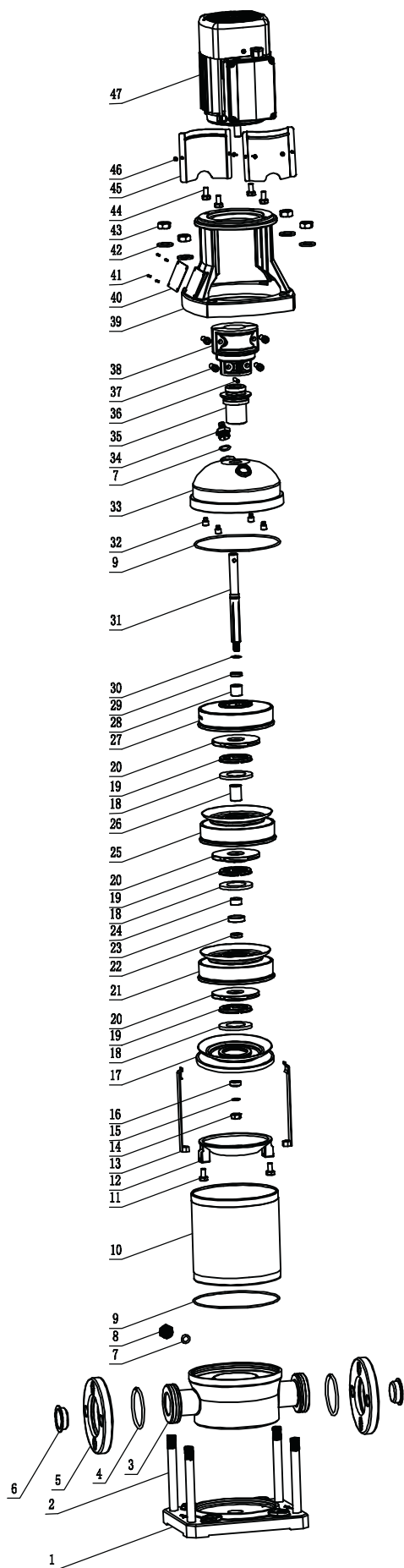


No	Description	Material	EN/DIN	AISI/ASTM
1	Motor	X	X	X
2	Shaft connector	X	X	X
3	Pump head	Cast iron	EN-JL 1030	ASTM25B
4	Mechanical seal	X	X	X
5	Shaft	S.S	X	AISI420
6	Outlet	S.S	1.4301	AISI304
7	Impeller	S.S	1.4301	AISI304
8	Hydraulic stack	S.S	1.4301	AISI304
9	Inlet	S.S	1.4301	AISI304
10	Pump body	Cast iron	EN.JL1030	ASTM25B
11	Neck ring	PTEF	X	X
12	Outer sleeve	S.S	1.4301	AISI316
13	O-ring	EDM/FKM	X	X

DIMENSIONS



Model	Dimensions (mm)					Weight
	B1	B2	B1+B2	D1	D2	KG
FCV-5-13VT	611	305	916	197	148	53



NO	Name	Spec.	Material	QTY
1	Base		HT200	1
2	Two-head bolt stud	M16XL	Chrome	4
3	Pump body		SUS304	1
4	Flange clasp		SUS304	2
5	Flange		SUS304	2
6	Dust cap	G1	PP	2
7	Ring	d18X2.4	NBR	2
8	Drain valve assembly		SUS304	1
9	Ring	d170X3.1	NBR	2
10	Housing		SUS316	1
11	Hexagon bolt	M8X20	SUS304	2
12	Fixed board		SUS316	1
13	Tension rod		SUS316	2
14	Type 1 non-metallic insert hexagon lock nut	M10	SUS316	1
15	Spring washer	ø10	SUS316	1
16	Impeller fix ring		SUS316	1
17	Inlet section		SUS316	1
18	Impeller fix ring		PTFE	N
19	Mouthpiece outer ring		SUS316	N
20	impeller		SUS316	N
21	Diffusor with bearing		SUS316	N
22	Short sleeve	L=4.5	SUS316	N
23	Bearing sleeve		Hard alloy	N
24	Short sleeve	L=10.5	SUS316	N
25	Diffusor		SUS316	N
26	Long sleeve	L=29	SUS316	N
27	Outlet section		SUS316	1
28	Short sleeve	L=19	SUS316	1
29	Snap ring seat		SUS316	1
30	Clasp		SUS316	1
31	Pump shaft		SUS316	1
32	Clasp seat shock colloidal		FI	4
33	Pump cover		SUS316	1
34	Discharge valve screw		SUS316	2
35	Mechanical seal	NJK-16-Q18FF	Carbon silicon carbide-FT	1
36	Cylindrical pin	ø6X25	SUS316	1
37	Hexagon socket head cap screw	M6X20	A3	4
38	Coupling			1
39	Motor base		HT200	1
40	Name plate		AL	1
41	Plate rivets	ø2.5	H62	4
42	Washer	ø16		4
43	Hexagonal nut	M16		4
44	Hexagonal bolt	M10X30	A3	4
45	Protective cover		SUS304	2
46	Cross recessed pan head screw and washer assemblies	M4X8	SUS304	4
47	Motor			1

- FCV-5-14VT
- Vertical Multi-Stage Pump

● Features :-

Stainless Steel 304

3 phase

220-380V

60HZ

IE3

2 pole

3500 rpm

Class 155(F)

Protection: IP55

● Product Photo :-



● Proformance Table :-

Model	Power		Max flow	Max head	Voltage	Inlet	Outlet
	KW	HP	m ³ /h	meters		mm	mm
FCV-5-14VT	4.0	5.5	10	131	220-380V	DN32	DN32

● Design

Vertical multistage in Line pump where Discharge and suction ports are located on one line to make it easier for connection access, it's either in Oval Victualic threaded to NPT cast iron discharge casing design or Slip on flanged and drilled to Universal standard which suite both DIN and ANSI standards.

● Applications: -

Water supply

Industry

Liquid transfer

Water treatment

Irrigation

● FCV Coding:

F: FUJIKA

C: Clean water

V: Vertical pump

● Working conditions: -

Temperature range: -20 Up to 104 °C

Ambient temperatue 50 °C

● Classifications:

FCV: Standard SS304

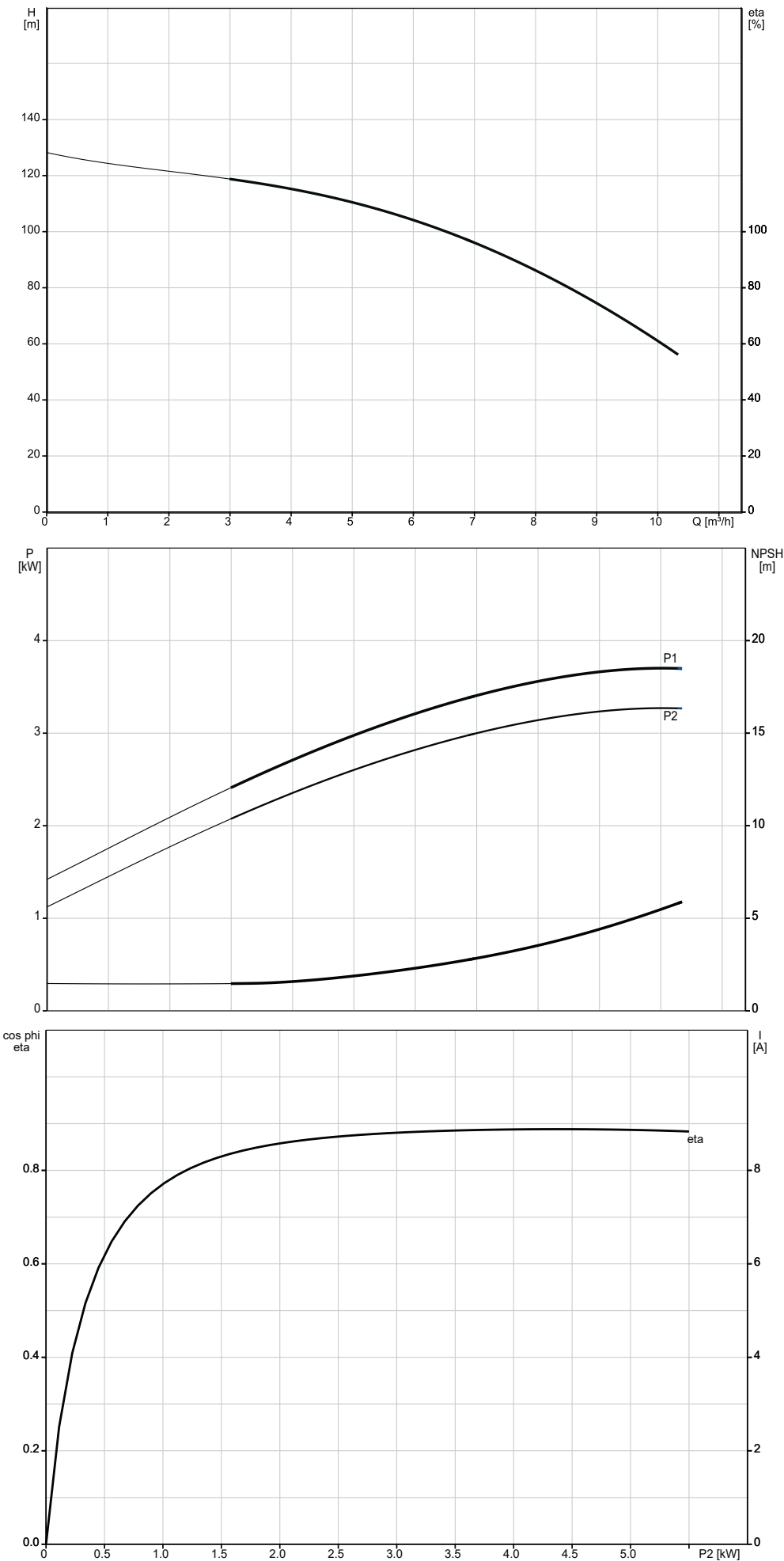
FCV-N: SS316

V: Oval flange

R: Round flange

T: Three phase

PERFORMANCE CURVE



Model: FCV-5-14VT



Design: Vertical multistage in Line pump where Discharge and suction ports are located on one line to make it easier for connection access, it's either in Oval Victualic threaded to NPT cast iron discharge casing design or Slip on flanged and drilled to Universal standard which suite both DIN and ANSI standards.

Construction of Material:

FCV basic design is made of :

Stainless steel imeller, sleeves and shaft , and cast iron discarge casing.

FCV-N

All wetted parts are made from SUS316 design.

The Shaft Mechanical seal is cartridge pull out type for easier replacement.

Mechanical Seal is EPDM /SUS304 for FCV and VITON / SUS316 for FCV-N Ensuring high reliability.

Coupling:

Shaft Coupling are rigid split type PJE Coupling.

Motor Type:

IE3 / IE2, 3-phase AC squirrel cage synchronous, high efficiency, TEFC fan cooled, IP55 Jetting protected.

General description :

The high efficiency ratings are in conformity to grade 3 efficiency value GB18613-2012 and IE of IEC60034-30 standard, but also the efficiency grade and installation dimensions are totally in conformity to IEC standard.

Motor shape is square motor as standard and round motor for larger sizes.

General specifications:

Operation factor: S1 (Continuous duty)

Insulation class: F

Protection class: IP55

Cooling ref.: IC411

Mount type: B35/B6

Ambient temperature: (-15 Up to 60 °C)

Altitude: ≤1000m

Motor paint finish is made with the most high technology with double painting and epoxy coating.

Motor housing construction are alaminum for motor rated up to 15HP and high grade cast iron for motors above rating.

The colour code: RAL 9005 deep jet black.

A standard split coupling connects the pump and motor shaft.

It's enclosed in the pump head/motor stool by means of two coupling guards.

The pump head and flange for motor mounting bracket is made in one piece (cast iron). the pump head cover is a seperate component (stainless steel).

Pump head cover stage has 1/2" priming plug and vent screw.

The pump is fitted with a balanced O-ring seal unit with a rigid torque-transmission system.

This seal type is assembled in a cartridge unit which makes replacement safe and easy.

In compliance with:

ENERGY EFFICIENCY ASIA JAPAN, IE3 IE2 IE1

MEPS (Minimum energy performance standard)

Current Regulation Energy saving act / top runner program standard

JIS C 4034-30

Power supply system

Three phase

Minimum energy performance IE3

Operated with inverter frequency: ok

Output (KW): 0.75 Up to 55KW

Number of poles: 2,4,6

Voltage (V): Up to 1000V

Frequency (HZ): 50HZ,60HZ,50/60HZ

Service duty: S1 ≥ 100%

Cooling method: tefc

Degree of protection: IP55

Area classification: safe area

Altitude: all

Ambient temperature: from (-20 °C and above)

Pump technical data

Rated flow: 5 m³/h
 Rated head: 106 meter
 Pump speed: 3450 rpm
 Nominal pressure PN: 25 bar
 Maximum pressure rating: 125% than shutoff pressure
 Pump mounting: vertical base mounting pump
 Shaft seal arrangement: single face cartridge type mechanical seal
 Seal code: Sic/Sic/Viton/SS316

Materials:

Wetted parts: Stainless steel EN 1.4408 SUS316
 Bearing: CU or SKF life sealed greased bearing
 Insulation: vertical
 Ambient temperature: 60 °C
 Maximum operating pressure: 25 bar
 Maximum pressure at stated temperature: 25 bar @ 0-90 °C

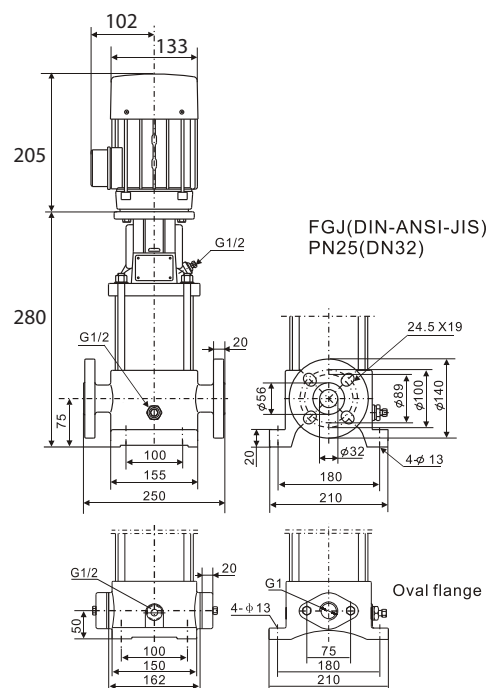
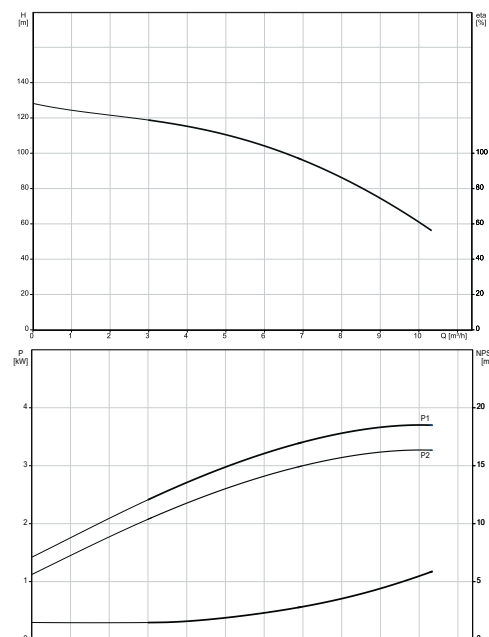
Motor type: by FUJIKA

IE efficiency class: IE3
 Rated power P2: 4 KW
 Power (P2) required by pump: 4 KW
 Frequency: 60HZ
 Rated voltage: 3 X 220-255V Δ / 380-440V Y
 Rated current: 13.6A / 7.8A
 Starting current: 1000-1470%
 Cos phi - power: 0.82-0.91
 Rotational speed: 3400-3450 rpm
 IE efficiency: IE3 87.3%
 Minimum efficiency: 84.7%
 Maximum efficiency: 88.5%
 Number of poles: 2
 Enclosure class: IP55
 Insulation class: F (IEC)

Dimensional:

Connection: UNIVERSAL flange drilled to both DIN & ANSI standards.
 Flange size for motor:
 Electrical data: IEC
 Motor standard: IEC DIN
 Approvals: CE
 Certified: IECEE, Coc
 Approvals for drinking water: WRAS
 Performance curve: ISO9906 2012

Gross weight: 54 KG
 Shipping volume: 0.13 m³
 Country of origin: PRC



Material resistivity to chemicals:

Stainless steel 316 material composition Up to :
18% chromium, 10% nickel, and 2% molybdenum.

Allows it to be used in variable applications and
that require superior corrosion resistance and
hygienic.

Applications such as:

Acidic water
Dissolved Chlorine
Sea water
Falls and Fountains
Water treatment
Diesel Fuel
Aggressive oil
Cooking oil
Drinking water
Juices
Water treatment
Food industry
Pharmaceutical Industry
Hygienic process
Fertilization
Fisheries
Irrigation

Characteristics:

The 316 stainless steel is used in salt and other
chlorides applications exposure and lasting
durability
steel over 304.

316 stainless steel's added molybdenum, makes it
much more corrosion resistant than 304 or other
stainless steel alloys. This allows 316 stainless steel
to be used in highly acidic environments that
other steels can't tolerate. 316's corrosion-resist-
ant properties also make it ideal in food process-
ing, pharmaceutical and medical applications due
to its ability to withstand the disinfectants and
high temperatures media.

Exposure to salt and chlorides can cause havoc on
304 stainless steel alloys. Salt can cause 304 stain-
less steel to rust by attacking the protective oxide
layer, and 304 steel is also more easily wear from
the effects of salt and chlorides. However, 316
stainless steel has all of the desirability properties of
304 stainless steel, but with the added benefit of
molybdenum, which allows it to withstand expo-
sure to salt and chlorides for more durability.

Austenitic stainless steels generally show high
corrosion rates when exposed to concentrated,
de-aerated alkaline aqueous environment at tem-
peratures above 60 °C.

Permissible corrosion rates of 0.05–2 mm/year.

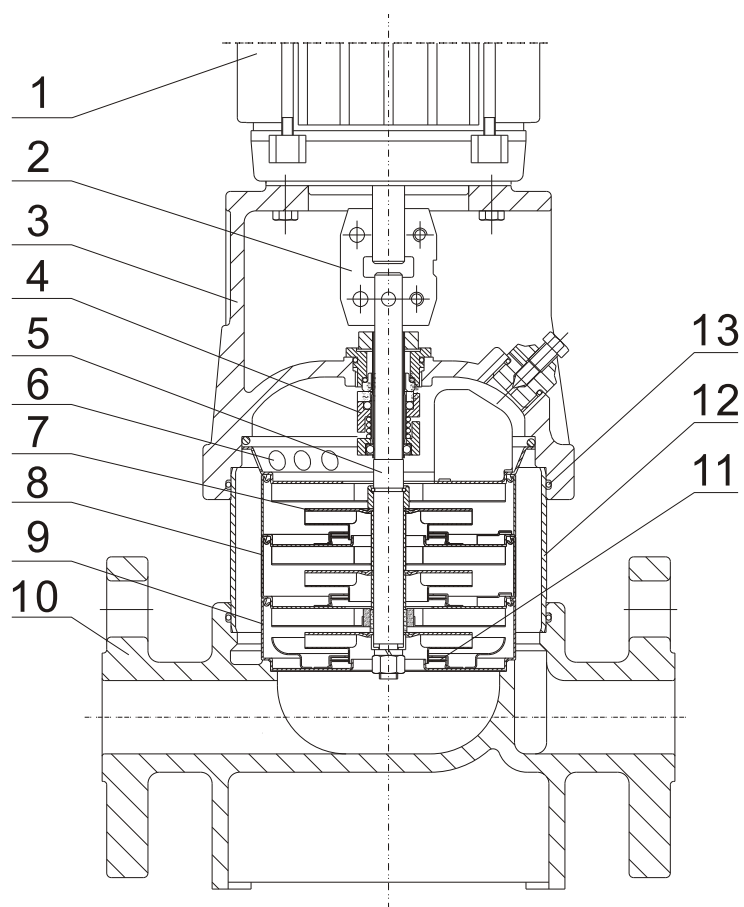
Caution:

The common stainless steel sheets types, 304 and
316 pumps should be considered non-resistant to
hydrochloric acid at any concentration and temper-
ature.

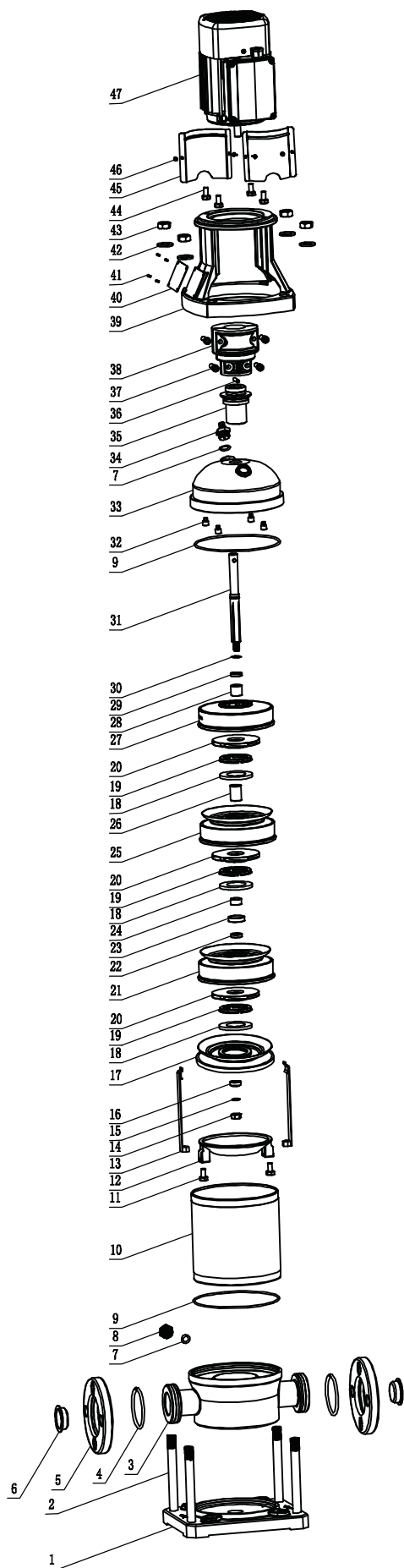
For Sulfuric acid:

SUS304 is only resistant up to 1% acid while type
316 resistant is up to 10% acid at 50° C temp.

SECTIONAL DRAWING



No	Description	Material	EN/DIN	AISI/ASTM
1	Motor	X	X	X
2	Shaft connector	X	X	X
3	Pump head	Cast iron	EN-JL 1030	ASTM25B
4	Mechanical seal	X	X	X
5	Shaft	S.S	X	AISI420
6	Outlet	S.S	1.4301	AISI304
7	Impeller	S.S	1.4301	AISI304
8	Hydraulic stack	S.S	1.4301	AISI304
9	Inlet	S.S	1.4301	AISI304
10	Pump body	Cast iron	EN.JL1030	ASTM25B
11	Neck ring	PTEF	X	X
12	Outer sleeve	S.S	1.4301	AISI316
13	O-ring	EDM/FKM	X	X



NO	Name	Spec.	Material	QTY
1	Base		HT200	1
2	Two-head bolt stud	M16XL	Chrome	4
3	Pump body		SUS304	1
4	Flange clasp		SUS304	2
5	Flange		SUS304	2
6	Dust cap	G1	PP	2
7	Ring	d18X2.4	NBR	2
8	Drain valve assembly		SUS304	1
9	Ring	d170X3.1	NBR	2
10	Housing		SUS316	1
11	Hexagon bolt	M8X20	SUS304	2
12	Fixed board		SUS316	1
13	Tension rod		SUS316	2
14	Type 1 non-metallic insert hexagon lock nut	M10	SUS316	1
15	Spring washer	ø10	SUS316	1
16	Impeller fix ring		SUS316	1
17	Inlet section		SUS316	1
18	Impeller fix ring		PTFE	N
19	Mouthpiece outer ring		SUS316	N
20	impeller		SUS316	N
21	Diffusor with bearing		SUS316	N
22	Short sleeve	L=4.5	SUS316	N
23	Bearing sleeve		Hard alloy	N
24	Short sleeve	L=10.5	SUS316	N
25	Diffusor		SUS316	N
26	Long sleeve	L=29	SUS316	N
27	Outlet section		SUS316	1
28	Short sleeve	L=19	SUS316	1
29	Snap ring seat		SUS316	1
30	Clasp		SUS316	1
31	Pump shaft		SUS316	1
32	Clasp seat shock colloidal		FI	4
33	Pump cover		SUS316	1
34	Discharge valve screw		SUS316	2
35	Mechanical seal	NJK-16-Q18FF	Carbon silicon carbide-FT	1
36	Cylindrical pin	ø6X25	SUS316	1
37	Hexagon socket head cap screw	M6X20	A3	4
38	Coupling			1
39	Motor base		HT200	1
40	Name plate		AL	1
41	Plate rivets	ø2.5	H62	4
42	Washer	ø16		4
43	Hexagonal nut	M16		4
44	Hexagonal bolt	M10X30	A3	4
45	Protective cover		SUS304	2
46	Cross recessed pan head screw and washer assemblies	M4X8	SUS304	4
47	Motor			1

● FCV-5-15VT

● Vertical Multi-Stage Pump

● Features :-

Stainless Steel 304

3 phase

220-380V

60HZ

IE3

2 pole

3500 rpm

Class 155(F)

Protection: IP55

● Product Photo :-



● Proformance Table :-

Model	Power		Max flow	Max head	Voltage	Inlet	Outlet
	KW	HP	m ³ /h	meters		mm	mm
FCV-5-15VT	4.0	5.5	10	140	220-380V	DN32	DN32

● Design

Vertical multistage in Line pump where Discharge and suction ports are located on one line to make it easier for connection access, it's either in Oval Victualic threaded to NPT cast iron discharge casing design or Slip on flanged and drilled to Universal standard which suite both DIN and ANSI standards.

● Applications: -

Water supply

Industry

Liquid transfer

Water treatment

Irrigation

● FCV Coding:

F: FUJIKA

C: Clean water

V: Vertical pump

● Classifications:

FCV: Standard SS304

FCV-N: SS316

V: Oval flange

R: Round flange

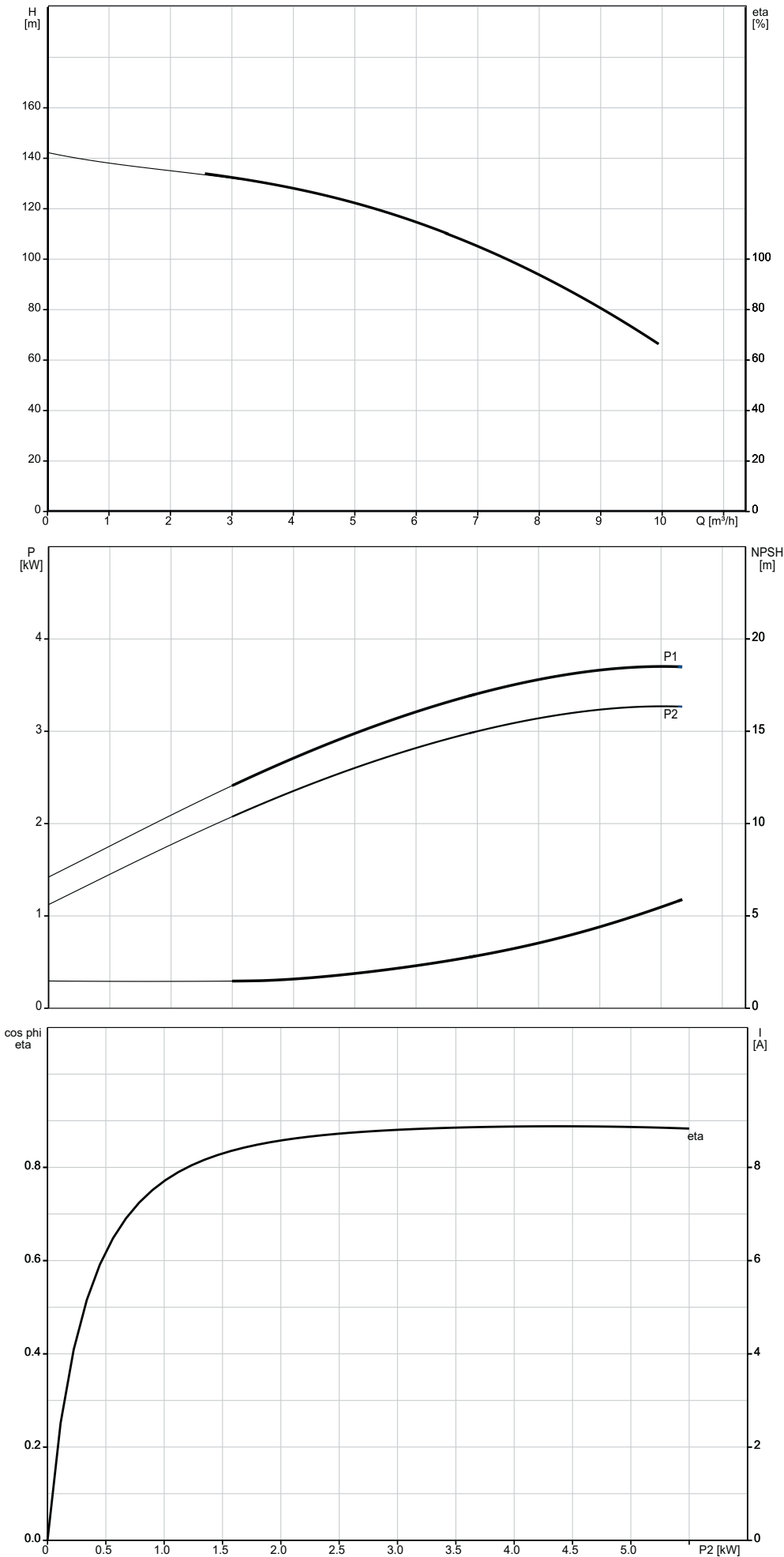
T: Three phase

● Working conditions: -

Temperature range: -20 Up to 104 °C

Ambient temperatue 50 °C

PERFORMANCE CURVE



Model: FCV-5-15VT



Design: Vertical multistage in Line pump where Discharge and suction ports are located on one line to make it easier for connection access, it's either in Oval Victualic threaded to NPT cast iron discharge casing design or Slip on flanged and drilled to Universal standard which suite both DIN and ANSI standards.

Construction of Material:

FCV basic design is made of :

Stainless steel imeller, sleeves and shaft , and cast iron discarge casing.

FCV-N

All wetted parts are made from SUS316 design.

The Shaft Mechanical seal is cartridge pull out type for easier replacement.

Mechanical Seal is EPDM /SUS304 for FCV and VITON / SUS316 for FCV-N Ensuring high reliability.

Coupling:

Shaft Coupling are rigid split type PJE Coupling.

Motor Type:

IE3 / IE2, 3-phase AC squirrel cage synchronous, high efficiency, TEFC fan cooled, IP55 Jetting protected.

General description :

The high efficiency ratings are in conformity to grade 3 efficiency value GB18613-2012 and IE of IEC60034-30 standard, but also the efficiency grade and installation dimensions are totally in conformity to IEC standard.

Motor shape is square motor as standard and round motor for larger sizes.

General specifications:

Operation factor: S1 (Continuous duty)

Insulation class: F

Protection class: IP55

Cooling ref.: IC411

Mount type: B35/B6

Ambient temperature: (-15 Up to 60 °C)

Altitude: ≤1000m

Motor paint finish is made with the most high technology with double painting and epoxy coating.

Motor housing construction are alaminum for motor rated up to 15HP and high grade cast iron for motors above rating.

The colour code: RAL 9005 deep jet black.

A standard split coupling connects the pump and motor shaft.

It's enclosed in the pump head/motor stool by means of two coupling guards.

The pump head and flange for motor mounting bracket is made in one piece (cast iron). the pump head cover is a seperate component (stainless steel).

Pump head cover stage has 1/2" priming plug and vent screw.

The pump is fitted with a balanced O-ring seal unit with a rigid torque-transmission system.

This seal type is assembled in a cartridge unit which makes replacement safe and easy.

In compliance with:

ENERGY EFFICIENCY ASIA JAPAN, IE3 IE2 IE1

MEPS (Minimum energy performance standard)

Current Regulation Energy saving act / top runner program standard

JIS C 4034-30

Power supply system

Three phase

Minimum energy performance IE3

Operated with inverter frequency: ok

Output (KW): 0.75 Up to 55KW

Number of poles: 2,4,6

Voltage (V): Up to 1000V

Frequency (HZ): 50HZ,60HZ,50/60HZ

Service duty: S1 ≥ 100%

Cooling method: tefc

Degree of protection: IP55

Area classification: safe area

Altitude: all

Ambient temperature: from (-20 °C and above)

Pump technical data

Rated flow: 5 m³/h
 Rated head: 114 meter
 Pump speed: 3450 rpm
 Nominal pressure PN: 25 bar
 Maximum pressure rating: 125% than shutoff pressure
 Pump mounting: vertical base mounting pump
 Shaft seal arrangement: single face cartridge type mechanical seal
 Seal code: Sic/Sic/Viton/SS316

Materials:

Wetted parts: Stainless steel EN 1.4408 SUS316
 Bearing: CU or SKF life sealed greased bearing
 Insulation: vertical
 Ambient temperature: 60 °C
 Maximum operating pressure: 25 bar
 Maximum pressure at stated temperature: 25 bar @ 0-90 °C

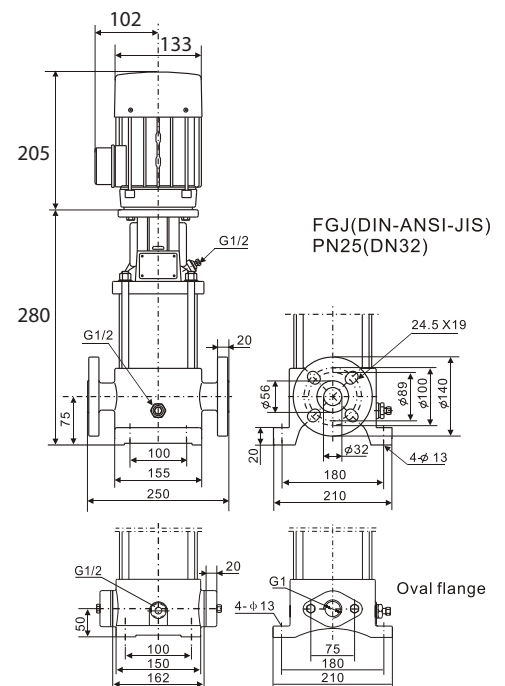
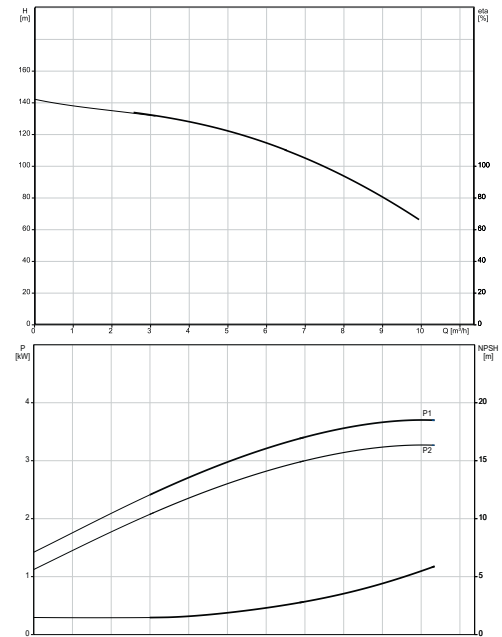
Motor type: by FUJIKA

IE efficiency class: IE3
 Rated power P2: 4 KW
 Power (P2) required by pump: 4 KW
 Frequency: 60HZ
 Rated voltage: 3 X 220-255V Δ / 380-440V Y
 Rated current: 13.6A / 7.8A
 Starting current: 1000-1470%
 Cos phi - power: 0.82-0.91
 Rotational speed: 3400-3450 rpm
 IE efficiency: IE3 87.3%
 Minimum efficiency: 84.7%
 Maximum efficiency: 88.5%
 Number of poles: 2
 Enclosure class: IP55
 Insulation class: F (IEC)

Dimensional:

Connection: UNIVERSAL flange drilled to both DIN & ANSI standards.
 Flange size for motor:
 Electrical data: IEC
 Motor standard: IEC DIN
 Approvals: CE
 Certified: IECEE, Coc
 Approvals for drinking water: WRAS
 Performance curve: ISO9906 2012

Gross weight: 55 KG
 Shipping volume: 0.13 m³
 Country of origin: PRC



Material resistivity to chemicals:

Stainless steel 316 material composition Up to :
18% chromium, 10% nickel, and 2% molybdenum.

Allows it to be used in variable applications and
that require superior corrosion resistance and
hygienic.

Applications such as:

Acidic water
Dissolved Chlorine
Sea water
Falls and Fountains
Water treatment
Diesel Fuel
Aggressive oil
Cooking oil
Drinking water
Juices
Water treatment
Food industry
Pharmaceutical Industry
Hygienic process
Fertilization
Fisheries
Irrigation

Characteristics:

The 316 stainless steel is used in salt and other
chlorides applications exposure and lasting
durability
steel over 304.

316 stainless steel's added molybdenum, makes it
much more corrosion resistant than 304 or other
stainless steel alloys. This allows 316 stainless steel
to be used in highly acidic environments that
other steels can't tolerate. 316's corrosion-resist-
ant properties also make it ideal in food process-
ing, pharmaceutical and medical applications due
to its ability to withstand the disinfectants and
high temperatures media.

Exposure to salt and chlorides can cause havoc on
304 stainless steel alloys. Salt can cause 304 stain-
less steel to rust by attacking the protective oxide
layer, and 304 steel is also more easily wear from
the effects of salt and chlorides. However, 316
stainless steel has all of the desirability properties of
304 stainless steel, but with the added benefit of
molybdenum, which allows it to withstand expo-
sure to salt and chlorides for more durability.

Austenitic stainless steels generally show high
corrosion rates when exposed to concentrated,
de-aerated alkaline aqueous environment at tem-
peratures above 60 °C.

Permissible corrosion rates of 0.05–2 mm/year.

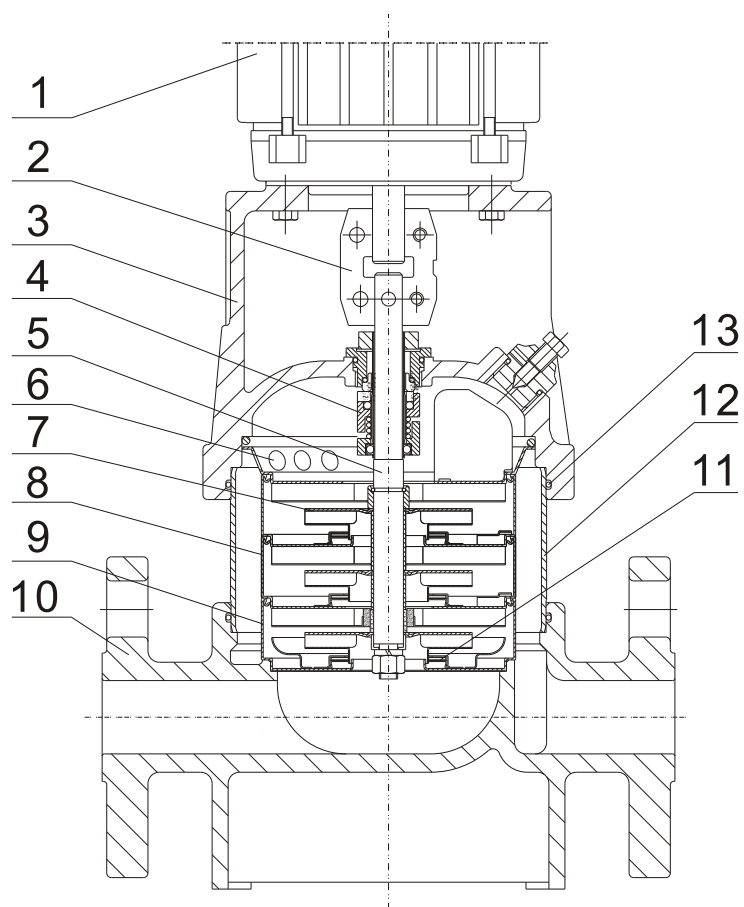
Caution:

The common stainless steel sheets types, 304 and
316 pumps should be considered non-resistant to
hydrochloric acid at any concentration and temper-
ature.

For Sulfuric acid:

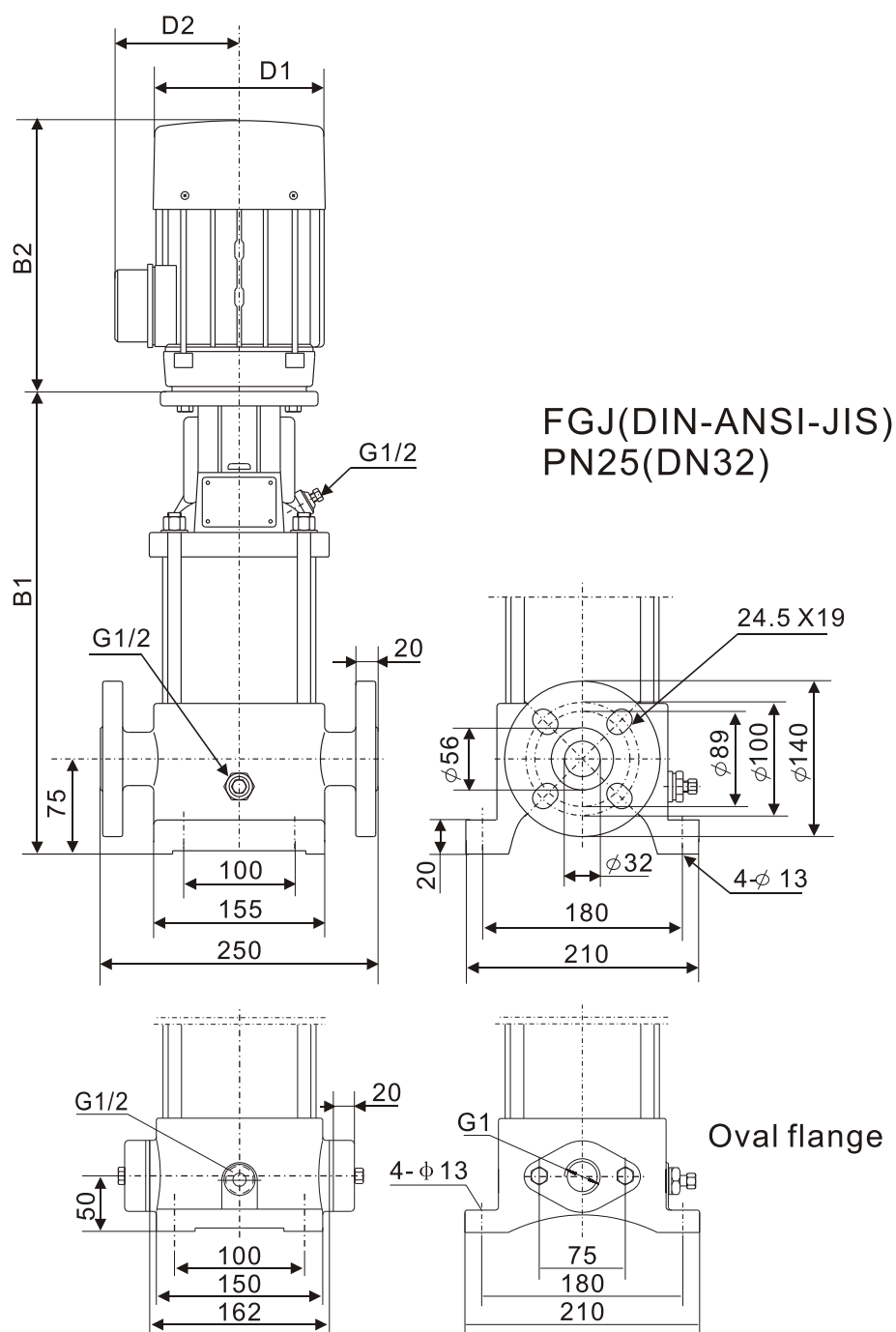
SUS304 is only resistant up to 1% acid while type
316 resistant is up to 10% acid at 50° C temp.

SECTIONAL DRAWING

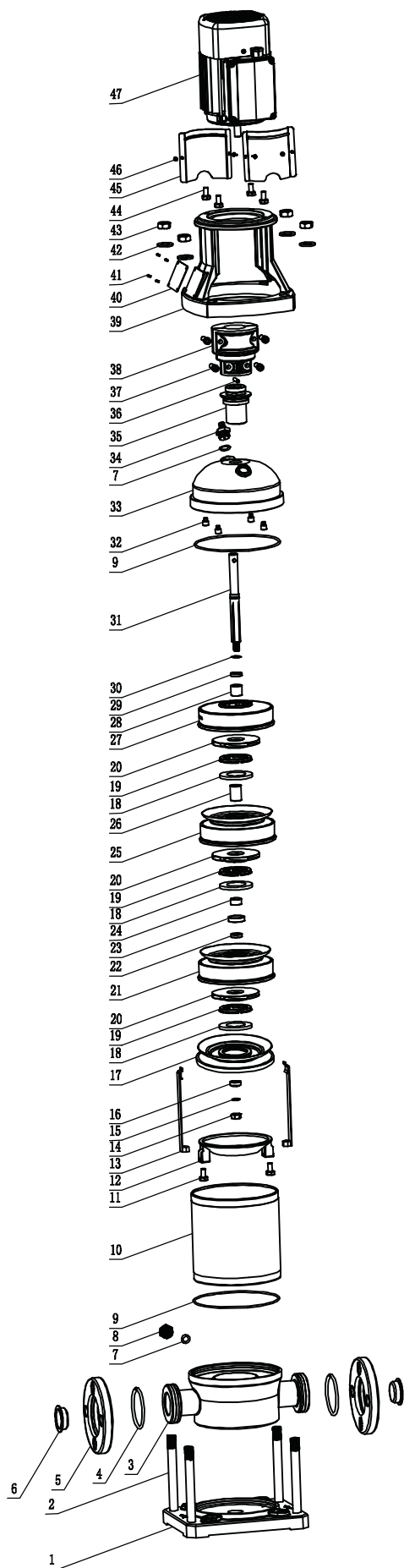


No	Description	Material	EN/DIN	AISI/ASTM
1	Motor	X	X	X
2	Shaft connector	X	X	X
3	Pump head	Cast iron	EN-JL 1030	ASTM25B
4	Mechanical seal	X	X	X
5	Shaft	S.S	X	AISI420
6	Outlet	S.S	1.4301	AISI304
7	Impeller	S.S	1.4301	AISI304
8	Hydraulic stack	S.S	1.4301	AISI304
9	Inlet	S.S	1.4301	AISI304
10	Pump body	Cast iron	EN.JL1030	ASTM25B
11	Neck ring	PTEF	X	X
12	Outer sleeve	S.S	1.4301	AISI316
13	O-ring	EDM/FKM	X	X

DIMENSIONS



Model	Dimensions (mm)					Weight
	B1	B2	B1+B2	D1	D2	KG
FCV-5-15VT	665	305	970	197	148	55



NO	Name	Spec.	Material	QTY
1	Base		HT200	1
2	Two-head bolt stud	M16XL	Chrome	4
3	Pump body		SUS304	1
4	Flange clasp		SUS304	2
5	Flange		SUS304	2
6	Dust cap	G1	PP	2
7	Ring	d18X2.4	NBR	2
8	Drain valve assembly		SUS304	1
9	Ring	d170X3.1	NBR	2
10	Housing		SUS316	1
11	Hexagon bolt	M8X20	SUS304	2
12	Fixed board		SUS316	1
13	Tension rod		SUS316	2
14	Type 1 non-metallic insert hexagon lock nut	M10	SUS316	1
15	Spring washer	ø10	SUS316	1
16	Impeller fix ring		SUS316	1
17	Inlet section		SUS316	1
18	Impeller fix ring		PTFE	N
19	Mouthpiece outer ring		SUS316	N
20	impeller		SUS316	N
21	Diffusor with bearing		SUS316	N
22	Short sleeve	L=4.5	SUS316	N
23	Bearing sleeve		Hard alloy	N
24	Short sleeve	L=10.5	SUS316	N
25	Diffusor		SUS316	N
26	Long sleeve	L=29	SUS316	N
27	Outlet section		SUS316	1
28	Short sleeve	L=19	SUS316	1
29	Snap ring seat		SUS316	1
30	Clasp		SUS316	1
31	Pump shaft		SUS316	1
32	Clasp seat shock colloidal		FI	4
33	Pump cover		SUS316	1
34	Discharge valve screw		SUS316	2
35	Mechanical seal	NJK-16-Q18FF	Carbon silicon carbide-FT	1
36	Cylindrical pin	ø6X25	SUS316	1
37	Hexagon socket head cap screw	M6X20	A3	4
38	Coupling			1
39	Motor base		HT200	1
40	Name plate		AL	1
41	Plate rivets	ø2.5	H62	4
42	Washer	ø16		4
43	Hexagonal nut	M16		4
44	Hexagonal bolt	M10X30	A3	4
45	Protective cover		SUS304	2
46	Cross recessed pan head screw and washer assemblies	M4X8	SUS304	4
47	Motor			1